

Math 320 An Introduction to the Mathematics of Finance

Numbered Transcript

MATH 320 • Dr. Paul Wilmott

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Lecture 1: The Mathematical Finance Toolkit

At a glance — Wilmott lays out the mathematical finance toolkit.

- 1.1 Hi, everybody. Thank you for coming to my course on the Introduction to Mathematical Finance. Before I tell you the things that we're going to be talking about, I do want to talk about myself, a little bit about my background, and what brings me here.
- 1.2 I guess I've been a mathematician since I was so high. I've actually been running businesses since I was about this high, not very serious businesses, but little tiny businesses. And I was probably a centimeter or two taller when I started working in mathematical finance. But I've reached an age where you start to shrink, sadly. My background is in mathematics, as what is called an applicable mathematician. That means I used to work on all different branches of mathematics applied to real-world problems, often industrial problems. And this means I was very lucky to see different types of mathematics, different types of mathematics in different fields. I did a lot of stuff in fluid mechanics, for example.
- 1.3 Then, sometime in the late '80s, I was introduced to finance. In those days, the only people who did maths of finance were in business departments or economics departments. No one in maths departments knew anything about finance, nothing at all. It's interesting, the politics of mathematics within universities. You have the hierarchy. At the very top of the hierarchy are the very pure abstract mathematicians, or mathematics, Fermat's last theorem, that sort of thing, the sort of thing you win medals for.
- 1.4 Lower down, you have applied mathematics, but applied in a way that isn't applied. It doesn't really acknowledge the existence of the real world. Further down, you have what I was working in, the applicable mathematics or industrial mathematics. You're kind of looked down upon by all the other people, but you have the most fun. It's much more interesting. You'll see lots of different types of problems.
- 1.5 And then, when finance started to appear in maths universities, they had to invent a whole new category, which was way down here, politically speaking, because it involved money. And in universities, they have this love-hate relationship with money. But it's even more fun than fluid mechanics. So that's where I'm coming from, is just enjoying the mathematics of finance without trying to make it too complicated.
- 1.6 And that's something, if you study this subject at university, you will realize that, very often, they try to make the subject really complicated, so that it's up here. It's respectable if it's complicated. Sometimes you buy books on mathematical finance, and they're down here, dumbing it down too far. I believe that's the right level for the mathematics of finance. And that's what we're aiming at in these next few lectures, the right level. Not too fancy, not too basic.

- 1.7** Anyway, the subject is mathematical finance, otherwise known as quantitative finance or financial engineering. Who here is a mathematician? Do we have any mathematicians? No mathematicians? Okay, that's fine. We're going to start very, very basic. We're going to get to slightly above basic. Nothing complicated, nothing too sophisticated. If you get into mathematical finance, you'll hear things like stochastic calculus and things like that. Really complicated stuff. I'm not going to go anywhere near those. This is a taster, really, for how you can apply mathematics to finance.
- 1.8** And by finance, I don't mean mortgages or credit scores or student loans, that sort of thing. I'm talking about things they do in banks and hedge funds. To give you an idea of how big the subject is, apparently there's something like a quadrillion dollars' worth of business done in these instruments. So we're going to be talking about a quadrillion dollars. That's even more than Elon Musk is going to earn. A quadrillion dollars is 1 followed by 15 zeros. So it's a big business, this; big subject.
- 1.9** Right. What are we going to talk about? First of all, the ideas behind quant finance go back to the early 1900s. There's a guy called Louis Bachelier who started work on applying the ideas of introducing randomness to the behavior of share prices. So rather than try to predict what share prices are going to do, he just said, "Let's assume they're random." That work disappeared for 50-odd years. He was French, and nobody knew about the work. It just disappeared. Anyway, it was rediscovered.
- 1.10** And then in the 1950s, people started working on what's called portfolio theory. Again, applying ideas of randomness to the way shares behave. And then, there were these things called options. I don't know if you know what options are, but we're going to be talking about them later on in the course. These are quite sophisticated financial instruments— sophisticated because they threw up problems in how to value these instruments, things that it took people 20-odd years to figure out.
- 1.11** There was actually three gentlemen called Black, Scholes, and Merton, who came up with the Nobel Prize-winning idea for how to value these things called options. So that's something that we're going to be talking about. And then, as I say, it became an academic subject, where what used to be fun, they made respectable by making too complicated. And I specialize in, what do you call it, the Goldilocks zone, getting the right level of mathematics. And that's what we're aiming for here.
- 1.12** So the overview of my lectures, we're going to start off with some mathematics, really basic mathematics, really embarrassingly basic mathematics. Then we're going to move on to fixed income, interest rates, then equities, et cetera, modeling equities, and this idea of randomness. We'll do some simulations. I'm going to do some work on spreadsheet stuff, live, before your very eyes; I shall be building up spreadsheets. Simulations, portfolios, a thing called modern portfolio theory. Then options, which is the really fun bit where you've got some really clever ideas.

- 1.13** Then, we end on two lectures, one on behavioral finance. Behavioral finance is a fun subject, but at the moment, it's mainly for dinner party conversations. It's not really applied that much in practice. And then, of course, AI, artificial intelligence, machine learning, which one has to do, although as I'm speaking, everything I've just said is immediately out of date. Anything we cover in AI is going to be redundant straightaway, I'm afraid. It'll just be an overview thing.
- 1.14** Okay, first bit. The first lecture starts now. We're going to do some calculus and some basic probability, and the three things we need from calculus—oh, who likes mathematics? Do you like mathematics? You love mathematics. One question I always ask audiences, do you prefer doing things with symbols or numbers? So who likes to do things with numbers? You need to see the concrete, the 5s and 10s and the 27s. Half the audience. And who likes symbols? Okay, it's always that way. It's always 50-50, which makes things a challenge.
- 1.15** If you're going to do anything, make any real progress with any mathematical subject, you've got to do the symbols. Numbers are great because you can't fool anyone with numbers, right? I could fool anybody with symbols, and I could write all sorts of nonsense down here. Maybe I shall. But you'd be far too polite, I suspect, to say, "Paul, that's nonsense." I'd get away with it for a while. But with numbers, I can't get away with anything. $5 * 7 = 35$, whatever I say it is. What's $7 * 8$, though? That's a problem. That's a difficult one, isn't it? $7 * 8$. Sorry, I get easily distracted.
- 1.16** The symbols, though, is where you see the structure. You can't see structure if you're just doing numbers. If I have some calculation over here, and there's a 5 over here and there's a 5 down there, you don't know whether they're the same 5, a volatility in interest rate, or two volatilities, or a volatility in an interest rate. If you see the symbols, you know that they're different things. So symbols are better. As soon as I see some mathematical problem, if they're using symbols I'm familiar with in a field I'm familiar with, I immediately know what we're talking about. But ultimately, if you are going to work in a bank or a hedge fund, you've got to come up with the numbers. You've got to say, "We need to buy however many millions of something to optimize our portfolio." You need the numbers.
- 1.17** So there's two things: calculus and basic probability. We're going to start off with two special functions, and the thing called Taylor series. All right, really easy, embarrassingly easy. The first thing is the exponential function: e^x , a very, very important subject, function e^x . We'll come back to e^x in a second. What's 2^x ? e is just a number, but let's get things that you're used to: 2; 2^x . What does it look like, 2^x ? Well, here's x . Here's 2^x . If $x = 0$, it's 1. If $x = 1$, it's 2. [If $x = 2$], it's 4, and so on. 2^0 , 2^1 and so on. It grows. And you can imagine, if you haven't got integers here or you've got halves, you've got square roots, et cetera, basically, it's this function. It looks like that. It grows very rapidly.

- 1.18** What's the slope of this thing? The slope or the gradient. We're always wanting to know in mathematics, what's the slope or the gradient of something? To calculate the gradient of this thing, you just pick some point, x , just some arbitrary point x , and then you go to another point, which is just a teeny bit further away. Let's call it $x + \epsilon$. We like ϵ in mathematics. We like Greek letters. And what you do is, you say, what's the value at here? That's $2^{x+\epsilon}$. What about here? That's 2^x . So what's the slope? Well, you take that difference, though, which is, if I'm not mistaken, $2^{x+\epsilon} - 2^x$, which is that, divided by that distance, which is ϵ , which is equal to, you can take the 2^x out, you get $2^x(2^\epsilon - 1) / \epsilon$.
- 1.19** Then you ask, mathematically, we always like to do, the limits of things go to 0. What happens as ϵ goes to 0? Well, this gets closer, this point here gets closer and closer to this point here, and the slope here becomes the same as the slope at that point. So, rather than being a slope between these two points, it becomes, in the limit, as ϵ goes to 0, the slope at that point. What does this become as ϵ goes to 0? I haven't got a calculator, but if I had to calculate, I seem to remember that, if I put, say, $\epsilon = 0.0001$, that becomes something like, obviously, it's 2^x , this converges to something like ≈ 0.6 . I can't remember exactly. Not very interesting.
- 1.20** What is interesting is, you could ask the question: what number should go in here instead of 2, to make it so that this calculation here $[(2^\epsilon - 1) / \epsilon]$ becomes 1? In other words, let's call it e , e^x . Everything is exactly the same. Get down to here, you have that the slope equals e^x times the limit of some number, it's just a number, nothing special, to the something small, minus 1 over ϵ [slope = $e^x(e^\epsilon - 1) / \epsilon$]. As ϵ goes to 0, this tends to some limit. What is the limit, or what choice of e makes that limit 1? That number is 2.718. If you have $e = 2.718\dots$, goes on forever, then that becomes 1, and the slope is equal to e^x .
- 1.21** In other words, that function is the only function which, when you differentiate it, when you look at the slope, the slope is the same as the function itself. And then obviously, the second derivative. So this is, mathematically, we write dy —sorry, if $y = e^x$, then $(dy / dx) = e^x$, and so on. The second derivative, every derivative is the same. So very, very special function. This thing here, this e , crops up all the time, all over the place, for no apparent reason, just appears magically. You'll see it shortly. So that's the first thing you need to know.
- 1.22** The second thing you need to know is, suppose I looked at $y = e^x$, and I said, I put in—let me get some numbers here. I don't usually do numbers. That's why I'm cheating by having numbers in front of me here. If I just said, for example, $e^{3.5}$, suppose I said, 3.5. What's $e^{3.5}$? It turns out to be 33— whoops, not there, over here—33.11.
- 1.23** Now, what if I turned the picture around, and I said, suppose I had $e^x = 33.11$, et cetera, et cetera. What is the x that makes that so? What I do is, I flip that to a mirror image of that. Look at this picture here, which has now got y along here and x along here, and it looks like that. So that's the same picture, except a mirror image through the 45-degree

slope. So then I say, well, if $y = 33.11$, what is the x that makes it so? 3.5. That is another function. So this is the e^x function. This is the function, the log function. So, logs and exponentials are kind of the opposite of each other. You take a log of something, they go backwards, you take the exponential of it.

- 1.24** What else can I say here? I can say that—let me change color just for a bit of variety. When we do some things in spreadsheets, this function is written as $\exp(x)$ in a spreadsheet, in Excel, and this function is $\log$, $\ln$, the log function.
- 1.25** What else can I say about these things? Because they're in—so yes, so sometimes you write x , and sometimes you write e . So, $e^{\ln x}$, for example, is just x because you go from here to here, which is the same thing, then go from here to here, and you get back to where you started from. Equally, $\ln(e^x) = x$ because they invert each other. They're inverses. So that was really basic stuff. You need those functions. They're going to crop up all the time.
- 1.26** Now, another thing you need to know, and that is, we're always differentiating things, finding the slope of things. I'd like to know, we know what the slope of this function is. Slope is e^x , second derivative is e^x , they're all e^x . What about the slope of this thing? So let me show you what we can do here.
- 1.27** Suppose, if we've got, let me write it as, let me write $y = \ln(x)$. Then, because the log and exponential inverse, that's the same as $x = e^y$. So differentiate, do $dx / dy = e^y$, which is equal to, which is the same as x , where we started out. Turn it upside down, $dy / dx = 1/x$ —well, turn it upside down— $1/x$. So, what have we got? We've got two special functions. The exponential function, the derivative is itself, and then the logarithm, which is the opposite function, and that has the slope $1/x$; $dy / dx = 1/x$.
- 1.28** The next thing you need to know is a lovely little idea called Taylor series. Taylor series. This is about approximating functions, really. So suppose you have some function, $f(x)$, here. Apologies for my handwriting, as I say, I haven't written, used pen and paper for decades. Suppose you've got some function. It looks like some random function like that, and you're sitting at some arbitrary point, x .
- 1.29** Very often, you want to know what's going on around here. Suppose x , I know this is $f(x)$, the point f of x here. Well, what if I move a little bit further away, to the point $x + \epsilon$? That's a distance ϵ . Well, obviously, this is $f(x + \epsilon)$. Can we somehow estimate what that value is, knowing only information about what's going on at this point? That point there. Obviously, approximately.
- 1.30** Now, if ϵ is really tiny, then this $f(x + \epsilon)$ would be really close to $f(x)$. That's obvious, isn't it, I think: $f(x + \epsilon) \approx f(x)$ where ϵ is tiny, when ϵ is tiny. For reasons which will become apparent, sometimes we need to know what the tiny bit is. It's going to be some plus something. Well, one thing is—again, I think it's pretty obvious—is that, if we know what the slope is at that point, df / dx . Suppose we know what df / dx is, the slope at the point x , not at the point $x + \epsilon$, but just at that same point.

- 1.31** Then, as you go along a distance ϵ , you go up how much? Well, you go up, it depends on what the slope is. The further you go along, the more you go up, so this is a good approximation, just ϵ times the slope. You're walking this direction, there's a slope like that. The further you walk in a direction, the higher up you go. It depends on the slope. But it's a slope at that point. That's the key thing.
- 1.32** Now, that's great. But there's some curvature going on in there. This isn't perfect; it's approximately. There's going to be some curvature, and maybe the curvature is going to matter, or maybe ϵ is not really tiny. If it's a little bit bigger, you need to have a better approximation. So what I'm going to do is, I'm going to do this properly, rather than waving my hands and say, "Well, it's sort of obvious, isn't it?" I'm going to do this: I'm going to write this as what's called a power series, powers of ϵ . So it's going to be a linear thing, linear $[a_1 \epsilon]$, then we've got quadratic, some sort of curvature thing $[a_2 \epsilon^2]$, then cubic $[a_3 \epsilon^3]$, and you can imagine it goes on forever, potentially.
- 1.33** So, the question is: what are these coefficients? I've told you what this one is. It's the slope, df/dx , at that point. But what about the next term, and the next term, and the next? Is it obvious what the 17th term is going to be? Not at all. It's not even totally obvious what that term is going to be. But there's a trick. And the trick is, differentiate with respect to—this is all mathspeak—diff, w.r.t, with respect to, ϵ . Differentiate. So the left-hand side becomes $df/d\epsilon$. Well, x and ϵ are the same. If you differentiate with respect to x , it's the same as differentiating with respect to ϵ , because $x + \epsilon$ are symmetrical like that. So rather than write $df/d\epsilon$, you might as well write df/dx . So that is equal to—remember, I'm differentiating with respect to ϵ , not x . There are no ϵ s here—0, plus, differentiate with respect to ϵ , that's 1, so you've got plus a_1 plus differentiate with respect to ϵ here. You get $2a_2\epsilon$, then $3a_3\epsilon^2$, and so on $[df/dx = 0 + a_1 + 2a_2\epsilon + 3a_3\epsilon^2 \dots]$.
- 1.34** All right. Now, you just set $\epsilon = 0$. If you put $\epsilon = 0$, that becomes df/dx , at that point x . So that is just $df/dx|_x$ —that vertical line with x at the bottom means evaluated at the point x , not anywhere else, not at $x + \epsilon$. At the point x , that equals $0 + a_1$ —well, all of those are 0. So, I'll write it up here: $a_1 = df/dx|_x$, which is exactly what I said to start with. My intuition was right about going along there and going up a bit. All right, now we go back to here, and we differentiate. So we forget that line there.
- 1.35** We go back to that line and differentiate with respect to ϵ . Again, we get: $d^2f/dx^2 =$ —well, there are no ϵ s, so that's 0, that becomes $2a_2$, because differentiated ϵ gives you 1, plus now 3×2 , differentiate that. I'm going to write that as $3! - 3 \times 2 \times 1$, et cetera—times ϵ plus, more terms. So $\epsilon = 0$, gives $d^2f/dx^2|_x = 2a_2 + 0$. So, $a_2 = 1/2(d^2f/dx^2)|_x$.
- 1.36** Anyway, go back here, differentiate again. With respect to ϵ , you're going to see a $3!$, then set up, and everything else is going to be 0 when you set up $\epsilon = 0$. What you end up with is $a_3 = 1/3!(d^3f/dx^3)|_x$. And so on, it goes like that. The n^{th} term is $1/n!$, meaning $n, n-1, n-2$, et cetera— $a_n = 1/n!(d^n f/dx^n)|_x$. So that gives us, we can go back to here and write this whole expression here as $f(x + \epsilon) \approx f(x) + \epsilon(d f/dx|_x) + 1/2 \epsilon^2(d^2 f/dx^2|_x) + \dots$

might as well write it as $1/2!$, but that's the same— $e^2(d^2f/dx^2|_x) + e^3 +$ the general term $1/n!(d^n f/dx^n) +$ and it goes on forever. That's going to become important in a second.

- 1.37** Right, example. Two examples. I'm just going to write it down. You can check this yourselves if you like. But e^x , for example. Now, backtracking a little bit. Here, I expand. When you do Taylor series, it's called expanding, expanding, but you're expanding about a certain point, x here. I've picked the point, a random, arbitrary point. But very often, we expand e^x about the point $x = 0$. It's very common to expand this function about here: x, e^x . In which case, because of this, because all of these derivatives are the same as e^x , and because $e^0 = 1$, $e^x = 1 + x + x^2/2 + x^3/3! + x^4/4! +$ —it goes on forever.
- 1.38** Another thing you need to know is: what about expanding log? Well, we have a problem with log because the log function $\ln(x)$ looks like this. It actually goes to minus infinity when $x = 0$. So you can't expand log about the point $x = 0$ because everything is infinity. It's no good. So what we usually do, what we find convenient to do, and we will shortly, is expand about the point 1: $[\ln(x+1)]$, in which case that becomes 0, because $\ln(1) = 0$. So, $\ln(x+1) = 0 + x - x^2/2 + \text{et cetera}$.
- 1.39** Some of these things are going to become important shortly. Actually, in what we're doing these few days, I think it's only the first couple, up to the x , that matters. But I thought I'd throw in the rest just for completeness. So that's the two functions: exponential, logarithm, and Taylor series. That is all you need to know for now. When you do this properly, you'll need to know about things like stochastic calculus and all sorts of things.
- 1.40** Now, we're going to do all the probability you need. We don't need to know much probability at all for what I'm going to do now: distributions. Distributions. You have two sorts of distributions. You have discrete and continuous. It's all about probabilities. For example, you might have a distribution for eye color: brown, blue, hazel, et cetera. So, brown. I have no idea what the probabilities are of these things, but I think brown is the most common. So what do we need? What do you know about probabilities? They have to be positive, and they would all add up to 1. As long as you covered all the different possibilities, they would add up to 1. That's called discrete.
- 1.41** We don't tend to use that in finance very much because we tend to work with things that have numbers. Rolling a dice. For example, rolling a dice. You'd have the numbers 1, 2, 3, 4, 5, 6, and they're all going to have the same, if it's unbiased dice, they've all got a probability of $1/6$. Again, all that must be positive. It must add up to 1.
- 1.42** Continuous might be a shoe size. My shoe size, as you buy shoes, they come in integers or half-integers. But your actual foot size, as opposed to shoe size, is going to be—no idea what the centimeters is. I have no idea what the distribution looks like. But now when you're dealing with—so that's continuous. It's not all foot sizes possible within

reason. The area under the curve equals 1, and all probabilities are positive. That's easy. You know that. That much you do know.

- 1.43** What else do you need to know? You need to know about averages. There are three main sorts of averages. There's the mode, the median, and the mean. These are the averages. Different sorts of averages. Mode just means the one, that's the mode. That is the mode. It's the highest distribution for brown eyes with that size foot. So that's the one with the highest probability. The median is where there's 50% on one side and 50% the other.
- 1.44** Now, there's a difference here between things. Whenever you read about averages of incomes, et cetera, whatever, in the US, very often, or house prices, they're often talking about the median. For some reason in the UK, we tend to not use the median as much, although the median is really useful, because it's 50-50. It's not affected, for example, by extremes. If someone's got clown feet, then that wouldn't change the median at all.
- 1.45** But it would change the other one, which is the mean. The mean is when you weight all of these numbers, sizes, by the probability. That's a 50-50 split. For example, with the dice, the mean is going to be, you take the dice, you take $1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6}$. What's that equal to? No, not the $\frac{1}{6}$, the mean, the average. What's the average number of spots on a dice? When you roll a dice, what do the opposite sides of dice add up to?
- 1.46** 3
- 1.47** 6
- 1.48** 7
- 1.49** 7. It's 1 opposite 6, or 2 opposite—yeah, yeah, yeah. So the average is actually 3.5. You just take all of the numerical events, sorry, the 1, 2, et cetera, and multiply by the probabilities. You get 3.5. Now, that is important. This is a complicated one, right? This one's easy. You just look at that, and you can see what that is. This one you can eyeball. This is harder to do because you actually have to do a calculation. But it is the most important because this is mathematically nice.
- 1.50** Because if you have— the mean is often written like this. The expectation of x , $E[x]$. If x is the random number, 1, 2, 3, 4, 5, 6, you write expectation, mean or expectation. But it's nice, it's popular because, or useful because, if you have two random numbers like x and y , maybe you roll a dice and you win. If you roll a 1, you win \$1; if you roll a 2, you win \$2, et cetera. At the same time, you're also tossing a coin. And the coin, heads, you win \$5, tails, you lose \$5. So what's your expectation? It's easy because you just take the $E[x + y]$. You can just do them separately. You can do this: $E[x] + E[y]$. You've got a dice over here and a coin over there. You roll one, you toss the other. You want to see how much you're going to win, what you expect to win. You can do it separately. That's

obvious, I think. From what I can see of your faces without my glasses, you're not overwhelmed by that.

- 1.51** But it's going to become important in a second, because this does not tell us—it tells us what the average is, which is great if you're going to be playing Snakes and Ladders a lot, which we're going to be doing later on. Not really. Do we have any gamblers in the audience? What do you, sir, what do you? A poker player. Very good. Very good. It's complicated, game theory and reading people and whatnot.
- 1.52** Anyone goes to casinos? Are these things legal in the States? I never know what's legal. Yeah, it is? Anyone played Blackjack ever? Roulette? I wouldn't play roulette. We can talk about Blackjack a bit later on. But the idea is, if you're playing Blackjack, roulette, poker, whatever, in the long run, what happens on average is going to become important. You're playing lots and lots of different games. They add up, and you can average. But that doesn't give you an idea of the spread possibilities. What's the spread here?
- 1.53** Now, there are lots of different ways of measuring the spread. You've got the range from 1 to 6. But there's one particularly useful one, and that's called the variance, the variance, bar of variance. Variance is a way of measuring the spread of things. What you do is, you take the average. Let's use, for the dice, let's use, let's call this $\bar{x}$. Often, you use $\bar{x}$ to simplify, to make it prettier. The variance is, you take the average, the expectation, the mean of the distance: $E[(x - \bar{x})^2]$. So here, for the dice, it's going to be 3.5. So you take $(1 - 3.5)^2$, multiply it by the probability, the $\frac{1}{6}$. So it becomes this. It becomes: $\frac{1}{6} (1 - 3.5)^2 + \frac{1}{6}$ —the probability, we're taking expectations, so you multiply it by the probability— $(2 - 3.5)^2 + \frac{1}{6}$ —you see what's going on, $(3 - 3.5)^2$, and all the way up to $\frac{1}{6}(6 - 3.5)^2$. That is the thing called the variance.
- 1.54** Now, that is a measure of the spread. It's really nice, but it's strange because if you were doing it in, doing foot size, for example, these would be all centimeters. So this answer will be centimeters squared, which is not intuitive—it's an area. So what's more useful is a thing called the standard deviation, which is simply the square root of the variance. And the nice thing about that is, it gets you back to the centimeters. So you can say that the average foot size is this, with a variance of this. So you've got between, from that to that size. It's maybe not that, but—
- 1.55** So, that's useful. And again, a lot of the reason why people use this measure for the spread of things is because it has nice properties. It has nice properties that, if you've got two coins, you've got two coins, and you've got a dollar riding on each one, plus win or lose one dollar, then it's easy. You can do the calculation. It's an unbiased coin, so it's going to be expectation; if you win one dollar or lose one dollar, 50-50, zero. Same for the second coin.
- 1.56** But what about the sum of those two? What about the sum? Because, again, it's very important for you. Snakes and, sorry, it's Chutes and Ladders, or Chutes and Snakes.

Snakes and Chutes? It's Chutes and Ladders. You cannot add standard deviations. Let's call that SD. You cannot add standard deviations. What you can add, you can add variances. So, the $\text{Var}[x + y] = \text{Var}[x] + \text{Var}[y]$, but only if x and y are independent.

- 1.57** Let's do the coin toss. Let's do the coin toss. You win, it's $\pm \$1$ depending on heads or tails. So the expectation of a single coin toss is going to be $\frac{1}{2}$ if it's unbiased. That's $E[x] = \frac{1}{2} * 1 - \frac{1}{2} * (-1)$, if you lose \$1. That's 0. So if you have n coins, suppose you got n coins, or you're tossing the same coin n times, then $E[x_1 + x_2 + \dots + x_n] = 0$. You expect to make 0. Reasonable.
- 1.58** What about the variance, though? Or rather, I'm building up to get to the standard deviation. Well, because the coins are independent, or the tosses are independent, $\text{Var}[x_1 + x_2 + \dots + x_n]$ is going to be the same as $\text{Var}[x_1] + \text{Var}[x_2] + \dots$ because they're independent, all the way up to x_n — $\text{Var}[x_n]$. And the variance of each one of these is going to be 1. I can do this calculation. It's $\frac{1}{2}(1 - 0)^2 + \frac{1}{2}(-1 - 0)^2$, which is going to be 1.
- 1.59** So, that is going to give us: how many of them are there? There are n of them. So the variance is n , which means the standard deviation is $\sqrt{n}$. The standard deviation is the real measure in dollar terms. How many dollars? Now, that's interesting. Already, this is fascinating because, betting on this coin toss, the more often you bet, the risk—the standard deviation is kind of like a risk, upside and downside risk—it doesn't grow like the number of coins you toss. It grows slower. Because you suppose you toss 100 coins or toss the same coin 100 times, the variance would be 100, but the standard deviation would be 10. Then do it 10,000 times, and the standard deviation becomes 100: $\sqrt{10,000}$. So it's not growing that fast.
- 1.60** So what you actually see happening, if you're tossing this coin or if you are the dedicated Chutes and Ladders people that I suspect you are, or at least will be by the end of these three days, then what you're going to get is something like this. You start off with a small number of coin tosses, and it's going to look like that. As you're getting more and more coin tosses, things spread out. But actually, the spread, as you'll see later on, when we do this in a little bit more detail with portfolios, it doesn't spread as much as you might fear. It only goes like $\sqrt{n}$, not n , which is going to be profound.
- 1.61** Part of that is the following, a theorem. Now, I don't do theorems, not this early in the morning, anyway. I don't do theorems. But we're going to do a theorem, and it's an interesting theorem. All this is relevant. Believe it or not, all this is relevant to what we're going to do. It is the minimal amount of maths you need to get started in the subject of quant finance: so a thing, a theorem called the central limit theorem, the central limit theorem. Now, the way this theorem works is, suppose you are—I'm going to roll the dice. Go to our dice again. 1, 2, 3, something like, I need 4, 5, 6 on the other side. You roll your dice, and we get a number. Think of a number. 1, 2, 3, 4, 5, 6.
- 1.62** 4.
- 1.63** Very good. Very good. If you can, speed things up a bit.

- 1.64 1.
- 1.65 I like it. I like it.
- 1.66 3.
- 1.67 3.
- 1.68 6.
- 1.69 2.
- 1.70 We add those up. I shouldn't have done this, should I? Is it? 21? Very good. 21. Okay, take that number 21 and put it over here. Then, I'll believe you, we do the same again. We won't do the same again. But we did six numbers, and it comes to 25. And then, we start again, and it's 29, and then it's 17, and so on. Now, what we've just done is, we started from this dice roll here and added up— the key thing is added up, the symbol for adding up, the summation, Greek capital Σ , the summation. We've added up lots of numbers.
- 1.71 We've rolled it six times, added them up, and it's given us a new distribution. You get the idea that this distribution is numbers 1, 2, 3, 4, 5, or 6 with equal probabilities. From those random numbers here, we've generated different random numbers here. It's a different distribution. And what this distribution looks like, well, we know what this distribution looks like, right? It's just 1, 2, 3, 4, 5, 6. Add them up. Add up six of them—sorry. Bad Paul. I shouldn't have said six, because there are six sides to a dice. You might think there's something significant about me adding up six. There's not. It just so happened that, sitting on the front row of this lecture, there are six people, six wonderful people. So I should have maybe added up 10 numbers. You see, that's why symbols—so I did that on purpose, actually. What are you thinking? I did it on purpose. This is where symbols come in. Because then you'd realize, the number of things I'm adding up is different. It doesn't have to be the same as the number of possibilities. So, sorry about that.
- 1.72 Anyway, add them all up, and we get another distribution. Our new distribution, first of all, is going to have a mean of its $E[x_1 + x_2 + \dots]$, it's just adding up all the expectations, which is $6 * 3.5$. So it's here somewhere. The variance is harder. I won't ask you to do that. Anyway, what you get is the distribution that now looks like this. The highest it could possibly be is going to be 36, and the lowest it could possibly be is 6. Anyway, the idea is, we started from this distribution. By adding up some of these, we get to that.
- 1.73 Now, what if we, instead of adding up six of them, rolled the dice 20 times, and added them up? We get numbers over here again, but they'd be further this way, and they'd be a bit more spread out; the lower and the upper would be a bit bigger. The central limit theorem says that, if you add up more and more— that's not mathspeak; you can say that mathematically properly, but I'll just say more and more—add up more and more random numbers from any distribution, meaning it doesn't have to be rolling a

dice. It can be a coin toss. It can be a deck of cards. Pick a card, and each one represents some number. Any distribution.

- 1.74** As you have more and more of them, what you end up with is that shape, what's called a probability density function. Probability density function. Some variable x , the probability density function. That's a special function, another special thing. Does anyone know what it's called? It's the bell curve. This is a bell curve, otherwise known as the normal distribution. The normal distribution. That has a formula. And the formula for that, I do know it, but—this much I do remember.
- 1.75** What's the most common special number that you know, mathematically? It crops up in areas of circles and things. Pi [π]. Yeah, him. π has just appeared for no apparent reason. This has got nothing to do with circles, has it? It appears. Sigma [σ]. Sigma is some standard deviation. This is the formula of this thing, some standard deviation, times e — I said it just occurred for no apparent reason— $e^{(-1/2\sigma^2)(x - \mu)^2}$, where μ is the mean. So, that is a formula for the normal distribution: $[1/\sqrt{2\pi\sigma^2}] e^{(-1/2\sigma^2)(x - \mu)^2}$. That's important because we're going to be seeing that quite a bit. It's not complicated. π , e , exponential, $\exp$ in Excel.
- 1.76** It's important because it crops up in nature a lot, because whenever you're adding up things, like we were doing, we took some distribution, like the distribution of rainfall every day. Do you get much rainfall here? You take this amount of rainfall every day, add it up over a month or so. If rainfall is random and things are independent, which you can argue about these things, then you're going to get another set of random numbers. And because you're adding up days of rainfall, you're going to start seeing normal distributions occurring for the amount of rainfall you get in any particular month. Or think of a share price movement as being made up of lots of trades over a day. So that's like adding up lots of random numbers. So maybe the distribution of share prices from one day to the next looks a bit like this.
- 1.77** So the normal distribution crops up all over the place in nature, in science, and to some extent in finance. So there we go. That's it: exponential, logarithms, Taylor series, expectations, variances, standard deviations, the central limit theorem, and normal distributions. Thank you very much.

Lecture 2: Core Concepts in Investment

At a glance — Core concepts in investment — fixed income and interest rates.

- 2.1 Now we're going to be talking about a subject called fixed income or interest rates—there's my not-so-good one—interest rates. Again, really, really basic stuff. You can get a fixed income. When you do it properly, it's really, really bitty, detailed stuff. When your mortgage is whatever percentage, tiny percentage, you're paying for your mortgage is now compared with the 16% I was paying in the 1980s; you're paying peanuts.
- 2.2 So you talk about percentages, but you also talk about basis points. A basis point is one-hundredth of a percent, which is nothing. One percent is 0.01, so we're talking about 0.0001, which is tiny: one hundredth of 1%. It's the numbers people talk—bandy about in fixed income. But you have to understand those 0.0001 numbers are being multiplied by hundreds of millions, and people's profits and salaries et cetera, are linked to those 0.0001 numbers. So it becomes a real details game. Whereas equities and commodities and currency things are bouncing around all over the place, interest rates don't do very much, and they're, as I say, quite small anyway.
- 2.3 So if you get into fixed income, you've got to be a real details person. If you get some tiny little thing wrong, some day count—they talk about day count conventions. How many days are there in a year? Well, it depends on which country you're in. Some countries don't have as many business days as other countries. I shun't name any. And things like that and bank holidays and whatnot, it's all fiddly, fiddly, fiddly. I hate it. But we're not getting into that detail; that's not something you want to hear me talk about.
- 2.4 I'm talking about more big picture stuff. I'm still going to be using the mathematics we've talked about, in particular, the exponentials and logarithms and Taylor series, those things. And I'm not going to be worrying about anything probability. I'm not going to be assuming interest rates are random. That gets into—that's a can of worms as far as we're concerned here today. When it comes to shares, everything is going to be random. But for fixed income, interest rates, we're going to assume, things are what's called deterministic, nothing random at all. Keep things manageable because we'd go insane if we tried to get down to the details.
- 2.5 So fixed income. A couple of things you can do with your money. You can put your money in the bank, just rolling it over day to day. If you do that, the interest you get is called the floating rate because it will vary from day to day. So the money in the bank will always go up—okay, maybe not. There might be inflation and taxes and whatnot, but it will keep going up. But the amount you get each day is going to change depending on the bank or whoever's setting interest rates. So that's good. That's good because you can get your money out anytime you want if you need the money. Just go to the bank and say, "I want to take all my money out." That's floating rates.

- 2.6** Then you have a thing called fixed rates, which means you take your money and you lock it up for a year, say. Lock it up for a year, you know exactly how much interest you're going to get at the end of the year, but a couple of things. One, it could be that the day after you lock your money away, interest rates rise, and you just, "Ah, I only had to wait a day!" Also, you can't get your money out. Maybe there are some penalties for taking your money out early.
- 2.7** So there are swings and roundabouts. Do you keep your money in a floating rate? You can access it anytime you'd like, but you don't know what you're going to end up getting, or a fixed rate?—fixed rate also like bonds, government bonds. Here, rather than taking your money and putting it in the bank, what you do is you buy government bonds, and each bond might give you \$100 in a year's time or 10 years' time or whatever, and so you pay less than the \$100 to get the guaranteed \$100 sometime in the future. So these things are all similar ideas.
- 2.8** The way interest rates are, it's embarrassingly trivial. You've got an interest rate of 4.3%, say, and you have \$100, then the amount you're going to have at the end of one year is just that, right? That's the thing we're talking about, just that, assuming we got the numbers right, which is always—
- 2.9** Of course, you can have different amounts here, and you can have different interest rates, and you get different numbers. That is just interest rates, a fixed rate. So if you tie your money up for a year at 4.3%, then that's what you have at the end. Or you could say, suppose I've got \$100 and I put it in the bank, and I end up with \$104.3, what was the interest rate I got? Just turn it around. Okay, you're dividing by 100, you take off that 1: $r = 4.3\%$. Okay, that's the thing we're doing. It could not be easier than that, but it does get a little bit trickier.
- 2.10** One thing that people obsess over, but I'm not going to obsess over, is conventions. I've already briefly mentioned day-count conventions: how many days are there, business days are there in a year? Another aspect of day-count conventions is when you're talking about months. Do you worry about the specific number of days in the month: 30, 31, 28, 29? Or when you're doing calculations, do you assume there are actually 30 or 31 days? There are whole books on this stuff, which, since I'm not a details person, would drive me crazy.
- 2.11** But another convention is if you're putting your money away for, say, a quarter of a year, three months, and somebody quotes an interest rate of 4.3%. Well, what do they mean? When they say 4.3%, they're talking about an annualized number: what you get in one year if you put your money away for one year. But what if you put it into something which only lasts three months, and they're quoting 4.3%? One possible convention, and I'm sure there are lots of conventions, one possibility is to say, actually, if it's only for three months, what you're going to get, your \$100 is going to turn into—you can guess; it's only for a quarter of a year—it's going to turn into whatever that is.

So you might take the interest rate that's quoted in annualized terms, turn it into something which applies for just the three-month term and do your calculation.

- 2.12** So what I'm saying, really, there are conventions. And if you do work in fixed income, if you move from the jobs that you've got now, which are lovely jobs, and I'm sure much better than working in fixed income, but perhaps not as lucrative, then that's the thing you have to obsess about. Good luck with that.
- 2.13** Then there's the question of—you have heard of this, I think—simple versus compound interest. Compound interest. Simple interest versus compound interest. You put your money in a bank, and you're going to lock it away for five years. Well, you go to your bank, and you've got to check some box saying— you put in how much money you're going to lock away, you put in how many years you want, it'll give you some interest rate, and you'll get to choose: do you want the money to go into the same account, or do you want the money with the bond, with the fixed term thing, do you want the money to go to your checking account? Do you want to come out with a checking account? Those two options. And then you have to check the box saying you've read all the rules and regulations, which nobody has, and enter.
- 2.14** So you have the choice between: do you want to take the money out every year, say, or do you want to keep it in the bond or the fixed rate account and let it build up? Now, that becomes the difference between simple, meaning, do you want to—by taking the money out, you get to spend the money—sorry, to take the interest out; this is when the interest comes out—you get to spend the money if you want. Whereas here, you effectively get interest on interest. All right? Simple versus compound. So this is where the money comes out. You don't get any interest on the interest, future interest.
- 2.15** So let me show you what I mean. Suppose you've got \$100. I've got some numbers here. I'll keep it simple. I'll just assume the interest rates are 10%, because then I can almost do a few calculations. If $r = 10\%$, then if you have simple interest, you start off with \$100, after one year, you'll have how much? 110. Very good. You have 110. In two years? With simple interest, you'll have 120. This is simple. So red is simple. Red equals simple interest. So you're getting what? This is the interest you're getting. This is the difference between—you're getting that much money, and then the next year, you're getting that much money. But this is the accumulation of it, effectively. And so it goes like this. All right.
- 2.16** Now, green for compound. How to spell green? Green is the compound interest. Here, first year, you're going to get the 110 again. But the second year, you're going to get interest not just on the 100, on the principal, on the 100, you're going to get interest on interest. So you will have $(1 + 0.1)^2$. That's how it works. Instead of a 0.1 every year, effectively, it goes like this: after one year, you've got $[(1 + 0.1)]$. That's after one year. That's how much money you're getting interest on. So the next time, you're going to be doing the same thing again after two years: $[(1 + 0.1) * (1 + 0.1)]$, which is 121. You get

an extra dollar. Wow. Don't spend it all at once—121. And so it goes; you get interest on interest.

2.17 Now, okay, that's one dollar. After two years, you get an extra boopy-doupe. But if you zoom out from that picture and look at what happens, I've got some numbers here. For example, suppose you compound at that rate over 30 years. Let's do an example of 30 years. At 10%, you're going to have, with simple interest, you're going to have the 100 + 30 years times 100 times the 10%, which gives us 400 [$100 + 30 \cdot 100 \cdot 0.1 = 400$].

2.18 But if it's compound interest, you guess. Anyone get any guesses on how much we got? Oh, I haven't actually got a number written down here, and I can't do this in my head. But from my graph, it looks to be, wow, a ballpark—and I'm just reading off a graph here—ballpark. I feel I have to check this. Simple. Compound. That's a big difference, and apologies if that number is wrong, but it's still going to be really big. Admittedly, 10% interest is big, but you get the idea. And the important thing really is that even if the numbers are—if we're not talking about 10%, if we're talking about 2% or whatever, it won't be quite so extreme. But remember that point about people in banks and hedge funds are multiplying these numbers, it's not 100, it's 100 million or billions or whatever. Remember that number I started off with, the one quadrillion dollars worth of financial instruments? Yeah. So then it becomes serious money after a while. So you do have to worry about these things.

2.19 Now, the mathematics of compound interest is very simple. We'll go over to symbols. Remember, we need to go to symbols if we're going to be doing these things for real. I think everything I'm doing from now on will be compound interest, I think. The thing you need to worry about is suppose you start off with some principal P , the amount of money you put into the bank, and it's a fixed rate of interest, and the interest rate is r , and you put it in the bank for T years. Let's suppose it's just one payment, so that's one payment per year. You get the interest at the end of one year, and I'll generalize that in a moment. Then, the amount of money you're going to have with compound interest is going to be $P(1 + r)^T$. Okay, that's very easy.

2.20 So you could either, if you know what the interest rate is and the principal is and the maturity of your fixed thing, you can calculate how much you've got, you're going to have at the end of T years. Or you could say, for example, if you're buying a government bond and you know you're going to get a hundred dollars at the end, and you know how much you have to pay for this government bond, and you know how many years it's for, you can turn this around to calculate the interest rate you're getting.

2.21 So for example, suppose, for example, you're going to buy a government bond and it's going to be a hundred dollars, then all you do, and this is how much the government bond costs us now, then you just do divide both sides by P , and then take the T th root of that to get to $(1 + r)$, and then take off 1 [$r = (100 / P)^{(1/T)} - 1$]. That is the formula that tells you what the interest rate is for that particular government bond. That's

something we're going to be doing in a second, where we look at different bonds, and they have different principals, and they have different maturities, and you get different interest rates. So we'll talk about the different interest rate you'll be getting shortly.

- 2.22** So this example is also basic. If the interest is paid once a year, then this is the calculation you do. But what if you have n payments in a year? You might see pictures like this where you have time like this. This is the maturity of the thing. Sometimes you draw pictures like this, where these are all the payments you might get or coupons you might get on the bond. When you get to maturity, you get the money back, a coupon. So you'll see pictures like that knocking about. But suppose this is one year, this is one year, how many interest payments are there? Okay, that would be the end. How many payments are you getting paid every six months? Are you getting interest every three months or every month, et cetera?
- 2.23** So all that happens to this, the way it changes, is just P times 1 plus—remember this convention that I'm using this r/n : $[P(1 + r/n)]$. But now the number of payments you get is going to be n times T : $[P(1 + r/n)]^{nT}$. So this number that goes up here is how many interest payments there are, rather than, let's say, time. So it would be that. So this is just once a year for the number of years; this is n times a year for n years. So if you wanted to calculate an interest rate for a government bond, you'd be taking the n^{th} root of this and so on. It's all very straightforward.
- 2.24** But the interesting bit, the mathematical bit about this, where the mathematics comes in because raising things to the power of three or five is not really mathematics. The interesting bit in this comes about because what happens if n gets larger and larger? So what we've got is a little sketch up here of an increasing number of payments while keeping r fixed. We're going to keep this at, say, 5% or something of one payment a year for a certain number of years or two payments a year for a certain number of years, while holding r fixed. What does that look like?
- 2.25** Well, because it's compound interest, you can expect this ever-increasing curve. It's going to go like this. Actually, let me do it with dots because this is a bit—no, I'll keep the line. But this is, say, $n = 1$. So you get one payment a year. So these are years, for example. So it starts off here, and then it becomes this. Next time, next year, it's that. The year after that, it's that. The year after that, it's that. So on. It keeps going like that. The line in between is really just so it's clear what's connected.
- 2.26** Then, if I have two payments a year, what's going to happen? If I have two payments a year, that. The next one is going to be ever so slightly higher, tiny, tiny bit higher because, for example, let's look at it with 10% again: $(1 + 0.1)$ versus $(1 + 0.05)^2$. That's the difference between 10% once a year and 10% twice a year, i. e., 5% twice. Which is bigger? That one is bigger because that's equal to 1, whereas this is 1.1. This is $1 + 0.1 + 0.05^2$, which is 0.0025— I'll get the number of zeros wrong, so I'll just leave it as that. The key thing is that is the difference between a payment of 10% once a year and a

payment of 5% twice a year. And again, as I keep saying, that's where you make your money. In fact, that you've got to track of.

- 2.27** And then, you go on to the third payment. That's 18 months. It'll be ever so slightly higher each time like this. And then, you move on to what happens if you do that four times a year, for example. So twice a year, four times a year. Then you got $(1 + 0.0025)^4$, which is going to be ever so slightly higher than this. Ever so slightly. We're going to do the dots in between as well. Yeah, I haven't done all my dots. I haven't done all my dots. A lot more dots than I've got. Something like this. Have I got enough dots? Anyway, every three months. Every three months and so on. Something like that.
- 2.28** What's the obvious question we could ask being mathematicians? We ask what happens as n tends to infinity and beyond. What does it do? What does it behave like? This is where we use what we did before to start. This is where we have Taylor series and whatnot. Let's see what happens if we do this. Well, forget about the P to start with. It doesn't matter whether it's \$100 or a million dollars, whatever. You just put that in front. We're not going to worry about that. It's this expression here. We're going to look at what that does as n gets larger and larger. Mathematicians are always doing that. They're always looking at how things tend to infinity or tend to zero.
- 2.29** There's a mathematical technique. It's not relevant here, but sometimes you use what's called, I think it's called, *reductio ad absurdum* or something like that, where you take extreme examples to prove a point, to prove something mathematical, which, when I use it on my wife, to give you extreme examples, it does not go damn well. She's not a mathematician. So what do we do?
- 2.30** Well, because we know that exponentials and logarithms are inverses of each other, this can actually be written as $e^{n \log(1 + r/n)}$ — I don't need that. I don't need that one there— $e^{n \log(1 + r/n)}$. Let me write that a bit bigger: $\exp^{n \log(1 + r/n)}$. That's exact. That equals that. It's exact. But we're going to be letting n go off to infinity. This r/n , as long as we're holding r fixed, this r/n is getting smaller and smaller. So this is where we use the Taylor series. Taylor series. Now, I don't even remember what I said: $\ln(1 + x) = (x - x^2/2) + \dots$ something like that, I think, when x is small. This is the Taylor series expanding $\ln(1 + x)$ around the point 1.
- 2.31** Well, that's exactly what we got here. So this is approximately $e^{n \log(1 + r/n)}$ —well, that's x now. That's $x = r/n$. It's approximately r/n . So it's $r/n + \text{dot, dot, dot}$, meaning all the other small terms. And what happens magically is the n 's cancel as n gets larger and larger. As n gets larger and larger, this thing gets smaller and smaller. You can forget about it. But this doesn't. That becomes e^r . It's E again. Exponential. That's what that function becomes as you have more and more dots, as you have more and more payments, that curve becomes— it won't be much higher, but as I keep saying, it'll be high enough.

- 2.32 So, whenever we're doing things in finance, or whenever I'm doing things in finance, as I say, I try to steer clear of fixed income because it's really, really detailed. I tend to always be working with that rather than that. I mean, to the mathematician, we'd rather work with that than with that. But to the fixed-income practitioner, that is just an approximation that's not good enough. They would work with things like that.
- 2.33 Okay, that's that. Another part of this is also a thing called discounting. Later on, we're going to be talking about options. Options are contracts—I won't go into details now—but options are a contract that you pay something now, and you get something back potentially in some time in the future, a year in the future, let's say. But you don't get anything until the year or whatever in the future. So knowing that you're going to get \$100 in the future, you have to build that into your calculations for today. What is \$100 in a year's time worth today? That's the thing called present valuing, present value, or the time value of money, and so on, all sorts of different expressions, phrases one might use.
- 2.34 So \$100 in the future is not worth \$100 now; it's worth less. So what you do, you've got to go working backwards, expect to see things like that, but like that. You can think of this going backwards in time, if you like, or the relationship between money here and what's it worth, what's the equivalent here? Instead of going this way and doing e^{rT} or this expression here. Going backwards is like e^{-rT} . So watch out for things like that when we're doing present valuing, discount factors, et cetera.
- 2.35 Now, when you see bonds quoted in the markets, you will see different values for different maturities. What I mean by that is—I've got to actually use some numbers. As much as I hate to use numbers, I have got to use some numbers now. Suppose we see the following. If you want to buy a government bond and you know you can get \$100 in the future, and suppose that the maturity, that is, how long you tie your money up for, you've got different possible maturities. So suppose you tie your money for one year, you're going to get \$100 back in one year's time. You're paying less for it now. How much are we going to pay for it? We're going to pay \$96.32 to get \$100 back in one year's time.
- 2.36 We look in The Financial Times, or wherever you look for your numbers, and you see that a maturity of two years, you're going to tie your money up for two years. If you get \$100 back in two years, the price of such a bond is \$92.08. I'll fill in a few more. If you tie your money up for three years, you're going to pay \$87.78 and \$83.86 for four years, and five years it's \$80.13. These are the numbers I'm going to be using. These are examples of what are called zero-coupon bonds, meaning you're going to get \$100 back in 1, 2, 3, 4, 5 years' time. But before that, nothing. There's no coupons. You're not getting anything back. So this is to get \$100 back in the future, with no coupons, nothing before that. All right? \$96.32. So you see, you're paying less and less because you're tying your money up for longer and longer. You're still only going to get \$100 at the end. What does it mean? How can you tell?

- 2.37 Should you? Do you want to tie your money up for one year and get \$96? Or do you want to tie it for two years and get \$92? Well, what's the interest rate I'm getting? One way of looking at this is you want to level the playing field in the sense of you can't really compare these numbers. They're going down. Well, great, but are any better value than others? Would it be better to lock your money away for one year and then take a chance on what interest rates are going to be doing thereafter? Or should you tie your money? You think interest rates are pretty high now, they're going to fall, so tie your money up for longer now.
- 2.38 Well, one way of doing this, comparing these, is to look at what's called the yield. So the yield is just using that formula: $P(1 + r)^n = 100$. So you put in the different P's. You put in the different n's, which is just the number of years. And so that's less than 100. So you're going to have $(1 + r)^n = 100/P$. Then take the n's through to that, n's through to that, and then subtract that 1. So that's what that formula is: $[r = (100/P)^{(1/n)} - 1]$. It's going to go in there. All right, that's the yield. That tells you what the interest rate is, the effective interest rate paid once a year.
- 2.39 And so we can just back that out. It's very simple. The yield is 3.82 for this one. It's 4.21. It's 4.44. It's 4.5 and 4.53. These are percentages. Which looks something like—well, the bond price is going down. That's a bit exaggerated. That's the bond price against the maturity. This one, the yield against maturity is—all right. That's a very typical picture. Not always like that, but it depends on what people think interests are going to do in the future. But this is typical. It tends to be increasing.
- 2.40 The idea, now, the idea—I'm not an economist. We have a Chancellor of the Exchequer, who is apparently an economist in the UK, allegedly. But the economists say things like, "You've got to tie your money up for longer. You can't access it." You jolly well better get a higher interest if you're going to tie your money up. If it was decreasing, why would you tie your money up for longer? Well, it depends on what people think interest rates are going to do in the future, because even though these numbers are fixed, in practice, they change from day to day. But we're not going to worry about that at the moment.
- 2.41 All right, that's great. That's the yield. Okay, increasing yield. But there's actually something else you can look at because this interest rate, this 4.21%. If you're getting 4.21% the first year and then the second year, if $n = 2$, it's two years, it's $(1 + 0.0421)^2$. But that's like saying I'm getting 4.21% each year. But hang on, didn't you just say you're getting 3.82% for one year? That's inconsistent because to get to two years, you've got to go through the first year. So this interest rate is applying for two years, even though for the first year, this is the interest rate that's applying. And this one, the interest rate for three years, assumes you're getting that for three years. In fact, you're getting that for the first year and something else for the second year. It's all inconsistent.

- 2.42 So what people do to make things consistent—sorry, I shouldn't have written that there because I need that space—are things called forward rates. Forward rates is just a way of making this all consistent. So I'm going to look at what is the interest rate? I'll start off with a picture down here. So this is going to be the forward rates. What interest rate applies for the first year, for the second year, for the third year, for the fourth, fifth year, to make all of these numbers consistent? When you do that is you must introduce a different interest rate for each period here. These yields, that yield there applies for all of that, even though it's going through these three or these two.
- 2.43 Now, this is interesting because this is a point where you have to make some assumption about what interest rates are going to do. For example, if we're now in the business of assuming the interest rates change from one year to the next—that's what's going to happen—and they're going to change in such a way that they're consistent with these. But if we accept the fact that interest rates are going to be changing, there's no reason why they should change on the first anniversary and the second anniversary. There's no reason why they should change every year. They might change after two months, and then stay constant, and then change again after 16 months.
- 2.44 So we have to do one of these things that people in fixed income do all the time, which is to make some unrealistic assumption. And the unrealistic assumption I'm going to do here is I'm going to assume that interest rates, that these rates that apply over this period, over this period, over this period, et cetera, are constant for all of that year, even though things might happen in the middle of the year. I'm going to assume it looks something like this: constant for that period, a different constant, a different constant. The question is: how do I find out what these constants are?
- 2.45 To do that, well, first of all, the first constant is really easy because there's no overlap. This thing here applies to the first year. So I'm going to do what's called bootstrapping. Bootstrapping, where I start from this one and ask: what interest rate must apply for the first year to make the bond-year bond, that? Well, the answer is rather trivial. It must be that. Okay, that one's easy. That's 3.82%.
- 2.46 Then, I have to ask the more complicated question: what interest rate must apply over the second year to make that [92.08] the bond price, given that for the first year, that [3.82%] is the interest rate. So what I need to do is, instead of having this and 100 in this calculation, I take this [96.32] and this [92.08] because this [92.08] is where we are after one year, and this [96.32] is where we become—is that a 2 or a 7? Oh, no, it must be a 2. So what we do is the same formula, but instead of getting 100, we have 96.32. So we do this. We have 96.32 divided by 92.08. It's only over one year, so there's no n^{th} rooting, that's quite straightforward. So you just subtract off the 1, and whatever that comes out to be, that is our interest rate that applies over the second year. If my calculations are correct, that should be 4.6%. So the first one is still 3.82%, and the second one is 4.6%. And then, you do the same. You do that [92.08] divided by that

[87.78] and subtract off 1, and this is what we end up with: [4.90%] . Look what just happened. It's gone down. The forward rates now look like up, up, down. It looks something like that. So that's called the forward curve.

- 2.47** That's what the market thinks—people always say that this is what the market thinks is happening. Nonsense. You have to understand these guys, people who are trading these things are 22 years old, and they don't know what they're doing. Their job is just to buy and sell stuff, and if they don't buy and sell stuff, they don't get a bonus, so they have to buy and sell. So the numbers come about because of supply and demand and things like that, people buying and selling stuff. But people say this is what the market thinks interest rates are going to do in the future. They're going to rise a bit, and then they're going to fall off a bit.
- 2.48** But that's interesting because then if someone says, "Actually, I'd like to tie my money off for two and a half years. I don't want one, two, three, four. For some reason, I need the money in two and a half years' time." Then you might say, in that case, a reasonable value for a two and a half year bond would be this, this, and then half of that. So if it's a two and a half year bond, you do something. Well, yeah. You do the obvious thing. I won't do it in case I get it wrong, but you'd effectively—
- 2.49** So that's forward rates. You see forward rates a lot, and it's just a way of making these numbers consistent with each other, but they're not consistent with what happens in practice, because in practice, this curve changes every day. It doesn't just change—here, it's changing. Yes, that's a good point. There's two ways of things changing with time.
- 2.50** One is the maturity. Clearly, you look at these numbers on a screen in Bloomberg thing, you take a snapshot of these numbers, and you can see this is how forward rates look like they're changing. But then you come back a day later, and these numbers will have changed because people, supply and demand, people are buying things, people's expectations have changed, some things have gone up, some things have gone down, and this picture will change.
- 2.51** So it's changing not just with maturity, but it's also changing with—I usually use a little t , a lower case t to mean calendar time. So these curves, you'll sometimes see people writing the forward rates, not just as a function of big T , how far in the future, but also a little t to represent the fact that they are changing constantly. And that's the point where people start assuming randomness, which we're not going to do, assuming things are random. Don't do that.
- 2.52** Now, the final thing I want to mention just very, very briefly. These calculations apply to putting money in the bank for fixed periods. They apply to government bonds. They also apply to corporate bonds or bonds issued by countries that are perhaps not as reliable as the good old USA, that maybe they don't pay people off. So there's always the issue of default risk or credit risk. Credit risk. What I mean by that is, you've tied

your money up, you've put money in the bank or in—I won't name any particular countries, but in some—and you find that you don't get the money back because the company has gone bust, defaulted.

- 2.53** This is the whole subject of credit risk, which is very interesting. The models that people use for default risk are, in my view, very naive. I was saying this in the run-up to the credit crisis of a decade and a half ago, and it turned out to be true. The models that people use for credit risk are like this. They go like this. This is now, and you've got some bond which will pay off \$100 in some time in the future, and you've got an interest rate of r , so you might expect something like $(1 + r)^{-1}$ is how much you might pay for that now. But then there's also a chance that you get zero if the company goes bust.
- 2.54** So people put in things like they introduce p for the probability of default, so $(1 - p)$ for them not defaulting. So you will see people do calculations like, well, actually, if the interest rate is r — this is where I talk about the thing I call the risk-free rate. Risk-free interest rate might be r . So that might be a government bond. The US is not going to default, so zero risk of default. But this particular company might default, so you want to discount like that $[(1 + r)^{-1}]$, but then take into account the p .
- 2.55** So you do things like you say, well, actually, the value of this bond now should be $100 * (1 + r)^{-1}$ because you're going backwards, you're discounting it. But then you've got to multiply it by $(1 - p)$. Well, you take the average. Effectively, you take the average of 100 times the present value, the present value $[(1 + r)^{-1}]$ of this bond now, if you're guaranteed to get it, multiplied by the probability of getting it, plus p times 0 to give you the value today.
- 2.56** So, a couple of things: if you see that the government bond has a certain interest rate, and then you find that some corporate bond has a different value—sorry, the government bond is using a certain interest rate, that gives you this $[100 * (1 + r)^{-1}]$, and then you observe in the market, actually, the value is less than that, well, from that, from knowing this [the value] and knowing this, the risk-free rate $[(1 + r)^{-1}]$, you can back out p . So people would then back out the probability of this particular bond defaulting.
- 2.57** Some complications, sometimes you get a recovery value. You do get something back from the corporate bond. It won't necessarily be zero. Very often, you will see that they use, instead of $(1 - p)$, a mathematical model for default says that actually the company is like this. The company is doing fine. So this is the existence of the company, if you like, existence. Does it exist or not? Well, yes, it exists. It exists, it exists, and then randomly, it goes to zero. So, often people apply models of randomness to say when the company goes bust, in which case these things $[(1 - p)]$ become a little bit more mathsy. They become things like e^{-pt} is the probability of the company still existing. And so that particular function looks like this. So it exists now, but then further and further in the future, it's unlikely to exist. So there's a slightly different p , but for small

p, because of the Taylor series, they become exactly the same. But this is more mathy, isn't it? Because companies don't just go bust at the last second just before they have to pay out a bond.

- 2.58** Or maybe they do. Actually, that's a good point. What makes a company go bust? Is it something like this, which is effectively spinning a roulette wheel? And when the green zero comes up, they go bust, but spin it, not bust, spin it, not bust, spin it, not bust. Eventually, they go bust. That would give us something like that. Or is it, actually, no, that's not what companies do. Companies don't spin roulette wheels to decide whether to go bankrupt or not, do they? As far as I know, it's a business decision to go bankrupt to some extent. So maybe they actually do wait until, "Oh, gosh, we've got to pay off some \$10 billion loan tomorrow. Well, maybe today is the day to go bust."
- 2.59** So that's one of my criticisms going back years was that the models are incredibly naive for credit risk. It's a funny business, quantitative finance, financial engineering. It is a balance between trying to be as accurate in your modeling as possible, but also making assumptions, which are completely dumb.
- 2.60** One of the problems is to do with the fact that when doing mathematical modeling, it's not like physics, right? Physics, decay of radioactive particles. Bang. It's very accurate, decay of these curves for how radioactive something is. It has exponential decay, half-life, and all that kind of thing. A very, very good model based upon sound physical principles. What's this based on? Well, we don't want to have the maths as too complicated. People have this weird balance between, on the one hand, using really complicated mathematics, but then really naive assumptions within the whole of finance.
- 2.61** Anyway, I'll talk about some of these things when I talk about options and behavioral finance and all sorts of other bits and pieces. But that's just an overview of the mathematics one sees in fixed income, a basic stuff or credit risk. What else can I say? Any questions about those ideas? Very basic stuff. The next level up from this is really to assume interest rates are random. Yeah.
- 2.62** I'm curious if you came up with a different model.
- 2.63** Oh, I've come up with loads of models. I'm not sure you wanted to broadcast this, but my interest has always been to do things that I find are realistic and fun. I've done all sorts of fixed income models, and credit risk models, and models for—you'll hear about some of my models for how to make investors happy, for example. But that's because of my background in doing fun things and liking variety.
- 2.64** The secret of success in finance, really, is to get your name attached to a model like the Black-Scholes option pricing model or the Hull-White interest rate model. To do that, you have to come up with something that's similar to what already exists, but—we have the expression—won't frighten the horses, i.e., not too revolutionary. And quite

frankly, I don't find that particularly interesting. So if you can come up with a new formula that's similar to other formulae, then you've got a fighting chance of—

- 2.65** It doesn't mean your model is any good. Actually, this bit you can record. Well, maybe you can record that bit, but this bit. There are models for—I said that the next level is assuming interest rates are random; it involves a thing called stochastic calculus, things like that. Don't worry about that. But the thing is, people have different models for interest rates. But how do they choose those models? What's the justification for choosing models? You'd think, wouldn't you, that maybe people look at data. Yeah, you're nodding. They'd look at data. Look at what have interest rates done in the past? Okay, the future might be different from the past, but it's a starting point, right?
- 2.66** The problem with that is that it leads to models which are a bit complicated. There's another route you could take, and that is, sometimes you end up with what we call differential equations, partial differential equations, that you have to solve. Depending on what the model is, that you assume for interest rates, you have a different differential equation and a different solution. Now, the problem comes about because often you can't solve these equations with formulae nicely. When you do your math as an undergraduate, there's an exam question, and you know that there's an answer. There's a formula at the end of it. Not in finance. Not if you look at data and you try and build up your model to be statistically accurate. The model you end up with is not an exam question; it might not have a solution.
- 2.67** You might have to do something numerical. You might have to write some long piece of code to solve it. Wouldn't it be much nicer if it was just a formula? A formula maybe in a font, a size 12 font that's about that big. We don't want it that big, right? We want it usable like that with maybe two or three parameters, not too many parameters.
- 2.68** Unfortunately, if you go the statistics route, you'll end up with something that there's no formula, you have to solve numerically, and it's just a pain in the backside to do that, people think. They're getting paid a crazy amount of money in banks for doing nothing, really. So what they tend to do is they say, Well, let's choose the model that has the nicest formula. We don't look at data. This model, if I just assume this, I will not name any professors' names, but I will use this model because I can solve this equation, and it's this big. It doesn't make any sense.
- 2.69** Imagine if you apply that principle to designing airplanes. The equations for fluid dynamics are called the Navier-Stokes and Euler equations. The Navier-Stokes equations are nonlinear. It's not like the variance thing where the value of A plus B is the same as the value of A plus the value of B. They're nonlinear. They're complicated things because there's a quadratic, there's a something squared term. If quants, people who work in quantitative finance, were designing airplanes, they'd go to this equation and they say, "Oh, I can't solve it if it's got 2 up there, if it's squared. Let's just cross it out, and then we can solve it. Well, then, planes will be falling out of the sky all the time. But that's what they do.

2.70 Remember, there are, as I say, quadrillions of dollars floating through these equations, these models, which are all quite frankly, nonsense. Models for credit risk are the worst, followed by models for fixed income, which are really bad. We're going to be doing equities next, stocks and shares and things like that. Those models are not too bad. It's still not great, but not too bad, as I should be explaining shortly.

Lecture 3: Stock Market Behavior

At a glance — Stock market behavior and its mathematics.

- 3.1 Now, it's starting to get a little bit serious. I'm going to be chatting a little bit about the principles behind quant finance applied to shares, etc. Then we're going to do a little bit of spreadsheet analysis, very basic stuff, but it inspires other things. We're actually going to start building up some randomness into our models.
- 3.2 For this lecture, I'm going to be talking about some of the principles behind modeling shares and, to some extent, currencies and commodities or indices and things like that. Looking at the statistics, I'll talk about the pros and cons of various approaches, and then we're going to do a little bit of analysis, and draw some curves and come up with a mathematical model. So this is important, this lecture. Not that the others aren't.
- 3.3 There are three main sorts of analysis that one can do, that one does, in finance. They are fundamental analysis, technical analysis or charting, and quantitative. It's not a very good name, not very descriptive, quantitative analysis. Fundamental analysis is where you look at your company, you're analyzing a company. It means you will look at some balance sheet of profit-and-loss accounts, do the thing that accountants do.
- 3.4 You will look at who is on the board, look at what lawsuits they're involved with, this being America, and—controversial—and look at who's on the board, what intellectual property they've got, the company's got, anything's expiring, patents. Basically, you do a whole load of groundwork that's really tedious, and you'd hope that, being really tedious, it would somehow pay off, right? Doing all that work. I mean, who can read balance sheets?
- 3.5 I was going to say something about my wonderful accountant, but it could be used against me. Not that I'm going to say anything bad about my accountant, but it's what I do with his annual tax statements. I pay all my taxes in advance, but I don't understand, I don't understand all these pages and pages of numbers that mean nothing to me. Running various businesses, I produce my own spreadsheets that tell me how my business works, whether it's successful or not. Completely different from what I see on accountants' documents. Anyway.
- 3.6 So, fundamental analysis would be, you look at the company, and you see that it's been doing this. This is the share price of some company. So you go to all this trouble analyzing the company, et cetera, as I mentioned, and you conclude that that's crazy. All these 22-year-old traders who are selling this thing, they haven't a clue. I reckon, and I'm going to assume that I'm the best fundamental analyst in the world, I reckon the share price should be there. So, what do you do? You buy the stock. You buy the stock because you know, after all your analysis, it's going to be out here. And then you just wait, sit back and wait.

- 3.7** Unfortunately, the stock keeps doing that, and you go bust. There's a famous quote by Keynes. "The market can stay irrational longer than you can stay solvent." He's like the Groucho Marx of economists. "The market can stay irrational longer than you can stay solvent." In other words, it'll go there eventually, but by then it's too late. Because, as Keynes pointed out, you're not trying to figure out where the stock price should be. You're trying to figure out where everybody else thinks the stock price should be, all these 22-year-olds. So that doesn't work. It really is a lot of effort for not necessarily any gain.
- 3.8** Technical analysis is the opposite, because it doesn't involve lots of reading of profit-and-loss accounts and whatnot. You just see that. The pure technical analyst doesn't even want to know what the company is. They just look at that and they go, "It's going down, innit?" That's it. That's your technical analysis. They get more sophisticated. They talk about different shapes, head and shoulders and triangles and whatnot, saucer bottoms and things like that. But the statistics, the people who analyze these things, they say that maybe there are a few things like some crossing of moving averages may have some information, but generally speaking, it's no good. But at least it was easy.
- 3.9** The one that's been really successful is quantitative analysis. This goes back to, as I said, originally the 1900s, but then disappeared, and then it came back in the 1950s. It just says, let's assume everything's random. We just assume that's random. Now, when I say random, it doesn't mean we can't have a model for it. It means the model has to somehow take into account what kind of random behavior. Are we talking about tossing a coin or rolling a dice? What numbers are on the dice, for rolling a dice? Is it 1, 2, 3, 4, 5, 6 or different numbers? Are some of them negative? That's the kind of thing we need to do.
- 3.10** Now, as I mentioned before, when people do modeling of interest rates, they choose models because they are user-friendly. They are very simple. You can solve equations using them. In the whole of finance, there are very few clues about how to do the mathematical modeling. Now, I mentioned earlier the Navier-Stokes equation and the Euler equation.
- 3.11** The Euler equation is a model for conservation of mass. When you're modeling gases or fluids, you know that if you have a box, just an imaginary box, the fluid can flow through. Whatever comes in one side has got to come out the other sides. That leads to an equation, ordinary differential equation, a partial differential equation.
- 3.12** The Navier-Stokes equation is similar. It's conservation of momentum. It's Newton's law. As long as you're not talking about quantum effects or nuclear explosions or whatever, momentum, energy, all these things are conserved. It leads to another set of equations. You solve those equations, you design airplanes. You don't even need to build a model aircraft to design a plane because these equations are so good. We have nothing like that in finance.

- 3.13** Conservation? Share price halves. What's happened? Money has disappeared, nothing's conserved. So we haven't got any clues about how to do modeling, except the following. It's a slightly weird clue, and that is, when we buy a share, we don't care what the price is. We don't care what the price is. The share price is irrelevant because, if we have, say, \$1,000 to spend, if the share price is 1, we buy 1,000. If the share price is 100, we buy 10. We actually don't care. What we care about is how much does the share price go up or go down by? And that's a thing called the return, the return being the change in the share price divided by some reference level. So that's what we're going to do now.
- 3.14** Here we have some S&P 500 index data going back to 1927. A lot of data, and we're just going to look at the return and see how that behaves statistically and see if that can lead to some mathematical model. But that is literally it. That is it as far as—this is why equity models are actually not too bad because we've got that clue, that the level doesn't matter. When you model interest rates, the level does matter. Whether it's a 2% or 16% interest rate, that matters. When you're modeling credit risk, all these things, the level matters, but not when it comes to share price, at least to a first approximation, if you like.
- 3.15** So we have here data, 100 years, almost 100 years of numbers. We have the opening price, the high price, low, close, and adjusted close. That means adjusted for any dividends, splits, et cetera, which is irrelevant. All that is irrelevant for the S&P 500. In fact, we don't need any of this.
- 3.16** What I'm going to do now is, I'm going to look at the return. And what I mean by return here is, I'm going to take the share price one day and subtract off the share price the previous day and divide it by the share price on the previous day. So that's the return from one day to the next. And I'm going to call that R_i .
- 3.17** If I were to plot the distributions, this is a lot of numbers which we're going to think of as being random, but I want to analyze this randomness. One of the first things we do is, we standardize or normalize the numbers. Because if I were to now— I could use some built-in Excel function, and I could look at how these numbers, whether they're all around the middle or large numbers, positive or negative, I want to move it to the center. So instead of having a distribution using those numbers, which might look like it might be something like that, what I want is I want to shift it to numbers which are kind of in the middle and not too fat and not too thin.
- 3.18** So I want to move it to a distribution that's in the middle, and it looks something like that. And that's what I'm guessing, I'm anticipating— sorry, I'm getting ahead of myself. I'm drawing something that looks like a bell curve. We don't know it's a bell curve. It might be bimodal; it could be anything. But anyway, what I'm saying is, I want it to be in the middle, not too fat, not too skinny.

- 3.19** So what I do is I standardize, which means I calculate. So I'm going to look at the mean $[\mu]$ of these returns. So that's your three basis points. That's the mean, and I'm going to look at the standard deviation $[\sigma]$. So this is the spread, just over 1%. And now I'm going to have a column here, which is called scaled return. It's still the returns, but now what I'm doing is, I'm taking from all of the numbers that I've got in that return column, I'm going to subtract off—so you imagine it's over here— I'm going to subtract off the mean. It all moves to the middle. The shape doesn't change; it moves to the middle. Then I divide by the standard deviation, which is this spread, which will then, it'll give it—so my random numbers will now be in the middle, centered around 0, $\mu = 0$, and they'll still have $\sigma = 1$. So it's like that, which is equal to— so that minus the mean. There we go.
- 3.20** So what I've got now is a column of numbers which are just as random as the returns, but they're kind of more user-friendly. That's all I've done. I've moved them into the middle and squished them up. One of the advantages of that is, if I wanted to analyze some other shares, some other company, instead of the S&P 500 Index, I just drop the data into here, and it would give me a column of random numbers, which again would be in the middle, so I could compare and contrast this with individual stocks and shares or exchange rates or whatever. I could overlay them.
- 3.21** So what I'm really interested in now is not so much what the mean and the standard deviation are. I'm more interested in: what is that shape? So, to that end, I want to now plot that. To plot this, I'm going to need a kind of range. What is it from here to here? How big is the spread? So what's the smallest number in there? That's the min. I'm going to do the same with the max. This is just, if I'm going to be doing the plot, I need to know from A to B. That's all this is. This is nothing fancy, except it is rather interesting because of something I'm going to tell you in a second.
- 3.22** The thing I'm going to tell you in a second is somebody—who was it who mentioned the bell curve earlier? You mentioned the bell curve. Well, these are numbers that have $\mu = 0$ and $\sigma = 1$. Now, if they're from a bell curve, what is the probability of getting a number -17? Imagine a hat full of normally distributed random numbers— a distribution of random numbers, normally distributed. They've got $\mu = 0$ and $\sigma = 1$.
- 3.23** So I reach in and pull a number out—0.7, -0.3, 1.4, and so on. At one time, I pull out -17. What are the chances of that? Well, the probability density function, remember $1 / \sqrt{2\pi}$ —well, I've made it $\sigma = 1$ — $e^{-1/2}$. $\sigma = 1$, $\mu = 0$. I'll use ϕ to represent the number that comes out of the hat: $[(1 / \sqrt{2\pi})e^{-(\phi^2/2)}]$. So if it is normal, this is the probability density function, which kind of gives you a ballpark idea of how likely a number is to come out of the hat.
- 3.24** So ϕ is, let's make it simple. ϕ is, let's say, 20. It's close enough, 20. $20^2 = 400$; $400 / 2 = 200$. e is 2.7—let's say it's $\sqrt{10}$. Let's just get ballpark numbers. So that means we got, that'll be 10^{-100} . There's a 10^{-100} chance of me pulling out of a hat, with normally distributed numbers, mean of 0, standard deviation of 1, 10^{-100} . That means— this is

where I usually make a joke that won't go down well in this audience, so I will skip that bit.

- 3.25** So the chance is, the lifetime of the universe is, what, 14 billion years? If you pulled out a number out of a hat every second for the lifetime of the universe, you wouldn't even get close to picking out the number -17. In other words, it's got what are called fat tails. This distribution has got fat tails. Extremes are far, far, far more likely than you get from a normal distribution. So, things are not looking good for the normal distribution, the classic bell curve. All right, never mind. Let's carry on.
- 3.26** I wanted to draw this. I need to divide up into buckets. So I'm going to have a bucket width, bucket width, and I'm going to make the bucket width equal to that minus—I'm going to have 200 buckets. And then, to draw this, to do this, I need to have buckets. And then what I do is, I count. So I have these buckets, and I count how many times has a random number appeared in any of these buckets? And that's going to give me a histogram.
- 3.27** And so, the buckets go from the left-hand end, adding the bucket width. Those are the buckets, one side of the bucket. And then I need, I'm going to do it over here. I'm going to have the frequency. So there's a function in Excel that does the frequency, and it's called frequency, equals frequency, the data array that would be these things. Yes, that is right, comma— that's probably better.
- 3.28** So you can see that there was one instance of this number, this extreme drop, which, incidentally, was in October 1987. Do any of you remember that day? And a lot of big numbers in the middle, which is when the—which is small. These are small returns, a lot of small returns. And then you can see, the larger return drops off, but there's one remaining one around 13, wasn't it? That one, that was a big bounce back. Something like that.
- 3.29** Now, this is a frequency distribution. So from this, we can see that there's not much action in the tails, although when there is action, it's big—20% fall in the market is big. In an index, it's big. Usually, however, the movements in the share price are in the middle, not very dramatic at all.
- 3.30** I want to plot, instead of frequency, I want to plot a probability density function. This is the thing that has the area under the curve of 1. So I need to do some scaling. But first of all, I also need— let me get this right. Sorry, I have to subtract off from this, these numbers, half of the bucket width. What I'm doing here, this is just a little bit of housekeeping, because when I do the plotting, this is the left-hand or right-hand or something of the buckets, which I'm counting in here, but I want to plot against the middle when I do the plotting, not against the left or right. So all I've done there is just shifted things a little bit.
- 3.31** The probability density function. To get the probability density function, this is the empirical one, not the normal one. This is what the data actually looks like. I have to

take the frequency—I need something else. I need to have how many data points have I got. I need to count how many points there are, because I need to do some scaling to make sure that the 24,000 data points, I need to make sure that the area under the curve is equal to 1. So I need to take the frequency, divide it by the number, and then divide it also by the bucket widths. Then, copy that down.

- 3.32** All being well, I now want to plot, to compare and contrast, with the normal PDF. The normal PDF—the probability density function—for this is $1/\sqrt{2\pi}$ — $\sigma = 1$, $\mu = 0$, so it's $e^{-1/2}$ —not that one, that one: $[e^{-x^2/2}]$. There you go: $[f(x) = 1/(\sigma\sqrt{2\pi}) * e^{-(x-\mu)^2/(2\sigma^2)}] \rightarrow (1/\sqrt{2\pi})e^{-(x^2/2)}$ Okay, 10^{-65} , so the chances of that happening again, it's not going to happen. And then I'm just going to superimpose. There you go. There's your normal distribution.
- 3.33** The empirical distribution is skinnier. It's got those things in the tails that you can't see. It's still symmetrical. That works. Be thankful for small mercies. It is symmetrical. There's no skew in there, but it's not normal. That's really annoying because we really like normal distributions. We really like normal distributions. Now, lots of reasons, mathematical reasons. They crop up, as I said, scientifically, all sorts of reasons. What do we do? What do we do?
- 3.34** Some people say, "Oh, we've got to be true to the statistics"—normally, I would say that—"and work with distributions that have got the real empirical shape, the fatter tails, the higher peak." The problem with that is that they are not exactly user-friendly. They're not so transparent. Sometimes they don't even have nice formulae. Sometimes they have standard deviations; they're actually infinite. Shortly, in a few minutes, we'll be seeing how standard deviation is related to a thing called volatility, which is crucial. Volatility is a measure of risk and all sorts of things.
- 3.35** Imagine you're a quant at a bank, and the trader comes to you and asks you, you're using one of these fancy fat tail distributions, and the trader says, "What's the volatility?" And you say, "Oh, it's infinity." That's it. You might as well pack your belongings, and escort yourself to the door. You're fired. They want the number 27.6, not infinity. So you have a choice. Do you work with normal distributions, even though they're wrong, or work with the fat tail distributions, which is statistically more accurate?
- 3.36** Well, the thing is, in practice, this 20% crash—these fat tails don't happen that often. Do you want every working day in a hedge fund to be governed by what may happen in 20 years' time? If you do that, if you go around obsessing about these fat tails, you'll be terrified to do any trading, which may be the scientific thing to do, but it's not going to be good for your bonus.
- 3.37** And also, it doesn't necessarily help you, because the crash might be totally random. So you're still going to suffer when the crash happens. Then we can get real, and we can say, when the crash happens, it'd be nice if you don't lose your shirt, but everyone else

has lost their shirt. The whole market has fallen 20-30%. Banks are collapsing left, right, and center, and you're the one person who didn't lose money. Fine, but your whole bank has disappeared. There's no bonus or anything for you. So what's the point?

- 3.38** If you want to be real about it, there are different levels of market crashes. The market crash, if it just happens, if you're the only person that is losing money, that's bad because you're fired. If everyone else is losing money, well, things will recover, you'll get a job. Then the market could fall 80%, in which case money means nothing. We're all living in caves, eating tins of cold baked beans. So there are real-world things besides the statistics.
- 3.39** So what I'm saying is, use normal distributions because they're really user-friendly, and you can understand them. And they have two parameters. They've got the μ and the σ . But never forget that the crash could happen, so maybe have some insurance in place for if/when a crash does happen. That's what I would say to that.
- 3.40** All right, we're now close to having a mathematical model. We're going to work with normal distributions, and the model is now going to be as follows. It's going to be, this is the thing that's going to be random. We're going to say this comes from a normal distribution, but with a non-zero mean and the non-unit standard deviation. So just rearranging this, you can rearrange this.
- 3.41** And what you get is S_i —take that up there and take that over that side—you get $S_i = (1 + R_i) * S_{i-1}$. In other words, and this is going to be random, that thing there, random. You take yesterday's share price, you multiply it by 1 plus a random number, and you get the next share price. That's the model. And that has this property that you don't care what the level of stock price is. This is a thing called a geometric random walk.
- 3.42** What I mean by geometric is that you're multiplying the previous thing by a number. You're not adding. It's not an arithmetic thing where you add a random number to yesterday's share price to get today's. You're multiplying by a random number. And this is a small number. We're going to say it's going to be something like this. It's going to be $S_i = 1$ plus— what was the mean we had? Did I say it was something like 3 basis points? 0.0003, for example, in the data we've got there—plus—what was the standard deviation? It was about, it was something like 0.011, I think. Let me do it in symbols, in words, so half words and half numbers: $[1 + \text{mean } 0.0003 + \text{standard deviation } 0.011]$ —times a random number—I'm going to use ϕ to mean a random number. ϕ is my hat full of random numbers from a normal distribution, with $\mu = 0$ and $\sigma = 1$. I'm sorry, and I've got to have the S_i on the other side, so times (S_{i-1}) : $[(S_i = 1 + 0.0003 + 0.011 * \phi) * S_{i-1}]$
- 3.43** So, to do a simulation, which we will be doing later on, to get the next share price, you take the previous share price and multiply by 1 plus something small, and the something small represents the growth. And a parameter here, which is a measure of the risk, the variability, multiplied by 5, which has never been pulled out of a hat, and that is the mathematical model.

- 3.44 It's not the greatest mathematical model as it stands, because it either has numbers, which are great, but those are the numbers for the S&P 500 index, and we don't really know what the numbers might be for other stocks, for example. Or it's got words in it. I want to write this using symbols because I'm a mathematician. There's a T-shirt in there somewhere, isn't there? Think about that.
- 3.45 So I'm going to show you how to do it. The trick is— it doesn't look very complicated, but believe me, this is what people do when they're doing portfolio, constructing portfolios or doing option pricing, which we'll be doing later on. This is the model but they don't write it like this. This is what they mean, but they don't write it like this. When they write it in maths, full-on maths, it looks more complicated than that, but it really is just this.
- 3.46 But what I want to do is, I want to turn it all into symbols. One issue is that I've used daily data in this spreadsheet, and I don't want to be constricted to daily data. For example, we started off with dy / dx and things like that, calculus. Calculus is all about infinitesimals. This mean is not an infinitesimal. That is linked to a time step of one day. I'd like that to be linked to just pure symbols, maybe connected to the time step I use, for example.
- 3.47 What I'm going to do is, I'm going to say: how does the mean, the average change in the stock price, which here is 3 basis points, how is that linked to the time step? For example, if I'm using daily data, I had a mean of 3 basis points. If instead I used weekly data, how big would that be? Well, it would be five times the size, five business days in the week. The means, remember, expectations add up. This is the random bit, with $\mu = 0$, $\sigma = 1$. This is the mean bit, and means and expectations add up. So what happens over five days is just five times what happens over one day. It seems plausible.
- 3.48 So what I would like is, I would like to have, the mean is proportional to— that doesn't look very good—the time step. I'm going to use δt as a time step. I'll try and write that better. δt . Now that's not $\delta * t$, in the same way that dy , dx is not $(d * y) / (d * x)$, it's dy , dx . That's δt . It's one symbol: δt . I claim that the mean ought to be proportional to the time step, and I'm using μ , the Greek letter μ , to represent this constant of proportionality. Now, let's look at this for a second and see what this represents. I've justified it by saying that expectations, you can add them.
- 3.49 Suppose I for a second cross off ϕ here, then I would be left with that $S_i = (1 + \mu * \delta t) * S_{i-1}$. That's in the absence of ϕ . Let's just do a little thought experiment. Suppose I start off as S_0 . The next one will be $S_1 = (1 + \mu * \delta t) * S_0$. So $S_2 = S_1 (1 + \mu * \delta t)^2 * S_0$. And you can see that $S_3 = (1 + \mu * \delta t)^3 * S_0$, and so on.
- 3.50 Now look at this. Suppose I say I'm going to—this is a time thing, time 0 here, and I'm going to go up to time t , and I'm going to divide it up into a lot of time steps. Each of these has got size δt . So how many of them are there? Well, I go up to t , and each one is size δt , so the number of them is $t / \delta t$. That's how many there are. So that means S , S

at time t , and I'll use that notation— $[S_t]$. S_t is going to be that $[1 + \mu * \delta t]$ to the power of that $[t / \delta t]$: $S_t = (1 + \mu * \delta t)^{t/\delta t} * S_0$. this is looking a bit like something we did before, which if I go up here, $S_t = S_0 * \exp$ — because $\exp$ and $\ln$ are inverses, it's going to be $\exp(t / \delta t) * \ln(1 + \mu * \delta t)$, which is approximately $S_0 * \exp(t / \delta t)$ —this is where good old Taylor series comes in— $\ln(1 + \mu * \delta t) \approx \mu * \delta t$ plus small terms. The δt 's disappear, leaving me with $S_t = S_0 e^{\mu t}$.

- 3.51** In other words, if there's no randomness, the stock price grows like that, which is geometric growth, exponential growth. And if that wasn't δt , if it's δt^3 or $\delta t^{1/2}$ or something, when I got to this point here, there'd be some δt 's left over, and you wouldn't get anything sensible. So the only way to get something sensible from this is to have that sort of behavior. So our model so far is $S_i = (1 + \mu * \delta t + \sigma * \phi) * S_{i-1}$, so it's getting there, isn't it? It's getting there in terms of parameters and whatnot. It's starting to look a bit more mathsy, except standard deviation. It's more like actuarial mathematics and proper maths. So what about the random bit? And I say one of the reasons you do this is because mathematicians really do like going to what's called continuous time, when there are no actual time steps.
- 3.52** Next one: I'm going to do the standard deviation now. The standard deviation is going to be—what's it going to be? It's going to be something to do with δt . Intuition is not so good for this. Standard deviations don't add up. Variances add up. If what happens day to day is independent, variances add up, but standard deviations don't. All right, let me try the same trick.
- 3.53** This is t , this is 0 . We're going to divide time up into intervals of δt . And the standard deviation over each day—let me call it SD , so that's the standard deviation. We can't add up standard deviations, but we can add up variances, which is standard deviation squared. How many of them are there? There are $t / \delta t$ of these things that we add up, so multiply that by that, and we want that to be independent of δt . In other words, we don't want any—when we do this calculation, we don't want any δt , which means the standard deviation squared must be proportional to δt . So it must be this: standard deviation must be proportional to $\sqrt{(\delta t)}$. Let's call the constant proportionality σ , and that's it. That is the job done.
- 3.54** Now, we have a mathematical model which is all symbols, no words, no numbers: $[S_i = (1 + \mu * \delta t + \sigma * \sqrt{\delta t}) * S_{i-1}]$ and that is important. It gets a red box being important because that is a very famous thing called a lognormal random walk. Geometric Brownian motion is another thing, named after the Scottish botanist, I think, who first came up with the idea of Brownian motion, which is, he dropped bits of pollen in some water and saw how it spread out. That's to do with the action of the water molecules on the pollen or something, and things bounce around.
- 3.55** So this sort of mathematics here is diffusion. Einstein did some work on diffusion. Actually, before Einstein worked on diffusion, this man I mentioned right at the start,

Louis Bachelier, used these sort of models in finance, so finance precedes whatever Einstein was doing. So that is the lognormal random walk for stock prices.

- 3.56** So this is the growth rate $[\mu * \delta t]$, as you saw here, it's the growth. The bigger μ is, the faster it grows. This thing $[\sigma]$ is called the volatility. Volatility. It is the annualized—meaning, we've scaled it— the annualized standard deviation of returns. That is what volatility is. Volatility is incredibly important because volatility is a measure of risk, the higher that is. So what you see here is, the stock price is growing day to day, because μ is positive, and it's bouncing around.
- 3.57** But look at the different sizes of the bouncing around versus the growth. In the data here, we had the, we had the returns. They're bouncing around all over the place, whether you're looking at the unscaled or the scaled, they're bouncing around all over the place, even though it's got positive growth, the bouncing—so the bouncing around is more important; it's more obvious than the growth. Keep it simple. Suppose δt is 0.01, 100. Then that's 1/100, but that's only 1/10. $\sqrt{0.01} = 0.1$. It's much bigger than that.
- 3.58** So what you see, the share price, it does this. What you're experiencing, your lived experience of this share price, is it's going crazy, up and down. You have to wait years before you realize, "Oh, actually, it's going up. I'm making money." That's why, when it comes to deciding where to put your money, what to invest in, both of these are important. Ideally, this would be big and this would be small. In other words, it'd be growing with not too much bouncing around. But they tend to go hand in hand. Bigger the risk, bigger the return, as I'll be explaining later on.
- 3.59** But it's very hard to even know what this is, because you have to wait a long time. Of course, if you wait 25 years, you've been through several business cycles. It used to be a bull market, then it was a bear market, then you're going sideways. That's very hard to measure. This is easier, not easy, but easier. When we do options, you'll see how incredibly important that thing is and that one isn't. We are going to be doing some simulations of this later on to see what this looks like, by generating random numbers, pulling random numbers out of a hat.
- 3.60** What else can I say about this? Yeah, if you divide both sides by S_{i-1} , you see that there's no S_{i-1} on the right-hand side, so the return is just that. Again, nothing to do with the level, which is where we started out. It does assume, there's one assumption here that's not totally correct, and that is, when I did all of this, for example, this bit here, the variances being additive only works if what happens from one day to the next are independent of each other.
- 3.61** Is that the case? Well, let's have a look. I'm going to go to October 1987. So that's a really big number. That's the number I'm looking. This is the famous October 1987 crash. Ballpark, 20%, so 17 standard deviations fall. Now, this point about what's called serial autocorrelation means, is there any relationship between what happens one day

and what happens the next day? And the big assumption in all the finances is that there's no relationship.

- 3.62** That is clearly not true here around this big crash because 2.2 standard deviation fall, is big. It's not that big, it's quite big. Bounces back 1.36, biggish. No, it's getting a bit big there. Big fall, another biggish fall. Oh, a big fall. Four standard deviations is a big fall, 17 standard deviations. So we've got a few big falls, followed by a really big fall. Then it bounces back, then it bounces back a lot, and then it falls a lot. There's another big—look, something's going on.
- 3.63** I'm not saying you can mathematically model this easily. You can't say a fall followed by a big rise, followed by a big rise, followed by a fall is going to lead to something. I don't know about that, but there's clearly a relationship. It's not like 17 came out of the blue. There was something going on. This is not uncommon. The bounce back is quite common, that what happens after a big crash is, there's a bounce back, and very often it's half the size of—don't quote me on that, but ballpark, half the size. So serial autocorrelation is a big thing.
- 3.64** There are other models. I won't go into it now, but we're going to see it later on, especially when we come to do options. Another way of doing this is actually even simpler. I'll just give you a flavor of this. Another possibility is to say, let's forget this hat full of random numbers. Let's just toss a coin. So you might have something like this. You've got S_{i-1} and it's going to go to one of two places. It goes up with a probability p or down with a probability $1 - p$. And then it goes to S_i .
- 3.65** Well, what is S_i ? Usually, to capture this geometric behavior, you would say something like, it's going to be proportional to the previous stock price. So the next stock price after the S_{i-1} , so here it's time t , this is time $t + \delta t$, what you often say is, I'll make it proportional, u , some u , which is going to be $u = 1$ plus something to do with the mean and the standard deviation, or it falls to $v * S_{i-1}$. So that's what would happen if the stock price fell, and $v = 1$ minus something.
- 3.66** This also captures the geometric nature because you're not adding the same number. You're not adding a random number. You're multiplying by u and v , which you've got the randomness. So within here, there's the μ and the σ and the δt and a random—not a random number— μ , σ , δt . This is governed by a coin toss, but a biased coin toss. Typically, more likely to go up than go down, so p might be greater than half. If p was less than a half and stocks were falling, then no one would buy the stock, and they'd just go down to nothing.
- 3.67** So watch out for the following sort of picture that we'll be seeing later on. That is, the stock price doing this, and the next day, so that's one time step. The next day it goes up some more or down, or it does this and that, so this point it goes like that or like that. And this point it goes like that and like that. Notice how it's curving because it's multiplied by the previous stock price. So again, it gives the kind of exponential growth.

Or if it falls, followed by a fall, followed by a fall, it has this exponential decay. So that's the picture you get, something like this, what are called trees, and up or down, or down and up, and up and up, or down and up. This has a similar kind of property to the other random walk. So that's the key thing, the lognormal random walk.

- 3.68** Wasn't it easy, the spreadsheet? Getting those numbers? So the μ that came out of that—actually, I suppose I should do that, shouldn't I? I should finish off with— yeah, so I should—maybe I'll finish off by saying: what would the mean be in annualized terms? It's that times 252. Why 252? Because people say that there are 252 business days in some years. So that, and then the standard deviation, equals $\sqrt{252}$ times— that's not times, times that. So that, if I wrote those in percentage terms, that would say that the growth rate, the S&P 500 index over 100 years is 7.7-7.8% with a standard deviation of 19%. And those are standard kind of numbers, growing at 7%, 8% a year, with a volatility of that kind of number. Any questions? Yeah.
- 3.69** Basically, this was off of the S&P 500 since 1928, correct?
- 3.70** Yes, I assume that's the data I've got.
- 3.71** Do you use different models compared to different indices?
- 3.72** This lognormal random walk, as I said, is the best thing we've got in finance because of that clue about the return. You would use this for different indices. You'd do it for stocks, other individual companies. You'd use it for currencies as well, which is a bit odd because currently the dollar-sterling exchange rate doesn't grow exponentially or fall exponentially. So there are some issues there.
- 3.73** You can use it for commodities, but then there are different commodities. There are commodities that are just for storage of value, and then there are commodities which we use, like orange juice or oil. So they have different behaviors, or maybe they've got seasonal behaviors. These can be captured to some extent by changing some of these parameters. You might have some seasonality you can throw in there. Often people don't bother with that thing. They just leave it as it is.
- 3.74** There are more complicated things. What sometimes happens is, if instruments are not very liquid or not very popular, then people use simple models. But sometimes, especially in the world of options, with so much mathematics, they find that just saying volatility is 19% is not good enough. They need to have something that's a bit more accurate. Now, accurate doesn't necessarily mean more correct or more predictive. Sometimes it just means more complicated. And so, often they have models for how this behaves.
- 3.75** For example, they may have volatility itself being random, pulling numbers out of a hat to give you what volatility is. For reasons which you'll see later on, this number is almost irrelevant. So this number is almost irrelevant for options. So nobody, any options trader, will not have a clue about this because they don't need it, whereas they

need this. They go to a lot of trouble to try and figure out what this should be. So that's where, there are complicated models for that. You'll be seeing some hint at those later on. But yeah, I'd give this 70% for accuracy, whereas credit risk, 10% for accuracy and interest rate is 20%.

Lecture 4: Financial Modeling

At a glance — Financial modeling, recapping the tools so far.

- 4.1 A slight, little recap: what have we done so far? We've talked about a couple of very special functions, the exponential function, the logarithmic function. Then we did Taylor series, didn't we? We saw exponentials appearing all over the place out of nowhere, it seems. We did a little bit of analyzing share prices, basic stuff, and we ended up with something along the lines of—what was it we had?
- 4.2 The next stock price can be related to the previous stock price by something along these lines: $[S_i] = (1 + \mu\delta t + \sigma\sqrt{\delta t} * \phi)S_{i-1}$. That means that the return was normally distributed with a certain mean and a certain standard deviation. The mean scales with the time step: $\mu\delta t$ — μ for the S&P 500 index, we found, was about 8% or so.
- 4.3 This term: $[\phi]$, this is the randomness term. This $[\sigma]$ is the thing called volatility, which is incredibly important because this thing dominates. The stock price is doing this. That's the thing that dominates over short time scales. This is the number which we're saying is drawn from a hat full of normally distributed random numbers because it's simple, it's user-friendly, there are nice formulae, there are scientific reasons why we might get normal distributions. It's not totally accurate, but as I said, for pragmatic reasons, it's probably best to go around assuming that things are normally distributed, but always have at the front of your mind that there could be a stock market crash the next day.
- 4.4 So, going around assuming that there's going to be—working with those fat tail distributions is going to make your life a lot harder with no noticeable gain. You'll just be paranoid because you wouldn't trade anything. So trade as if nothing's going to happen, but always remember, there could be a big crash. It's the tails that are obviously important. But most of the time, nothing really happens. When we did the bell-shaped curve, the empirical curve, we saw that most of the time, share prices, indices don't move that much. You never see Wall Street Journal headlines saying markets move 0.16% today. It'll be something, "Oh, 20% crash." Most of the time, the vast majority of the time, nothing much really happens. But the volatility is still there, and then occasionally, boom. But that occasionally that collapse is relatively rare. So, that's the model. We're going to do simulations in a second of that, and I'll show you how you can use these simulations.
- 4.5 I do want to introduce you to another thing which will become important later on. I did hint at this earlier. That is, what about the model where the stock price starts off here and goes up to some multiple of the stock price, or down to another multiple of the stock price? Because I talk about coin tossing. Sometimes I talk about pulling random numbers out of a hat. Well, the coin tossing one, we're going to be using that a bit later on. So I just want to tie these two approaches in for later.

- 4.6 What we have here is the stock price goes up to some number u , and $u \approx 1$. It's going to be 1 plus something small because in one day, this is what happens over one day, or let's call it δt again. What happens over one day is that the stock price doesn't change that much. It's usually pretty much close to what it was before. So it goes up to 1 plus something, and the something is going to be related to the mean and the volatility. Or it goes down to 1 minus something. So it will go from 100 to 101 or 100 to 99, and you'll have some probability for this happening: probability going up, probability going down. So this is something that we're going to be using later on when we come to options and derivatives, it's that simpler model of just the two states. It's called a binomial model.
- 4.7 So, for example, you might have this being 60% chance of rising, 40% chance of falling, or it might be 50-50, it might be a coin toss. Anyway, that's a simpler model, but we don't really use that in practice so much. It's used in theory. When we come to options later on, you'll see how we use this as part of the buildup of a theory. But this is the model which captures the same randomness. This thing looks like this: when I do a simulation, what you'll see is, if it starts off at, say, 100, and it will do something like this. If it was this thing, if you do a simulation, it'll do something like this. But once you zoom out, that's only in two weeks, once you zoom out, meaning you do this over a year, you change the scales a bit, it actually looks, it does the same thing once you zoom out.
- 4.8 So let's do some simulations— this is of time. So the way this works is I'm going to start off by having a stock price. Let's start off with the stock price of 100. Anything in yellow, when I do this, anything in yellow is something you're allowed to change. It's just a parameter or variable or something. If it's not in yellow, don't touch it. So the stock is going to be 100. We need to have something like—we need to know: what is the mean growth rate of this stock? Let's do 8%. Actually, let me call that a percentage. Again, that's something we could change. Depending on what the thing is we're trying to model, what we're trying to simulate, you would use different values for that. So these are the numbers that we got earlier for the S&P 500 Index. So 0.19, I think it was, something like that, wasn't it, for the volatility of the S&P 500 index? So that's our starting position, starting data.
- 4.9 We want to simulate over a certain time step. Now, if you were to simulate, the smaller the time step you use when you're doing this simulation, the smaller δt is, the more accurate you get. If I'm interested in what happens over a year, suppose I'm curious to know: what might the stock price be in a year's time? Then I could do a simulation that goes from here over to here. It could be there, or it could be there, or it could be—I could do a simulation of where it could be. But really, I'd like to know what goes on in between.
- 4.10 Why would I like to know what's going on in between? Because I might want to know: what is the probability of this happening? Oh, gosh, that's a disaster. I'd have to sell at

this point. I'd like to know, rather than close my eyes and not look at my share price for a whole year, I might want to watch what's happening every day. So I would want to simulate with δt , the time step very small. Yes, because I don't know what your particular portfolio is like, what you've got, I don't know how frequently you look in The Financial Times or The Wall Street Journal to see how you're doing, but you probably look, if you've got a fair amount of money invested, you might look quite frequently, which is something I shall be talking about in a later lecture.

- 4.11** So I'm going to have a time step, and I'm going to have the time step—let me say it's going to be one day. Everything is annualized, so one day is ballpark, well, there are 252 [trading] days in a year, so $1/252$ is one day, it's a fraction of the year. Again, that's something you can change, but I'm keeping it as that. So that's my starting position.
- 4.12** We're going to have a column here for a time. It starts off at 0, obviously, and it's going to increase by—let me actually give these things names. I'm going to call this the vol., and I'll call this the time step, obviously. So, start off at 0. The next time step is going to be the previous time plus the time step. I'm going to drag this down. I'll go down to a whole year— that's too far, not far enough, there we go. So that's just time. Because I'm going to be doing plots, I want to see what the share price is. I want to plot this, essentially. So that is the time.
- 4.13** Then I'm going to be doing the share price. What I mean by doing the share price is I'm going to have a simulation one: [Sim 1]. Share price starts off at, obviously, starts off at that. This is where the fun happens. This is where I put this particular formula. So the share price on the day after the zero day is going to be equal to the previous day's share price. So that's that thing here: $S_i = S_{i-1}(1 + \mu * \delta t)$. If I just left it at that, then what you'd see is the thing that we did the other time, the other lecture, which is that's just exponential growth. It's the wrong formula.
- 4.14** We want to put the randomness back in, which is volatility times the square root of the time step, multiplied by—and this is where we need a random number drawn from a hat of normally distributed random numbers. Now, there are a couple of ways you can do this. One is to exploit the function in Excel, which deals with normal distributions. That, I believe, is NORM,S,INV— it's not quite finished—(RAND()). I believe that is the function. Sometimes I cheat because some people obsess—does that look right?—some people, not me, but other people.
- 4.15** The people who do this for a living, should we say, within banks and hedge funds, they have to make their code very, very fast. So they would have their own ways of generating random numbers and more accurate random numbers and things like that. Sometimes I cheat by exploiting the central limit theorem we did earlier, that if you actually add up a lot of numbers drawn from a uniform distribution, then it approximates a normal distribution, and that can be faster. But we're not interested in speed here, so I'm just going to go with that. So that is one simulation.

- 4.16 Let's see what that simulation looks like. There we go. Now, one simulation isn't much use because this is not a prediction. That's not going to be—this is not what's going to actually happen in real life. But I can press F9. So you can see, you can get an idea of what the share price might be doing. But since it's not a prediction, it's not much use. What is more useful is to run many simulations, and by running many simulations, you get a better idea of all the possible things that could happen. What I mean by that is I'm going to drag that over. I've done 126 simulations. That's it. That's 126 simulations. It gives you an idea of what the stock price could be doing.
- 4.17 Now, one of the fun things about this is, you can see, you can do a lot of things, all those random numbers I'm pulling out of a hat. You've got a whole range of things that could happen in one year's time. If I now do, I'm going to do a little quick calculation on the side just to show you the thing that people who do this for real are faced with, one of the problems they're faced with. I am going to do—where should I do it? I'm going to temporarily, in here, I'm going to calculate the returns. So what I've done is, I've done the simulation, but I want to now pretend that simulation one was what really happened. So pretend that simulation one actually happened.
- 4.18 Going back to what I did in the previous spreadsheet, I could calculate, I could say: what is the return each day? Copy that down there. Then I could ask myself: what is the mean that came out from my calculation? What is the standard deviation, using simulation one, which is the thing that I'm pretending for the moment is real? So the mean return is going to be the average of all of those returns. But then I want to multiply that by 252 to get it back to being the annualized thing, an annual return. I do the same with the standard deviation. So I do the equals standard deviation [=STDEV] of that. And then I multiply that by $\sqrt{252}$. Remember the standard deviation, the volatility scales with the square root of the time step.
- 4.19 Now, look what has happened here. I did a simulation where I told it, I want an 8% mean, and I want a 19% standard deviation—19% volatility, I should say. So a growth rate of 8%, a volatility of 19%, and that's what I put in. Let's turn that into percentage. Well, look what's happened. The standard deviation I put in was 19, and what actually happened in my simulation was that I had 18.2. That is pretty accurate. You might think, "Oh, if I had a smaller time step, I would get even more accurate." Yeah. But look at what's happened to the return. I put in an 8%, and actually, it's come out to 17%. Even though I told it to simulate an 8%, the actual simulation has given me a 17%. Now, if I press F9 again: a very good volatility. Now, the mean for that particular simulation is negative.
- 4.20 Now, that's illustrating—there's nothing wrong. I haven't programmed this incorrectly. Even I could make mistakes with something this simple. But it shows you what we are faced with when we're doing statistics of a share price or an index. The problem is that the growth is really dictated pretty much by what goes on between here and here. What happens in between doesn't really matter. And you can see that from this picture

that the range of future stock price values, whereas the volatility, the standard deviation, is governed by every single data point in there, which is a lot of data points. So for that reason, the volatility comes out to be pretty accurate.

- 4.21 That means anybody who takes, let's say takes, if that were the genuine [data] off what actually happened, then anyone trying to figure out what is the growth rate going forward has got a problem because these statistics are so awful. That's over a year. Now, if you had 100 years' worth of data, then things would be more accurate. But who's to say that the next hundred years is going to be the same as the last 100 years? Or even who can say that the next month is going to be statistically the same as what happened over the previous 100 years? There are all sorts of issues here.
- 4.22 Even asking the question: are we in a bull market, a bear market, or whatever? Clearly, we're in some bull market here because of the 8%, but we still have a stock price that went down. So even though the growth is supposed to be positive, just those random numbers that came out of the hat gave us the negative return. So that's interesting and worrying at the same time.
- 4.23 When we come to options soon, you will see that anyone who works in options is very lucky indeed because—and this is very clever, and this won a couple of people a Nobel Prize—it turns out that this $[\mu\delta t]$ is not very important when you're trying to value these things called options, which we will be doing. This $[\sigma\sqrt{\delta t}]$ is important, but not this $[\mu\delta t]$, which means anyone who works in what are called derivatives or options or these more complicated financial products, they are very lucky because they only have to worry about this fellow $[\sigma\sqrt{\delta t}]$. Whereas if you're in the business of allocating shares to a portfolio, you need to know what this $[\mu\delta t]$ is, and your job is much, much harder. So we're going to be talking about portfolio stuff shortly, but that's much harder.
- 4.24 Now, what have we done here? Why have we done it? More to the point, why have we done that? Well, we can ask all sorts of questions. We can ask questions about, for example, let's do an example. So that's the Sims sheet. Let's add another where we do a little bit of analysis. So let's do this analysis on this page here. I'm going to take all of these values here, and paste them over here for me to do a little bit of— I want to transpose.
- 4.25 Let's do the values. These are all the values from each of those simulations right up to one year. And this I can do, let's see: what is the largest and smallest? I can do the same trick as I did when I analyzed the share price. I can look at what is the minimum of all of these. What is the maximum—equals $\max [=MAX]$ —of all those. Look, even though it's got a nice 8% growth, we still have, we could go down as far as that. I want to do the same thing. I want to do the bucket width, exactly the same thing as I did before. I won't have so many buckets this time, though. Let's have just 20 buckets. And then, I will go from this. This is exactly the same. I won't put the boxes, the notes for this. It's exactly the same as I did before in the previous lecture. And then, I'll do the

frequency, the frequency function. The frequency function is a clever way of doing one calculation to output multiple things.

- 4.26 And then, I want to turn that into a probability density function. Again, I like probability density functions rather than histograms. Nothing against histograms, but they are the thing one does when one is seven years old in school before one moves on to probability density functions. Histograms and bar charts are great when you've got discrete things like shoe size. But if you're doing foot size, then you're probably dealing with continuous distributions.
- 4.27 I'll do the midpoint of this again. Then I want to turn that into a PDF by scaling, so it equals that, divided by the bucket width, divided by the number here, and that gives me a PDF. I can now plot these things—what's it doing down there? So plot these to see what it looks like. There we go.
- 4.28 That is supposed to—I mean, I've only done 126 simulations. When people do this for real, they will typically be doing tens of thousands of simulations to get a good idea of the statistics. But the picture that we now see, this is now the share price down here. These are the share prices at one year, which is how far out we did the simulation. And this is the probability density function. So what it looks like, is like that. That's what it should look like. It does look slightly like that, doesn't it? Except jagged version of it.
- 4.29 This is not a normal distribution. First of all, it can't go negative. We can't have the share price going negative because it's this multiplicative thing. It can go very high. It's what's called a lognormal distribution. Lognormal distribution. Okay. This idea that we've got this geometric behavior where you're multiplying share price by a random number, that's normal to the multiplication turns into lognormal.
- 4.30 You can see that if you know that, if you recall things like $\ln(A * B) = \ln(A) + \ln(B)$, then take logarithms of this $[(1 + \mu\delta t + \sigma\sqrt{\delta t} * \phi)S_{i-1}]$, you'll see that $\ln(S)$ behaves in a certain way, and it gives you the lognormal random walk. Anyway, so that's what that looks like.
- 4.31 Why have I done that? Because it'll give us an idea, for example. We can ask questions like: what is the probability of the share price being above, say, 120 in a year's time? So we could do things like—we don't even need that plot, really. We don't need that plot. What we can do is go back to our simulations, and we could have in here—I'm ignoring this column, this F column, because it's different. So I'm only doing everything from column G onwards.
- 4.32 So I could do, let's say, I'm going to ask the question: probability S greater than—let's put a number in here—greater than 120 [Prob. $S > 120$]. We can change that. Probability S greater than 120. What we do in here is, I'm going to put in here the number of simulations I've done: COUNT all the way along there. So that's how many simulations I'm doing: Numb-Sims, that's what that is. Let's call it Numb-Sims. So I could do something like this. I could do equals if [=IF] that closing share price and what years'

time is greater than that, then 1, else 0. Copy that all the way across. And then, I just put in here SUM of all of those divided by that. And that is the answer to that question: what is the probability of the share price being above 120? Or I could put in any number here now. The chance of it being above 150? It's possible. What about being above 80? 95. So there's a 5% chance you're going to be below 80, even though you've got this enormous growth.

- 4.33** Then, suppose you've got some stop-loss thing in place, or you'll sell when the stop the price hits, reaches some level. Suppose you want to get, you need for some reason, you need \$20. Then you could say, probability S greater than some number ever, any time. Let's do this. The number in here. What we do is we just need to put a formula in here that tells us what we're trying to—actually, let me put the formula in here, this one. Any test we want to do, we could say: equals if max is greater than that, then 1 else 0 [=IF(A1 > B1, 1, 0)]. Good.
- 4.34** So what's the probability of ever being above 120? We've got a 37% chance of being above 120 at some stage, whereas— there you see what's happening. As I press F9, the numbers are changing. The reason the numbers are changing is because I've only got 126 simulations. Let me put the formula in here. If I did 10,000 simulations, I would still have a picture that looked like that. It'd just be even more crowded. It would still look like that. But still, I would get the problem that any particular simulation would not necessarily have the statistics of what I put in, those things. Volatility would be good, the drift wouldn't be. But if I did 10,000 simulations, then any statistics based upon those simulations would be robust. So these numbers would not change.
- 4.35** There's a rule of thumb when you do these simulations, that the accuracy or how stable those numbers are, these statistics, probability of being above 120, et cetera, the change is roughly speaking proportional to the square root of the number of simulations you do. So if I do 100 simulations, then the $\sqrt{100} = 10$ or 10%, 1/10, 10%. So you'd expect a big change in these numbers. But now I've got 1,600. If I had 10,000, $\sqrt{10,000} = 100$, so you'd expect a 1 % error in these numbers. So that's why people do tens of thousands of simulations to get the error in the statistics down.
- 4.36** Because we're talking statistics here. I mean, only one of these, well, none of these paths are going to come to happen, but there will only be one, what's called a realization, obviously, when one experiences the next year. So these things come into their own when they're, for example, if you've got a portfolio of lots of different stocks. If you've got 100 different stocks in your portfolio, then the statistics will start making some sense. But for a single stock, one realization, it's not always totally useful.
- 4.37** Anyway, obviously, you can put down anything you can have a formula for. I mean, you could say something like—this is a little bit more complicated. Yeah, I won't do it, but I'll give you an example of the thing you could do if you've got a separate column of numbers. You could ask questions like the following. You could say, for example, you could say, get more sophisticated. You could say, I've talked about the probability of

being above a certain level or reaching a certain level. But you could ask the question: what's the probability of falling below before you reach some other level?

- 4.38 You can imagine when I was talking about technical analysis, fundamental analysis, and quantitative analysis, I said that the market can stay irrational longer than you can stay solvent. Well, suppose you think that actually the share price really ought to be here, but because of those traders, 22-year-old traders and whatnot, it's quite likely the share price keeps falling. And so you might think, well, "I can't—I'm here now. I could hang on until I get to this level. At that point, I've just got to cut my losses." I might say that if it falls that far, then clearly the 22-year-olds know more than I know.
- 4.39 So you could ask the question: what's the probability of falling to that level before I get up to where I want to be? So what's the probability I'm going to have to close out the position before I get the returns that I would hope for? So that's another question you could ask. It's a little bit harder to do. You can't do that in one formula because you have to—I could ask questions in one formula about the probability of falling to some level or reaching some other level, but it's a question of the probability of this happening before this happens. And that means I have to have—I would have to put in an extra column here, for example, in which I said, I looked at what the minimum is—the running minimum. I'd have to keep track of the running minimum and what the running maximum is to keep track of to see whether or not this happens before, if you see what I mean.
- 4.40 This is also used to do scenario analysis because I've done a simulation here, thinking I know what the growth rate is, what the volatility is. But people use these simulations to ask questions about: what if the volatility changes? What if it's not actually 20%? What if the growth is not 8%? You can change these numbers, and you get different distributions. As you increase the growth rate, what would happen to this probability density function here is it would be shifted to the right if the growth is bigger. If growth is smaller, then it goes to the left. If the volatility was smaller, then this would be more peaked. If the volatility were greater, it would be flatter. So it gives you an idea of the things that could happen in the future. I think that is enough of the simulations. Any questions about simulations? Yes.
- 4.41 If you go back to the Sims page and there's two of them that are really high, the blue one and the black one. If there was 10,000 simulations, I'm not quite clear why there isn't one on the other side going the other way.
- 4.42 The downside?
- 4.43 Yeah.
- 4.44 It could easily happen that there would be some on the downside, but because it's lognormal, things are more spread out on the upside. You can see this, can't you? The spread here, also in this picture here, you can see that the tail is bigger here. I mean, technically, it goes on—this is the lognormal distribution—it goes on forever; it just gets

exponentially small. This, on the other hand, this tail doesn't go on forever. It has to stop at 0. So there's a limit. There's an asymmetry in the distribution. And you're right, that is seen in that particular picture quite nicely, more happening up here. Here, they're more squished. Yeah. I mean, you might get the occasional one, but as you do more and more simulations, once you're up to the 10,000 simulations, you wouldn't see that. I mean, you see more on the upside, but they'd be spread out. Yeah. Good question. Any other questions?

- 4.45 During this example, you use Excel. What other programs are going to be replacing this?
- 4.46 I do Excel simply because it's good for teaching, because you can see the formulae. Nobody does this for real in Excel. Everyone's using Python, et cetera, to do things. I think that they're no good for teaching because usually this thing, you'd have one line of code in Python that did it, and you wouldn't be able to see what's going on. You have no clue. But people use, obviously, more modern. And Python seems to be the programming language of choice at the moment. But then there's been so many other ones as well over the years. Do you know Python? Okay, that's more than I do. I mean, I've tried it, but I hate it because you can't see what's going on.
- 4.47 For something like the S&P 500, where you have 100 years of data or more, it seems like this is easy. But for things that are maybe one-year-old, two-year-old, five-year-old or very low, does that just mean the volatility goes up?
- 4.48 It doesn't mean the volatility goes up. It just means—I was going to say you have less confidence in your data. Statistically, you have less confidence. If you believed that the numbers that we've got here, these growth rates and the volatility that came out of 100 years worth of data, if you believed they were some scientific thing—I would say, rainfall or something, but you've got climate change and all these things—something that never changes with time, then you'd feel confident the more data you've got, the more accurate your statistics are going to be. You still have the problem when you're simulating going forward, each realization might have completely different growth rate.
- 4.49 But in finance, how confident are you that what happened in the late 1920s is similar to what is happening now? You don't have any confidence. So the numbers do change, which makes you think, "Even though I've got statistics for what happened 100 years ago, it doesn't necessarily help me with the next year." So even if the company has only been around for a year, then still you haven't got much confidence in anything, really, which is one of the big things.
- 4.50 One of the things you do is you do experiments where you change the volatility to see: what effect does that have? You change the growth. How does that change these statistics going forward? So you do this scenario analysis or robustness and stress testing of any of your models.

4.51 But no, that's a big question. And that's what makes quantitative finance fun, because although you may have lots of data, loads of data, tick data, coming off every few seconds, you've got data. But it doesn't necessarily help you going forward. I mean, AI bubbles and all that and people are talking about, you're not going to see that in the data because it hasn't happened yet.

Lecture 5: Portfolio Theory & Investment Strategies

At a glance — Portfolio theory and investment strategies.

- 5.1 So, this is all about portfolio stuff now, investing and some mathematical strategies, just an introduction to things. So I will actually kick off with something that gamblers have known about for many decades, but quants in finance have only recently rediscovered. It's interesting to watch. There's a book, I think it's called Fortune's Formula, which talks about the two different strands of gambling versus investing and the mathematics. There's a lot of overlap, but the gamblers seem to know a lot more about investing than bankers do.
- 5.2 Anyway, this idea goes back to the 1950s, this first idea. You've got \$100, and you've got, let's say, a biased coin, or you've got some bet and you feel confident that you've got an edge. All right? How much of that \$100 should you bet? You've got an edge, 10% or something. You expect a 10% profit on average. Well, as I mentioned the other day, we'll take extreme examples. If you, for example, bet a tiny amount of this, tiny amount, then what will happen? Well, you've got an edge. So this is time. This is the number of times you bet, effectively.
- 5.3 You start off with \$100. You bet a tiny amount. You're probably going to make some money. You may lose some the next time. It's going to go like this, right? Gosh, you think you're going to be there. It's going to take you forever to make any serious money. It's maybe going. Is it going up? It should be going up, but it's such a tiny amount of money you're betting, you can hardly see it going up. What's the opposite extreme? You bet the whole, you bet the farm on the \$100. Or what happens? Great, you win. You've doubled your money. You've doubled your money again, and then you've lost everything. That's it. Game over.
- 5.4 Well, first of all, one fairly obvious observation is that, as your money grows, if it does grow, then you should be betting not the same amount, you should be betting the same fraction. It doesn't make any sense. Suppose you actually come up with some best way of investing \$100. It's not one cent, and it's not the full \$100. It's something, it's \$3 each time you toss this biased coin or place your bet. Your money will grow, but eventually, you're at \$1,000. You're not going to be betting \$3. You're going to bet \$30. It seems to make sense that you would scale things with how much money you've got, right?
- 5.5 So, can we do the mathematics of that? So let me introduce some notation. I'm going to have— we're going to bet a fraction f of our money each time. And I'm going to use ϕ again to represent random return on each bet. So if we start off—forget the \$100. Everything's going to be multiplied by 100 or whatever you start with. Let's just start with \$1. You place a bet, and you are going to—this is what you're going to have after

one bet $[1 + f\phi]$. So you started off with \$1, put a 1 in there. After one bet, you're going to have this: $[(1 + f\phi) * 1]$.

- 5.6** Why have you got this? Well, let's say it's a coin toss. So, actually, ϕ can be either ± 1 , if it's a coin toss, ± 1 . And suppose you bet a fraction, 1%. So this would be 1%, this $[f]$ would be 0.01, and this $[\phi]$ becomes ± 1 . So if it's a coin toss and you're betting f , then either you've got $(1 + f)$, if you get heads, or you've got $(1 - f)$, if you have tails. Simple.
- 5.7** What happens next? So that all depends upon what ϕ is, whether it's ± 1 . The $+1$ is going to have a probability p , and the -1 is going to have a probability $(1 - p)$. And this $[p]$ is going to be greater than half because you're betting on—you've got a positive edge.
- 5.8** So what happens to your money? Well, after one bet, you've got $(1 + f\phi)$, let's say ϕ_1 is the outcome of the first coin toss $[1 + f\phi_1]$. Then this is how much money you've got after the first coin toss. So you're going to bet a fraction of this. So after the next coin toss, you're going to have $(1 + f\phi_2)$. The idea being, for example, if you won the first one, you'd now have an amount $(1 + f)$. That's how much money you've got. If f is 1%, you'd have 1.01. Then you bet the same fraction of the 1.01, and you lost. OK, so it's down to that $[(1 + f) * (1 - f)]$. Next time you win $[(1 + f) * (1 - f) * (1 + f)]$; you win again $[(1 + f) * (1 - f) * (1 + f) * (1 + f)]$. Notice I'm multiplying again. I'm doing the multiplying because I'm betting the same fraction. It's exactly like the share price thing, where everything is scaling, multiplicative.
- 5.9** If you're betting the same amount, it would be $(1 + f - f + f + f - f)$ —it would be just a string of numbers added up. But because you're betting the same fraction, that's the sort of thing you get. So, you bet a certain fraction, and it would look something like this: this is how much money you've got. It would be doing the same random path, but hopefully going up.
- 5.10** Now, what mathematics can be applied to this? What's the logical way of choosing f ? Well, one way, and this goes back to a gentleman called John Kelly Jr., he said, bet so as to maximize your long-term growth rate, the growth rate. So if you're betting the same fraction, this is going to be doing the same kind of exponential growth that we had with the share price or interest rates. There's a lot of overlap between the mathematics.
- 5.11** So what have we got? To get from the how much money you've got to turn it into a growth rate, what you do is, you take logarithms in the same way we had e^{rt} . Remember, we had e^{rt} ? To get back to the interest rate, you just take logarithms. So in the corner here, we had e^{rt} , it was money in the bank. To access this, you took logarithms of the value of your bond and all that stuff, and you end up finding what r is.
- 5.12** So, all you have to do is take the logs of this $[(1 + f) * (1 - f) * (1 + f) * (1 + f)]$. So that means you've got to—and that gives you the growth rate. So what is this thing? Let me do it all in maths now. The amount of money you have is going to be—you know the symbol for sum $[\Sigma]$? Well, the symbol for product means multiplying. It's this fellow

here $\prod$. So you're just multiplying all of these $(1 + f\phi_i)$. That's the same as that. But this is from i equals 1 up to however many times you toss the coin.

5.13 All right, so that just means $(1 + f\phi_1) * (1 + f\phi_2) * (1 + f\phi_3)$, all the way up to $(1 + f\phi_n)$. Then we take logarithms of that to get to this return. It's like saying this is growing exponentially. What's the rate of growth? Take logs. So we take logs of that. And because of this thing I mentioned before: $\ln(A * B) = \ln(A) + \ln(B)$. Taking logs of this: $\ln(\prod_{i=1}^N (1 + f\phi_i))$, there's logs again, that turns into the sum: $\ln(\prod_{i=1}^N (1 + f\phi_i)) = \sum \ln(1 + f\phi_i)$ — N of those: $[\sum_{i=1}^N \ln(1 + f\phi_i)]$. If you'd like details, you could say, actually, I really should have $1/N$ in front of this, meaning this is the growth rate per cointoss: $[1/N \ln(\prod_{i=1}^N (1 + f\phi_i)) = 1/N \sum \ln(1 + f\phi_i)]$

5.14 Now, if it's the same coin over and over again, all of these things, the result, the ϕ_i 's, will be different, but the statistical properties of this will be the same every time. It's ± 1 with these probabilities, if it's a coin toss. So we start off with a coin toss, and then we'll advance beyond that.

5.15 So that's your actual growth rate: $[1/N \sum_{i=1}^N \ln(1 + f\phi_i)]$, but it's random. What do you expect the growth rate to be? What's the average growth rate? What you do is, you take expectations of that. You take the average of this, which means, what's the expected growth rate? You take the expected value of all of these sums. Well, since I said that the statistics of ϕ_i is the same every time you toss this coin, we're not changing the coin in this particular example, then that just amounts to taking the expectation of one of these multiplied by N . So what we end up with is the expected growth rate is the expected value of $\log(1 + f\phi)$, where $f\phi$ is just your general coin toss: $E[\ln(1 + f\phi)]$.

5.16 All right, we've got exponentials, we've got logarithms, we've got averages, all the things we did in the first lecture. What's so special about that? Well, the special thing about this is that it behaves rather unexpectedly. Oh, maybe not, given what I said here. So let's actually do it for the coin toss. So ϕ is either $+1$ or -1 with these probabilities. So let's take the probability. So this expectation, you just take the outcomes: winning is that $\phi = 1$: $[p * \ln(1 + f)]$ multiplied by the probability of winning, tossing heads with this biased coin, plus the other outcome, which is losing: $[(1 - p) * \ln(1 - f)]$, multiplied by the probability of losing: $[(1 - p) * \ln(1 - f)]$. There, that: $[p * \ln(1 + f) + (1 - p) * \ln(1 - f)]$

5.17 "So what?" I hear you crying. That's very important and really, really rather simple and special. I'll change color. I can plot that, and I can plot this as a function of f , the fraction. So this is the expected growth rate. Well, when $f = 0$, you've got $\ln(1) = 0$. So if you bet nothing, your expected return is 0. What if you bet everything? $f = 1$, you bet 100%. OK, got $p * \ln(2) + (1 - p) * \ln(0)$. $\ln(0) = -\infty$. So if you're betting, if you're down here, it starts to go negative. If you plot this, what it looks like is this. That's what that function looks like, as a function of how much you bet. Well, we can see it. If you bet not very much, this is what's happening. If you bet too much, if f is too big, that's what

happens. This is what happens if you bet exactly 100%, but if you bet 95%, it's going to go down and down and down like that. It's never going to go back up.

- 5.18** What can we say about this? We can say several things, one of which is that there is a maximum. It's always nice when you have some maximum of something. There. You can find out what the maximum is, differentiate with respect—it's where the slope is equal to 0—so differentiate with respect to f , you get $p / (1 + f) - (1 - p) / (1 - f) = 0$ and so on. What do I get? $p - fp - (1 + f - p - fp) = 0$. That fp and that fp disappear, cancel, and I'm left with this p and this p , double up, so we've got $2p - 1 = f$.
- 5.19** Anyway, the point is, if p is, say, 0.55, then that gives me $1.1 - 1$, it gives me $f = 10\%$. So if $p = 55\%$, then you're betting 10% of your money, and it maximizes your growth rate. And so that will give you a curve that goes— alright? That's known as the Kelly criterion, or Kelly fraction. It optimizes. This becomes important in things like if you're a card counter at blackjack. Now there are strategies. You've got to read the book by Ed Thorp, *Beat the Dealer* and other books on blackjack. But there is an optimal strategy in blackjack if you know how to play blackjack. You've got certain cards, and the dealer's got an up card. Whether you take a card or stand, that's the optimal strategy. That won't win. You won't win the optimal strategy. The bank, the house, still has an edge against you. But the trick comes in card counting, which is keeping track of what cards are left in the deck.
- 5.20** For example, if there are a lot of aces and tens left in the deck, then that's good for you. If all the aces and tens have gone, then the deck's bad for you. So the trick is to know what's left in the deck, roughly speaking. And once the deck moves in your favor, that's when this thing kicks in, and you can adjust your bet accordingly. So you're sitting there playing blackjack, you're betting some small amount until the deck gets favorable. Then you increase your bet along in this fashion, once you've got a positive edge. Of course, you do that these days, they kick you out of the casino because they know you're card counting, right? If you're betting the minimum, the minimum, then suddenly you start betting large, they know what you're up to. So, that's the thing called Kelly fraction, Kelly criterion.
- 5.21** In practice, people don't, unless you really know what the odds are, which you won't, I mean, even if you've got a biased coin, you're going to have to toss the coin many, many thousands of times before you know the statistics. It's like we did with the simulation. You do one simulation, and you toss the coin three times, you're not going to get the odds very accurately. So you need to toss the coin many thousands of times to get any idea of the statistics, knowing the p . But that's tossing a coin, and you can do that, if you've got nothing better to do, tossing the coin.
- 5.22** In practice, if we're talking finance, we have no clue about the statistics. So what people tend to do—it's not just finance, of course, it's horse racing, et cetera, anything where there's betting or investing—what people tend to do is, they tend to do what I think is called half-Kelly. Half-Kelly means that you bet half the amount, and then it

doesn't decrease your expected growth rate that much, maybe down to $\frac{3}{4}$, but it reduces your risk a lot.

- 5.23** So you can see the key things are, anything above here is just suicidal. Anything above here is unnecessarily risky. You might as well be this side, and typically, maybe half-Kelly is best, to allow for the fact that you don't actually know. If you don't know what p is, you don't know what this curve actually looks like. Is it shallower? Is it more focused at this end? That is the coin-tossing version of this.
- 5.24** When you do it for financial applications, then you've got something similar, but not with such a nice formula. Let me get rid of all of this and do the same maths, but in general, more general. And this is where we're going to—it's more logarithms and now Taylor series. This is why we did that tiny bit of mathematics right at the start.
- 5.25** So I don't know how this applies to poker. Obviously, there's an added dimension to the game of poker. But certainly, things like roulette, where you've got a negative expectation, what you should do if you're going to play roulette is, first of all, well, you shouldn't play roulette. That's the first thing you should do. But the second thing, the next best thing to do is, figure out how many times you're going to play roulette in your life, how much money you're going to bet in your life, and then bet it all on red, once, or black. I don't give financial advice. I'm not registered.
- 5.26** When we do it with finance, we have the same idea. We still have the $\ln(1 + f\phi)$. But now this is not ± 1 . ϕ is going to be going back to the distributions. You still want to maximize the expected return. And so this is where— if you're betting a small amount, which typically you would be, then $\ln(1 + f\phi) = f\phi - \frac{1}{2}f^2\phi^2$ plus smaller terms. Take the expected value of that, and what you end up with is— let me get the formula right. It's not complicated, but don't make any mistakes. You get $f * \mu$, the expected return on your investment, and then minus $f^2\phi^2 / 2$ plus smaller terms. So that's the expected growth rate $[f\mu - \frac{1}{2}f^2\phi^2]$, which then you see, well, it's in the same shape. It's a quadratic in f . It does exactly the same thing.
- 5.27** But you can maximize that. Again, you just differentiate with respect to f . So do d / df of that, and you end up with $\mu - f\sigma^2$. So you make that equal to 0, so $f = \mu / \sigma^2$. That being the standard deviation, σ being the standard deviation. So when you take expectations of this thing, you get something like that. So that's your critical investment.
- 5.28** Now, this is gambling, which is different, so I'm told, from investing. Although a little plug for the guy who wrote that famous book, Beat the Dealer, Ed Thorp. We're going to be talking about options next. He also did the theory behind options, years before the Nobel Prize winners did it, but he didn't publish it like the Nobel Prize winners did, so he didn't get a Nobel Prize. He used it instead for his hedge fund. So Ed Thorp was the guy who wrote the book. He didn't discover this, but he did discover card counting, etc., and the option pricing, formally. So he has experience of hedge funds and blackjack and all sorts of things.

- 5.29 We're now going to do something similar, which throws up similar kind of μ 's and σ 's, very similar ideas, but different, and now applied to cash in the bank and stocks and stuff like that. And it goes like this. I'm going to start plotting volatility or risk. The amount that the stock bounces up and down is a measure of risk, volatility standard deviation, all similar. And here we're going to have expected return, which, these are your μ 's.
- 5.30 So every stock has a—you can put indices or anything. You can put it as a dot in this picture. So what did we have for the S&P 500? It was about 8% expected return and risk of 20%. So there, that's the S&P 500. Other things. Every dot represents a different investment. So, what you would do with the spreadsheet that we developed, the first spreadsheet, is you'd download your data, drop it into those columns, and out would magically come the μ and the σ , and then you could plot it on this picture.
- 5.31 And we can say some obvious things from this. First of all, there are a lot of these things you wouldn't bother investing in, because why on earth would you invest in, say, this fellow here? Why would you put your money in that fellow when, for the same level of risk, you could put your money in whatever that stock is? It's directly above it. It's got exactly the same risk. It's got less growth. Forget it. So all those down here are ones you wouldn't bother with. You could also put into the picture a risk-free interest. This is your money in the bank, risk-free interest rate.
- 5.32 So that's the kind of pictures we're going to be dealing with now. I'm going to simplify that for a second. You already see the idea that things up there is good, down there is bad. That way is bad because of high volatility; that way is good. The best is if you go that direction, because you've got a higher expected return and lower risk if you head that direction. Heading that way is the worst. That's the thing we're going to be dealing with in a second.
- 5.33 But we're going to start off very simply. And that is just to say, suppose we've got this risk-free interest rate, money in the bank, and we've got one stock. So the goal here is to build up bit by bit, to talk about individual stocks, which is the dots. Stock plus bond. What happens if you have two stocks? And then what if you have multiple stocks? What if you have multiple stocks and the risk-free interest rate? So you've got a whole portfolio of things you could invest in. What's the mathematics behind that?
- 5.34 It's all about expectations, means, variances, standard deviations, correlation, we're going to see for the first time, and optimization. It's all very simple, and it all leads to a Nobel Prize for Harry Markowitz, who did this as a student in the 1950s, I think it was. All right, let's suppose we've got \$1 to invest. Now, because I'm a mathematician, I always make things—these problems are all linear. It doesn't matter where you start, \$1, a million dollars. You just multiply by whatever your starting number is, so I'll just say 1.

- 5.35 We're going to invest. Out of this, we're going to invest b , an amount b , which is somewhere between 0 and 1 in the stock, an amount b in the stock, b dollars, b is a dollar amount, which means we're going to put $1 - b$ in the risk-free. So we've got $1 - b$, there's dollars around all of this, in the bank, had some risk-free interest rate. We're going to keep that fixed, and b in the stock, the share.
- 5.36 Let's do the mathematics of that. Let's suppose the share price is currently S_0 . So, the number of the stock we're going to buy is going to be, or the amount that we have in the stock is going to be—the quantity of stock we're going to buy is going to be b / S_0 . And so, the value in the stock is going to be that. That's how much we've got in the stock. It's just scaled with S_0 . So that's a dollar amount. Those S and S_0 are both dollars, so those dollars cancel, but the b is also dollars. So that's in the stock. And we also have $1 - b$ in the bank.
- 5.37 So, looking at things like the expected growth in the share, I think I'm going to do everything over one year. Let's do everything over one year. So I'm not going to worry about δt 's and things like that, just to keep the number of symbols, not have too many floating around. So what we're going to have is, what happens to the $1 - b$ that's in the bank after one year? That is going to turn into $(1 - b) * (1 + r)$. If it's a one-year interest rate, that's simple.
- 5.38 What happens to this fellow here? Well, he's going to be b / S_0 , whatever this S becomes. And what is S going to become? It's going to be a geometric thing. Again, it's going to be $(1 + R)$ —I'm using R to mean the return over one year, the thing with the normal distributions, all that kind of thing, multiplied by the starting value: $[b / S_0 * (1 + R) * S_0]$. So let's cancel S_0 . So what we have is the amount of money we've got after one year in the bank: $(1 - b) * (1 + r) + b * (1 + R)$. So that is how much money we've got at the end of one year.
- 5.39 This much we know because we know what this r is, but we don't know what that R is. But we do have some statistics for R , and the statistics for R , we're going to use the same notation as before. I'm going to use μ , so $E[R] = \mu$, and we're going to have a standard deviation S.D. of $R = \sigma$, the same notation as before. There are no square roots, t and δt and all that, because we're talking about one year.
- 5.40 The change in the value of our portfolio has got a deterministic bit $[(1 - b) * (1 + r) + b]$ —you know exactly what it's going to be—and a random bit $[(1 + R)]$. What you find is that the expected return, the expected return is obvious from this, over one year, is just the expectation of this thing, which is—sorry, the expected change in the price is just $(1 - b) * r + (b * \mu)$. That's taking expectations of this, but subtracting off the \$1 we started with. So that's the change in things, that—a bit due to the money in the bank and a bit due to the random component in the stock, but that's the expected value.
- 5.41 And the standard deviation, what you have to do is, you take this, you square it, subtract off—sorry. This, subtract off this, square it, and take expectations. I'll cut to

the chase because it's just a little bit of—it's quite straightforward maths, the end result is obvious. It's just that: $[b\sigma]$. The more money you have in the stock, the more risk you've got.

5.42 So now I can go to this picture and say, if I were to put 100 % in the bank, that would be $b = 0$. That'd be 0 stock. So, putting $b = 0$, I end up with r and 0, which is that dot. If I make $b = 1$, that is, I put everything in the stock, that's 0. I have μ , $b = 1$, and σ , which is that. So that's obvious, right? Put everything in the bank up at that point, everything in the stock at that point. What about halfway between? If I do go halfway between, then this is $\sigma/2$, so we're here. Actually, this is linear. This is all you end up with this, that. Very simple. That's a straight line.

5.43 So the more you put in the stock, the further along you are here, the higher your return, and the higher the risk. And of course, this is assuming that $\mu > r$. It actually doesn't make any sense. You hope that the stocks have higher expectations than putting money in the bank, because why would you buy a stock? If a stock is risky, it's risky and what? It's got a negative growth or growth less than risk-free? Forget about it.

5.44 You can also, if you like, you could say, why does b have to be less than 1? Why can't I borrow some money and put it in the stock? You can do that. You can have $b > 1$. What does that mean? It just means that this continues like this, if you want. All right, so you can get as much risk as you want. Lots more return when you have that.

5.45 Now, this is interesting. There's another Nobel Prize winner, William Sharpe, and he came up with the idea of the Sharpe ratio. The Sharpe ratio means: what is the slope of this curve? This distance here is—well, this is μ . This is $\mu - r$, a distance. That is just σ . So the Sharpe ratio is $(\mu - r) / \sigma$. And this starts to address the question of, out of all the red dots I could put on there, which is the best red dot? If I've got other stocks around here, which is the best? The best one is that one there, because if you draw the line between it and the r , it has the highest Sharpe ratio. Think of the Sharpe ratio as being your expected return in excess of the risk-free rate per unit of risk. That's what that means. So that's the measure of how good a particular stock is. So you'd buy that one. That's the one you'd go for.

5.46 This doesn't answer all our questions, though. You'd buy that one. But where you are on that line, how much money you invest, whether you're down here because you don't like to take too much risk, or up here because you want lots of risk, you get compensated, you expect to get compensated, being up here. That's not guaranteed. Or maybe really risk-seeking, almost, you borrow money to invest in here. But whereabouts on that line becomes a personal choice.

5.47 The idea of a personal choice is going to become important in the next lecture on options, because there's only so far you can go with this mathematics. Some economists talk about utility functions. I will be mentioning them later on, utility functions being something here. It's a function which tells you which investments are

the same for you, for an individual. You might think, if I'm going to have a certain level of risk and a certain growth, then that's the same to me as a different risk and a different growth. And you might have curves which represent equal utility, and then you can talk about maximizing your own utility and stuff. But I mean, who knows the utility function? No one. So that's a bit academic.

- 5.48** So what have we got? We've got the idea that investing in risk-free and stock gives you this straight line in this picture. You get the idea that the Sharpe ratio is somehow important and that you like stocks with a high Sharpe ratio. Well, this is only one stock. This fellow down here, we've dissed him. We don't like him. He's got this great risk and this terrible return. But there's something else we can do. We can get a bit cleverer, and the way we get cleverer is to talk about portfolios.
- 5.49** It's going to be the same picture, but now I'm going to do it with two stocks. And then, I'm going to do lots of stocks and risk-free, as well. So let's do it with two stocks, and you'll see what I mean. So we get the risk-free for the moment. Here's the risk, the volatility. Here's the expected return. And we're going to have two stocks. We have this fellow, and we'll have—
- 5.50** Now, let's suppose there's no risk-free instruments. You still have the same idea about Sharpe ratios. But if $R = 0$, then you're just going through the origin here. Clearly, if there's no risk-free thing, then clearly the slope of that is less than the slope of that, so that's the guy to go for. But what if you're trying to build a portfolio, meaning you put some in this and some in that? You diversify, right? Diversifying—don't put all your eggs in one basket.
- 5.51** Is there any advantage to splitting your money between these two, rather than putting everything in the red one? Well, this is the Nobel Prize-winning—I think we're going to have three Nobel Prize winners, or four Nobel Prize winners, in this course. Not me, but we've got Markowitz. Who did I just mention? Sharpe, Myron Scholes, and Robert Merton are going to be talked about.
- 5.52** So what we need is, we need to have some way of representing the relationship between these two, the random behavior. That's measured by a thing called correlation. Now, I don't want to get into the business of correlation—well, I will get to the business of correlation, but I want to give you some numbers, an example.
- 5.53** These are made-up numbers. These are some test scores, children's test scores, child x and child y. Child x gets 1. That is a very hard, tough test out of 10 or something. Child y gets 4. Next test: 4 and 5, 9 and 9—oh, that must've been really easy with just one trick question right at the very end. 2, 1. None of this is significant, except I've done the maths of it [3, 4]. I think that's to do with four o'clock in the morning. [2], 5 and 5, 3, those are the numbers I've got. Child x has an average of 3.714. Child y has an average of 4.429. That's the mean. It looks like child y is cleverer than—

- 5.54** There's a greater standard deviation. Standard deviation means you take—this is your mean. Take each of these numbers, subtract off the mean, square it, and then average those, what you've got there. And then square root it, 2.49, which is more than—child y is a little bit more consistent. Slightly more consistent.
- 5.55** But the question we ask ourselves is: Is there any relationship between the scores of these two children? What we can do is, we can measure a thing, first of all, a thing called the covariance, which is the expected value of each of the x values minus the mean. So that's x bar times y minus y bar: $E[(x - \bar{x}) * (y - \bar{y})]$. So what you do is, you take 1 minus that: $[1 - 3.714]$, multiply by 4 minus that: $[4 - 4.429]$. That gives you one number: $[(1 - 3.714) * (4 - 4.429)]$. 4 minus that: $[4 - 3.714]$, 5 minus that: $[5 - 4.429]$. It's another number: $[(4 - 3.714) * (5 - 4.429)]$. 9 minus that: $[9 - 3.714]$, times 9 minus that: $[9 - 4.429]$, and so on. And then do the average. Divide it by however many, 1, 2, 3, 4, 5, 6, 7 numbers. And that gives you this number.
- 5.56** That's the thing called the covariance. It's like a variance in that it's something multiplied by something, but instead of $(x - \bar{x})^2$, it's taking the count of two things. Then you take that, and you divide it by the standard deviation of the x 's and the standard deviation of the y 's. That and that. And this is now, this thing is called a correlation.
- 5.57** Imagine that these numbers are, I don't know, centimeters instead of scores. Then this would be a centimeter, this would be a centimeter, this would be centimeters, this would be centimeters, that is dimensionless. Whatever units these have, this becomes dimensionless. That is a number. I'm going to call it ρ , ρ between -1 and +1. This is a thing called the—it's one example of a correlation. I forget what the correlation, but I haven't got the number down here, but for these numbers, there's some correlation, and it shows there's some relationship between the two.
- 5.58** The thing about the correlation is, it doesn't prove—you've got that thing, correlation is not causation. You don't know whether they've got similar scores, while the correlation is, I think it's something like 0.6 or 0.7. You don't know whether the correlation is quite high because one is copying off the other, but not very accurately, or whether they're just—some exams are harder, others are easier than others. So it might be a cause for both of them, or they might be related. So this number is between ± 1 . The closer to 1 it is, the higher the correlation. Negative means it's a negative correlation. -1 means they go in opposite directions effectively. That's a simplistic version of that.
- 5.59** Anyway, we need that. We need that for what comes next. And that is, if I were now to invest—we're just going to do these two things. We got stock A and stock B, and I'm investing b in stock A. Why did I do that? Why didn't I invest b in stock B? Anyway, if I do the same thing, b goes into stock A, and the remaining $(1 - b)$ goes into B, just like I did before when I divided amongst the stock and the risk-free rate. What I end up with is a growth rate, an expected growth rate, which is kind of obvious again. It's $b * \mu_A + (1 - b) * \mu_B$. That's the—let's call that p for portfolio, the growth of the portfolio.

- 5.60 And the standard deviation of the portfolio of the two stocks $[\sigma_p]$ is the square root of the variance of the portfolio, which, it looks very— it has obvious things in it. It's got $b^2\sigma_A^2$, which, if $b = 1$, there would be nothing else there, and that would just be σ_A . At the end, you've got $(1 - b)^2\sigma_B^2$. And then the middle is where you get your correlation, $2b(1-b)\rho\sigma_A\sigma_B$. The correlation.
- 5.61 Now, the way you'd calculate the correlation, we did in the spreadsheets, I had the column of returns for the S&P 500 index, and you'd put next to it a column of whatever you're interested in, whatever stock or other index, and you'd use the CORREL Excel function, which would calculate the ρ for you. Anyway, what does this mean, pictorially? Well, if $b = 1$, then we're here. If $b = 0$, then we're here.
- 5.62 What about b being between 0 and 1? What you end up with is the following. You end up with what's called the hyperbola. I think it goes like this. You get that. Yeah, you get that. That's the places you can reach in this expected return against risk plot, by giving yourself the freedom to invest partly in A, partly in B, without any risk-free thing. So this is interesting because several things. One, first of all, you would never, ever go down. You'd never—there's some turning point here at that point there. You'd never invest. You could do a little bit of differentiation. You could figure out what the value is, what the risk and return for that point is. But you'd never go anywhere near these.
- 5.63 Because you could get, using the same—sorry, A is still there. A is still there. The green dot should still be there. The green dot is still there. By investing in—you can get anywhere along that upper black line using A and B. So you would never invest anything that got you down there, again for the reason, if you fixed the risk, you'd be better off there than there.
- 5.64 Another thing is, if we put in the risk-free rate, you can see—I need a ruler for this, but if you put in the risk-free rate, which I'm going to do in a second, you can see that, actually, if you join this red dot to here, you can see some of this black line might be above. You might have a higher Sharpe ratio for something on this black line than you would for the red dots. So this is very simple mathematics.
- 5.65 Now, if I do this for any number of stocks, and then throw in the risk-free rate— let me clean some a bit. Let me start off by forgetting the risk-free rate again, and then I'll throw it in at the end. Then make a few comments about this. So if I have many stocks, then if I have stock_i, stock_i just means a nice stock, stock₁, stock₂, stock₃, then my expected return is going to be, very obviously again, it's going to be b_i , however much we put into stock_i, multiplied by μ_i , so i equals 1 up to N : $[\sum_{i=1}^N b_i \mu_i]$.
- 5.66 And the standard deviation is you take the square root of the same kind of formula, but now a double summation: $[\sqrt{(\sum_{i=1}^N \sum_{j=1}^N b_i b_j \sigma_i \sigma_j \rho_{ij})}]$ and it looks exactly the same as before, where ρ_{ij} is the correlation between stock_i and stock_j. So that's the expected return: $[\sum_{i=1}^N b_i \mu_i]$, and this is the standard deviation, the risk: $[\sqrt{(\sum_{i=1}^N \sum_{j=1}^N b_i b_j \sigma_i \sigma_j \rho_{ij})}]$. So what happens now is, you do the same picture. You're going to have the expected

return here. You have the risk here, and now all of your dots, all over the place, like this, some of the negative expected return all over the place, and you end up getting a curve that goes something like that.

- 5.67** And so what you have to do, and you can do this in a spreadsheet, beautiful and visual, is, you just say, I take this, or I've got to have the constraint. I'm going to have the sum of the b's. It's going to add up to 1: $[\sum_{i=1}^N b_i = 1]$. Now I just have to—in Excel, it has a nice optimization thing called Solver. I know what the μ 's are for my data. I know what the σ 's are. I know my spreadsheets. I've got my correlation, so I know everything. I just have to choose the b's. The b's are up to me.
- 5.68** So I just say something to Excel Solver, I say, if I want a risk of 15%, just go away, do your magic, and find the values of the b's, such that this constraint $[\sum_{i=1}^N b_i = 1]$ is satisfied at the same time we maximize the expected return. And so it will get, fix this point. And then we say we'll do the same for 10% and do it for every possible value of the risk that we want. So fix this. I've done that. So fix this, this expression here, and vary the b's to maximize this [the expected return]. So this is the thing you maximize [the expected return]. This you hold fixed, then, very later on, and it gives you this whole curve. That has a name. The name of this is the efficient frontier. It's all Harry Markowitz. The efficient frontier. So you would never choose anything down here. You wouldn't choose anything in the middle. You'd choose portfolios that maximize your return for a given level of risk.
- 5.69** And then you say, I'm going to include the risk-free rate. Let me do all that in green or anything to do with the risk-free rate. I'm going to throw in— forget that now, because I'm going to throw in the— I think it's something like that: $[(1 - \sum_{i=1}^N b_i) * r]$. Yes, it is. b's can be anything. Whatever's left over from your \$1 goes into r. In fact, this can be negative because you're borrowing money. So that's what happens when you put the risk-free rate in. And what happens to the standard deviation? Nothing. It doesn't change at all. It's exactly the same expression. So here's your risk-free rate, and what you find happening now when you do this optimization, you can get anywhere along this line, anywhere you want along that line.
- 5.70** And it's a combination of two things. It's a combination of the risk-free money in the bank and this tangency, whatever that portfolio is there. Every one of these dots is a stock. This line here is a portfolio. This is an efficient frontier. This is an optimal portfolio. But there's one portfolio that's better than the other ones, and that's the one that maximizes your Sharpe ratio, because if you invested in this portfolio, you think, "Oh, great, this portfolio is fantastic." It's all right. It's on the efficient frontier, but it hasn't got the best Sharpe ratio. You can do better by finding the point that is tangential, going through this point, a risk-free investment, and this point here.
- 5.71** Now, that's fantastic. That gets you a Nobel Prize, and it's just hyperbola. It's very simple functions, very clever, though. But then the economists get carried away, and they say, "Whoa, this is so exciting. Let's see how far we can push this." And they say

things like, "Clearly, everybody would be investing in this particular portfolio. So all these other things, everything must—" They say things, I don't know the details because it seems a bit nonsensical, that somehow everyone would be investing in this very special portfolio. Why would anyone invest in any other portfolio? And so, somehow this dominates, and everybody invests in it, and share prices move so that everything is—they get carried away, economists.

- 5.72** Of course, different people might have different estimates for the volatilities and the growth rate. People may even have different access to different interest rates. Different people might have different pictures, or people might have other goals besides maximizing the expected return. But then again, wherever you want to be on this green line, that becomes a personal choice.
- 5.73** This is great. Fantastic. Looks good on paper, but in practice, what parameters have you got? You've got the σ_i 's, the volatilities of all these stocks, which, as I showed you, I demonstrated, it's not exactly stable, but it's the least unstable of all these parameters, I'd say. Then you've got the μ 's, the growth rates, for each of the individual stocks, and I demonstrated that that's very difficult to measure and even harder to predict going forward. And then you've got these, these correlations, correlations between stock_i, or investment_i, and investment_j, which is all over the place.
- 5.74** Correlations are very, very unstable. You can imagine a situation where you've got, let's say, two companies selling running shoes, and you might find that they're in the same business, the same sector, the running shoes sector. You can imagine that both share prices are rising as more and more people want running shoes. But then one of the sports stars promoting one of the running shoes gets involved in some scandal, and the share price, people stop buying that particular running shoe, and the share prices fall.
- 5.75** You can imagine all sorts of business reasons why there might be correlations and why the correlations might change. It would be crazy to go around saying, "Oh, the correlation is 0.6 because they're in the same sector," when all sorts of things can happen. And also, when there are market crashes, if you look at the statistics around the '87 crash, for example, what happens to correlations during crashes? It all goes to +1. They all, everything moves together. So all the numbers you've calculated are all out of the window. Crashes.

Lecture 6: Options Trading

At a glance — Options trading, where the mathematics gets serious.

- 6.1** All right, we're now going to do options. Now, options is where the maths can get pretty heavy, but not with me, it won't. Options. Do we know what options are? Options are a special type of contract. Imagine a piece of paper. It says on it, "This piece of paper entitles the bearer to buy stock X in one year's time for \$120." Wow. That can be valuable. If the stock is more than \$120, then you've got a piece of paper saying, "I can buy it for \$120. I know it's \$300, but I can buy it for \$120." It's some sort of coupon or a voucher, like in the back of a newspaper. Of course, you can buy it for \$120 in a year's time. When you get to the year's time, the stock might be worth \$80, in which case, "I've got to pay \$120 for this? Oh, it's only worth \$80. In that case, forget it." If I want the stock, I'll just go and pay \$80. All right.
- 6.2** So this piece of paper may be valuable, or when you get to one year, and you can decide whether to, what's called "exercise" the option. It may have some value, or it may have no value. It depends on two things. It depends on what the stock price is when you get to, it's either called expiry or expiration, and it depends on the amount you can pay, you're allowed to pay, which is called the strike, or strike price or exercise price. All right, so there's a couple of—I'm going to use—this is going to be K , and I'm going to call the expiry T . All right, the expiry date, expiry, expiration.
- 6.3** So this is an option. You have the option to buy the stock in the future. There are other options, things called puts I'll mention right at the end, which is the option to sell something. So, what you will see, you'll often see in books on options, if you are brave enough to buy such a thing, you would have the value of the option $[V]$. There's the share price $[S]$. And the share price at expiration would look something like, the option would look something like this. All right, there's the strike $[K]$.
- 6.4** Why would it look something like that? That's the relationship between the value of the option when you get to expiration, so the day on which you're going to exercise this option—oh, yeah, exercise, there's exercise, another word. That is the exercise, the act of taking advantage of the piece of paper, the contract, buying or selling the share.
- 6.5** So first of all, if the stock is less than the strike, if the stock price is \$80 and the strike was \$120, why would you pay \$120 for something that is actually only worth \$80? You wouldn't. So you'd just throw the piece of paper away, which means the piece of paper is worthless. On the other hand, if the stock price is higher than the strike, then you can go through the following process. You can say, I can buy it for \$120. It's worth \$300. In the market, it's \$300. I can buy it for \$120. So I hand over the \$120 and get the stock worth \$300, and I could immediately sell it for \$300. I don't have to sell the stock, but if I did sell it for \$300, that would give me a profit of \$180. It would actually be the profit,

the difference between whatever the stock price is now, \$300, and whatever the strike is, \$120, gives me \$180.

- 6.6 So that's this linear relationship: stock price $[S] - K$. Sometimes we write that as $\max(S - K, 0)$, whichever is bigger. If $S > K$, that's positive: $[S - K > 0]$. If $S < K$, that's negative: $[S - K < 0]$, so that kicks in. That's called the payoff. This is the payoff. The payoff is something you get in the future, but this piece of paper has value now. And that is the goal, to figure out how much should this piece of paper, this contract, this option, be worth now? How much should you pay for it?
- 6.7 The sort of pictures that come out—I'm jumping ahead to give you an idea of what happens— is that you get pictures looking like this. That is, say, a year before expiration. The value of the option—value of this piece of paper, if you wanted to sell it to somebody else, if somebody else wanted this contract, they would pay something for it. The bigger the share price is now, then obviously, the more this piece of paper is worth. It still allows you to pay \$120, the strike. But if the stock price is already \$300, this is a valuable piece of paper, because the higher the stock price is now, surely the higher it's going to be in a year's time. So the higher it is now, the more this piece of paper is worth.
- 6.8 On the other hand, if the stock price is really low, it's only \$20, there's a chance the stock is going to rise above \$120 by expiration, but not much chance. So this piece of paper is probably not worth much. It's still got some value because there is a chance that the stock will rocket. So somebody will pay something, but it might only be pennies. So that's the picture we're trying to get to. But how do we get to that picture?
- 6.9 This stumped people for a long time. The problem was, people like to take expectations. We've done a lot of stuff with expectations, but we also know from the Markowitz thing, the portfolio stuff, the modern portfolio theory, where we looked at diversification, et cetera, we know that the higher the risk, the more people want to be compensated. Well, we know exactly what the stock price is because we just open up The Financial Times or The Wall Street Journal. We can see this is the stock price, or we look at a Bloomberg screen, that's the stock price. We know what the stock price is.
- 6.10 But how do we figure out what the option value is? It must somehow be linked to the stock price—as I say, the higher the stock price, the higher the value of this option. But it's got risk. The stock is risky, bounces up and down. So as the stock is bouncing up and down, you can imagine the option price becomes worth more, worth less, et cetera. How do you figure it out?
- 6.11 Surely, I could do the spreadsheet where I did the simulations. I could have had a row at the bottom, which looked at what would happen. For example, I could have done this kind of calculation. Notice the axes. I'm labeling my axes, which is a first for me. This is expiration. That's when you can exercise the option. The stock is currently here. Here's the strike price, say. That's K , that's \$120. We're currently at \$100. We can do

simulations where we say, like I just did, and I could say, if I end up here, I get paid that much. I get that much. That's how much this option is worth. If the stock does this, it's worth zero.

- 6.12** In the same way, I looked at the probability of the stock being above, say, \$120. We had that row. I could ask the question, "How much do I expect to get in the future?" So the formula in that bottom row would be, instead of having— before I had things like, if this stock is greater than \$120, then 1 or 0, that was to calculate the probability being above \$120. But I could say, well, if stock is bigger than K, I'm going to get a payoff of $S - K$, that; otherwise, 0: $[\text{if}(S > K, S - K, 0)]$.
- 6.13** If it ends up here, that's called "in the money," expires in the money. If it ends up down here, it's called "out of the money." If the stock price is currently the same as the strike when you first entered into this contract, that's called "at the money." Anyway, you can imagine putting that formula in that bottom row, that spreadsheet, looking at the average, and that would tell you what your average payoff is going to be, which might be useful for calculating the price of the option, maybe. Who knows?
- 6.14** This is the things people were talking about in the 1950s and '60s. Two things about that. One thing is, remember—I'm not going to go into this detail at all, but I'll mention it. You don't get the payoff until expiration. When you do this properly in all the gory details, you have to allow for the fact that you get it here. So it's going to be a discount factor because you're valuing it now, something you're getting in the future. So there's going to be some e^{-rt} floating around, discount factors. Anyway, that's what people used to do. They used to do expectations.
- 6.15** But you got an expected payoff, that's fine. But we also know that you need to calculate variances and things, risks, because there's this relationship between expectations and between value, and the higher the risk, the more you expect to get. Anyway, for years, people were stumped until along came Black, Scholes, and Merton, who applied a subject called stochastic calculus, which is another thing we're not going to do. Stochastic calculus.
- 6.16** They invented a very clever concept, which we are going to do, but we're going to do it without the stochastic—it's a calculus with random stuff in it. We're not going to do that. We're going to do something simpler. We're going to do something called the binomial model, invented by gentlemen Cox, Ross, and Rubinstein. I'm going to ask you for some audience participation. Not much. I'll give you an example of how to approach this.
- 6.17** OK, so the stock price is currently \$100. That's the stock price today. We're going to use that simple idea I mentioned earlier, that we didn't do the details of, but a simple version of the random walk that we had, essentially a coin toss. I'm going to say that it can go up to \$101 or down to \$99. Everything is going to be done in symbols—sorry, in numbers to start with, then it will go to symbols. That's the model. But we have a

probability p of going up and a probability of $(1 - p)$ of going down. That is over a period of one day.

- 6.18** To keep things simple, even though that's in the future, I'm going to assume interest rates are 0, so I don't have to do any discounting and present value and all that stuff. All right. So that's the stock. That's what the stock does. It's a coin toss, but a biased coin. What am I doing? I've introduced a symbol. It's far too early in the day for symbols. I want to have a probability of 0.6, $p = 0.6$, going up, and so a probability of 0.4 of falling. So it's a biased coin, if you like.
- 6.19** Now, introduce an option. I'm going to introduce an option. And this is going to be this thing. This contract I mentioned, an option to buy something, I don't think I said it. It's called a call option. Call option. There's a thing called the put option, which I did mention, which is the option to sell something—forget that for now. Call option, option to buy. It's going to have a strike of \$100 [$K = 100$], and the expiration, expiry or expiration, is tomorrow. In other words, here, this day here. This is today. Tomorrow, the stock is either \$101 or falls to \$99. This is when we, "Please, sir, I have a piece of paper which allows me to pay \$100 for the stock."
- 6.20** Now, if the stock rises to \$101, then this piece of paper allows me to buy it for \$100, so that's worth something. If the stock falls to \$99, then being able to buy it for \$100 is worth nothing. So I can tell you right now that the option value, which I will do in red, the option value here is \$0. It ends up "out of the money." This piece of paper is worthless.
- 6.21** If the option ends up "in the money," because the stock has risen above the strike at expiration, then the payoff from that picture, that hockey stick shape, the stock is worth \$101; my piece of paper says I can buy it for—so the stock is \$101. This piece of paper says I can buy it for \$100. So I hand over \$100. I get something worth \$101. I sell it immediately, and then the amount of money I've just made is \$101 that I get for selling the stock minus the \$100 that I paid for the stock.
- 6.22** So the payoff, how much this option is worth at the moment of expiration tomorrow, is, altogether, \$1. These are all dollars. It's \$1. Okay? It's \$1. That's how much the option is worth tomorrow. So, we know in this simple model that the option tomorrow is either worth \$1 or \$0, and it's got a 60% chance of being worth \$1, and a 40% chance of being worth \$0. So 60% of the time it's worth \$1, 40% of the time it's worth \$0.
- 6.23** Sitting here [at \$100], how much do you think the option might be worth? I'll give you a clue. I'm looking for a wrong answer, but I'm looking for the right wrong answer, and you're all giving me the wrong, wrong answer. 60 % of the time, it's worth \$1. You get 0.6. And 40%, 0.4 of the time, you get \$0. So what might be the option's value here [at \$100]? This is the thing that we like to use in finance: our expectations, averages. You take a probability multiplied by an outcome and so on, like we did with rolling a dice, $\frac{1}{6} * 1 + \frac{1}{6} * 2$, et cetera, gives us an expectation. Here we've got something similar. We

got probability of an outcome. That's like rolling a dice or a biased coin, multiplied by an outcome. And then we've got another to do.

- 6.24** So what might be the value of this option? Currently, what might be the value of this option? First of all, it can't be— it's worth something because some of the time you do get \$1, so it's worth something. It can't be worth—so it's not negative. It can't be worth more than \$1, because the most you're ever going to get is \$1. It can't even be worth \$1 because all the times you might get zip. So it's somewhere between \$0 and \$1.
- 6.25** Now, there's an obvious calculation to do here. You've got a probability, 0.6, and outcome, \$1; and probability, 0.4, and the outcome, \$0. So you could say, what is the expected value for this option? You take the payoff weighted by the 0.6, that's $1 * 0.6$ plus the probability of $0.4 * 0$. So $0.6 * 1 + 0.4 * 0$ gives us the expected payoff of 0.6. Very good. 0.6, that's a very good answer. It's a very sensible answer. Lots of very clever people before you came up with exactly the same answer. But it's wrong, because this was a trick question. And this only works because you're so good at giving me smart answers, but wrong. It's wrong.
- 6.26** The reason it's wrong is because you don't take expectations. We talked about that picture over here with the growth rate and the volatility, and it only shows, et cetera. Generally speaking, the more risk you take, the more return you should get. You never just take expectations and say that's the end of it, because you might have two stocks with exactly the same expectation, but one is much riskier than the other. So you'd obviously go for the one with less risk. So that's wrong. But as I say, lots of clever people did exactly that.
- 6.27** Now, this is what you do, and this is a simple version of what Black, Scholes, and Merton did with stochastic calculus. This is what's called the binomial model by Cox, Ross, and Rubinstein, and I'm going to show you how this works. It's really sneaky. Let me start again. Exactly the same. Same picture. Go up to \$101, down to \$99, and the probability is 0.6, and the probability of falling is 0.4. And in red, I've got the option. The option payoff is \$1; the option payoff is \$0. And this is the thing, I'll call it V , that we're trying to find.
- 6.28** Now, the trick that Black, Scholes, and Merton came up with was to construct what's called a hedged portfolio, meaning that they, instead of just looking at the option on its own, they said, what happens if we construct a portfolio that is part option and part stock? So what I'm going to do is, I'm going to imagine I own the option, but I'm selling short a position half of the underlying stock.
- 6.29** Now, what do I mean by selling short? I'll tell you what that means. You know about buying a stock. You go out, you buy a stock, you've got the stock, a piece of paper stock. What about doing the opposite, selling the stock? Okay, if I've got the piece of paper, I can sell the stock. What about selling the stock when you don't actually own it

to sell it in the first place? Can you do that? That's called selling short. You can do that, and I'll give you an example using a lawnmower.

- 6.30** You think that lawnmowers are going to go down in value, and you don't have a lawnmower. If you could somehow get a negative position in lawnmowers, you could make your fortune. You haven't got a lawnmower to sell. So you go to your neighbor and you say, "Can I borrow your lawnmower?" So you borrow his lawnmower. You borrowed it. You now sell the lawnmower to somebody, without telling your neighbor, obviously. So you've sold a lawnmower that you didn't have. You borrowed it, and then you sold it. So you've now got some money from selling a lawnmower.
- 6.31** It turns out you're right. There's a sale on lawnmowers the next day. So you go out, and you buy the same model as the lawnmower for much cheaper than what you got from selling the first one, your neighbor's lawnmower. And so you bought it. You've got the money from selling the lawnmower. You've now bought exactly the same lawnmower, the make and the model of the lawnmower, back for less than you sold the other one for. You made a profit of the difference between what you sold the lawnmower for and what you paid for the new lawnmower. That's your profit. And then you can just say, "Thank you, neighbor," and give him his lawnmower back.
- 6.32** It's exactly the same with shares. You can borrow shares temporarily to sell them, selling short. It's called selling it short. But you have to buy the share back. So that's what we're going to do here. We are going to sell a portfolio which consists of the option and a position in minus half of the underlying stock: $[V - \frac{1}{2} * 100]$. So, what happens over here? If the stock goes up, then the option is worth \$1. We have a short position in half shares. In practice, it's not half the share. All these numbers are multiplied by 1,000. It would be 500, but mathematically, we say minus a half, times whatever the share is worth at that point: $[1 - \frac{1}{2} * 101]$.
- 6.33** The share has gone up, and we were short. We've just lost money. We made some money here because the option pays off, but because the stock has fallen, we've lost money on that. What happens down here? Well, the options end up out of the money. It's worth \$0, but our short stock position has actually made us a bit of money because it's gone from $(V - \frac{1}{2} * 100)$ to only $(V - \frac{1}{2} * 99)$. So what are these? $[0 - \frac{1}{2} * 99] = -99/2$. This $[1 - \frac{1}{2} * 101] = 1 - 101/2 = -99/2$. What do you notice about those two numbers? They are the same. Hence, my use of the word "hedged". This position, long the option, short the stock, one-half the stock, gives me the same amount of money regardless of what happens in the future.
- 6.34** This is brilliant because I know exactly how much I've got in the future. It's not that I'm holding just the option I'm nervous about. "Am I going to make money or lose money? Make money, lose money." No, I've got this position where, whatever happens, I sit back, I know I'm going to have $-99/2$, which means the portfolio, in the absence of interest rates and present value in discounting, we must have $V - \frac{1}{2} * 100 = -99/2$. Take that over to the other side. $V = \$0.5$. All right, $V = \$0.5$. The option value is \$0.5.

- 6.35 Now, that is interesting because there's something about that model that I did not use. What, in all of this, did not get used in my calculation? That probability. I used that, I used that, I used this, I used everything. It has nothing to do with probabilities—and that was the Nobel Prize-winning breakthrough, that you could construct this hedged portfolio. Suddenly, risk return, all that, wanting a positive excess Sharpe ratio, blah, blah, blah, irrelevant. The option value comes from doing this hedged portfolio and a thing called no arbitrage, which means, if you know you're getting $-\frac{9}{2}$ tomorrow, you know it must be worth the same today, because there's no risk involved. That is to say, if there were interest rates, then you'd have to discount this present value, etc. So that is the idea behind option pricing.
- 6.36 Now I'm going to do some other—I'm going to introduce some symbols now, which is going to be fun, to show you how this works in practice. I'm going to do it with symbols because there are other interesting—well, interesting. Some people think of what's coming up next is interesting. It's confusing. I mean, if you're confused by any of this, you will be confused by some other things coming up. I'm going to do it all in symbols now. As we know, I like symbols. Let's do the same model, but I'm now going to use all symbols.
- 6.37 The stock price is S . This is over a period of δt . This is still going to be what happens at expiration. We're going to have the stock going to uS or to vS ; that's the example I did before, but we want to have this multiplicative thing because we want this geometric behavior. I'm going to have p here, $(1 - p)$, even though it seems it doesn't matter. I'm going to keep it in there. And I'm going to have an option with a strike of, strike is going to be K . I'll tell you what, I'll do it more in generality. I'm going to be vague about what kind of option we're dealing with. I'm going to have an option whose payoff is V^+ —plus just means it's the up branch—and the payoff of V^- .
- 6.38 So in my example, this was \$101, this was \$99, this is 1, and this was 0. And then we're trying to find V here, so I'm going to construct a portfolio as before, which is going to be $V - \Delta S$. This thing Δ is the symbol we always use in quant finance, which is the number of shares that we sell short. So what happens to this portfolio of $V - \Delta S$? It becomes, this is only an expiration, it becomes V^+ —from the option—minus Δ times whatever the stock price is tomorrow or tomorrow, tomorrow after time δt : $[V^+ - \Delta uS]$. Down here, it becomes V^- —the option payoff if the stock falls—minus Δ times vS $[V^- - \Delta vS]$.
- 6.39 Don't be confused by the multiple uses of V . I'm using little v , that's a lowercase v , to mean how much it goes down, and uppercase V is something to do with the option. So these are the option payoffs. That's the V , that's the thing we're trying to find now. What is that?
- 6.40 So hedging is all about making the option value—sorry, the portfolio value, this is the same regardless of whether the stock goes up or goes down. So we just have to say: $V^+ - \Delta uS = V^- - \Delta vS$. So that would give me $\Delta = (V^+ - V^-) / ((u - v)S)$. So that tells me, if I have an arbitrary option, and I know what the V^+ and the V^- are, because I know what

the payoff is for the option, and I've got some model for the underlying stock, then I can calculate this thing called Δ . For example, we had $(1 - 0) / (101 - 99) = \frac{1}{2}$.

- 6.41** I should have said, remember that before, when I hedged with one-half of the stock, and I found that the option value is one-half, those are different halves. Again, symbols versus numbers. I had half twice, but they're different halves, as you will see. So this tells me how to do this hedging. Now, it looks very abstract, this Δ symbol. This is what people do in practice. You might find a bank or a hedge fund has a client who wants some fancy contract, fancy option, an option that has a certain payoff. It'll be, if the stock trebles in the next year without falling below a certain value, and its average is whatever, divided by the maximum—there's some complicated formula that the client wants an option to pay off for whatever reason they've got.
- 6.42** The quant in the hedge fund, the bank, would go through some calculation like this, and he would calculate this Δ . It would tell him several things—how to value the option, which you're going to see in a second, but also how to hedge it. So what happens with the guy in the bank? He constructs, he builds up this contract for the client, and then he adds a profit margin, a hefty profit margin. They've got to make a living. They add a profit margin.
- 6.43** But the client is taking some risk. They're buying the option isolated. They're not doing the hedging. They're buying the option. They may make money or lose money. They're happy with that. They're happy with it. They will actually experience that 0.6 that you gave me before, because they're not hedging. They will be sitting there, hoping the stock rises. But the bank is not in the business of taking risks. They've just sold something worth 10, theoretical number, which we'll calculate in a second. They sold it for 15. They have just made 5, but they've only made 5 if they can lock in that difference between the 15 and the 10. And the way they do that is hedging.
- 6.44** So they go through some calculations to value the option. They sell it plus a profit margin to the client. The client sits there hoping they're going to make some money, whereas the bank, by hedging, doing this, what we're doing now for real, has immediately pocketed the difference between the 15 and the 10, the amount they've sold it for and the amount that it's worth. So this is not just some abstract thing. This is big business, what they're doing here.
- 6.45** Anyway, so this is the Δ . This is the amount that you have to sell short, which means that the option value here—sorry, not the option, the portfolio value—it's the same whether you go up or go down. I have made it so. So it's now $V^+ - (uS) / ((u - v)S) * (V^+ - V^-)$. The S's cancel. And that's going to be the same down here because I've made those two things equal to each other. If I rearrange this a bit, I'm going to have $(u - v) V^+ / (u - v) * -uV^+ + uV^- = uV^- - vV^+ / (u - v)$. Is that right?
- 6.46** So that's what the portfolio value is. And in fact, to put the numbers in that I had before, that should become $-\frac{99}{2}$. Is it? Anyway. So that's what the portfolio is worth

tomorrow. And of course, that must be equal to the portfolio value today. Why? Because if it's not, you know you're going to get $-99/2$ tomorrow, so it must be worth that today. So that's $V - \Delta S$, which is $(V^+ - V^-) / (u - v)S$. So that's that: $[V - (V^+ - V^-)] / (u - v)S$.

- 6.47** So this is the formula. This is the thing I had before. I rearrange it, and I can write V equals—and this is the punchline. I don't want to get this wrong; this is too important to get wrong—this can be written as the option value is $(1 - v) / (u - v) * V^+ + (u - 1) / (u - v) * V^-$. Now, that formula there has caused countless people misery. And I suspect you're not dissimilar. First of all, this is— $v < 1$ because it's going down. So this thing is positive; $u > 1$ because it's going up, so that's positive.
- 6.48** I can rewrite this as some, I'll call it p' , is equal to—I'll call it $p' = (1 - v) / (u - v)$. I just want to simplify the way this looks. So this is $p'V^+ +$ —now, if you add these two numbers, these two things together, you've got $(1 - v) + (u - 1)$ on the top, which is $(u - v)$ on the top, divided by $(u - v)$. So the sum of those two things is 1. So what goes in here is just going to be $(1 - p')$: $[V = p'V^+ + (1 - p')V^-]$.
- 6.49** That's the thing that causes untold misery. The reason why that causes untold misery is because that looks—you remember the calculation that you did, the way I tricked you into doing, what was it, p times—what was the calculation I got you to do that was wrong? I got you to do the following: V equals— and this is wrong. What did we do? We did $V = pV^+$, didn't we? Remember that, the 0.6 times 1?— $+ (1 - p)V^-$. It's the same formula. I told you it was wrong. I tricked you, but it's not wrong. That's what I just calculated.
- 6.50** Oh—sorry, I'm acting now—oh, it's a different p . It's still an average. It's still an expectation, the same kind of calculation, but it's a different p . That's the 0.6. That's not 0.6. It's p' . It's this thing. You have no idea. You think I'm being dramatic. Well, I am. But, and this is the easy version. This is where I have to say that, usually, this is done by the subject called stochastic calculus. And I did start off by saying people try to make quant finance complicated in order to make it respectable. This leads to a thing called what's called risk-neutral pricing. That has confused so many people. You could even say it was the cause—you could say, maybe this is exaggerating it a smidgen, but this is the cause of the 2008 credit crisis, in that people don't understand this. Let me show you what's going on.
- 6.51** I want to introduce you to a hypothetical world very much like the university, a world in which they have no interest in data. They love abstract probability theory. They like symbols; they don't like vague numbers. They don't know what the concept of risk is because they can't lose their jobs, for example, because they've got tenure.
- 6.52** Imagine that we visit this risk-neutral, this planet. So here's Earth. And normally, I draw pictures of the United Kingdom at this point. That's England. This is the EU. There would be dragons. Imagine that we wanted value options. And we need some input

from some really clever consultants, academic types. Well, they live in their own world. They live in their own universe, don't they? So here they are, in what I'm going to call the risk-neutral planet. And this is real. This is the risk-neutral planet.

- 6.53** Now, the characteristic of the people in the risk-neutral planet is that they don't appreciate risk, as I say, because they can't lose their jobs, et cetera. They don't care about data. They want symbols. They want as much abstract probability theory as possible. We know they're very clever. We don't know how to value options. So we go to them to do some consultancy work for us. So we get in some rocket thing, and we head to the risk-neutral planet for advice, probabilistic advice. And what we do is, we say this.
- 6.54** After introducing ourselves, we say we have a problem in finance. Finance, money. But it's got a probabilistic aspect to it. So it goes like this. We have a share that can go from \$100—oh, numbers, numbers. They don't like numbers, but to \$101 and falls down to \$99. This is what we're telling the people on the risk-neutral planet. And then this is a key thing. We say we've looked at the data, and the probability of the stock rising is—stop there. You don't need to tell us what the probability is. We are famous, Fields Medal-winning probabilists. We have enough information here to calculate the probability.
- 6.55** OK, we've just spent billions in artificial intelligence, whatnot, trying to figure out what the probability is. I was only going to say 0.6. We can figure out what the probability is. And the calculation they do is, they look at—because remember, they don't care about risk. They don't need to be compensated for taking risk like people here do.
- 6.56** So they say, "Well, clearly, if we introduce some p ," and I'll call it p' , "clearly, we have the following situation, that if the stock price today is \$100, that must be the expected stock price in the future, which would be $p' * 101 + (1 - p') * 99$. From which I can calculate that p' , and this is easy because of the symmetry, you can see this happening, the probability of the stock rising must be equal to 0.5 because that is what makes this so. So $p = 0.5$."
- 6.57** So the people in the real world say, "No, it's not. You said it was 0.6. I know it's 0.6. But they're saying it's 0.5, and they're cleverer than us. Well, let's be polite. We're visiting their planet, so let's not say anything. Okay, 0.5." And then they say, we need to tell them by options. We explain how there's a payoff of 1 or a payoff of 0, and we're trying to figure out V . So then they say, "Well, clearly, the probability of getting a payoff of 1 is 0.5, because it's the same as this thing, and the probability of it falling is 0.5. So the expectation, the value, must be \$0.5. Doing the expectation calculation, the option value must be \$0.5."
- 6.58** It's the right answer. They've got the right answer for all of the wrong reasons. All the things I said about, you need compensation for taking risk, that you mustn't take expectations. They've taken expectations. They've done it twice, and they've got the

right answer. How annoying is that? Anyway, that's called risk-neutral pricing, because risk-neutral means they don't need to be compensated for taking risk. You've got risk-averse, means the more risk you take, the more you expect to make. Risk-seeking is when you're looking for risk for whatever reason. Risk-neutral means you don't take into account risk at all. So that's risk-neutral stuff, and that leads to a whole industry of maths books based upon that.

- 6.59** What it means over here is the following: the calculation $[V = pV^* + (1 - p)V^-]$ is wrong because that would give us 0.6. The calculation that we would do, what was it? For the stock goes up to uS , [down to] vS , let's introduce p' , meaning the probability that makes sense if we don't care about risk. So we would have the current $S = p'uS + (1 - p')vS$. The S 's cancel, leaving us with $1 = p'(u - v) + v$. So, $p' = [(1 - v) / (u - v)]$, I'm looking over there to check it's right, it's that, which is the same p' that went into the option value thing.
- 6.60** Now, the reason why that is called untold misery is because the way it's presented using stochastic calculus is presented in a lot more complicated way than that. And it's one of those things where you can confuse people by having too many symbols. This is nothing. This is nothing compared to the stuff you might see in some books. But also it makes you think that, oh, actually, all you need, every time you want to do a calculation, you want to value something, you just need to take an expectation.
- 6.61** Now, I want to emphasize this by, I'm taking the wrong expectation. The reason this works is because they've taken the expectation twice. But the first time, they took the prices, the values, to get a p , and then they used a p to get a value. It's doing the same thing, the same wrong thing twice, but reversing. It's like two wrongs make a right. Isn't that great? Two wrongs make a right. But it means that people in this industry think that going around taking expectations is a good thing, when it's not. And this only works because of this idea that we had before, which was hedging, this Δ , hedging, which got rid of risk.
- 6.62** See, what's happened here on this risk-neutral planet is fantastic. They've come up with the right answer. It always works, but they haven't discovered hedging. There's no mention of Δ . We calculated the Δ as part of our process. They didn't do Δ . They just ignored all of that side of things. If they actually started investing in options because they've done this, they'd get the right values, but they wouldn't know how to hedge. We know how to hedge.
- 6.63** So there's two sides to this. There's the speculator who buys options, and the speculator who buys options would pay this much for it: $[V = pV^* + (1 - p)V^-]$, but they'd actually, because they're not hedging, they'd experience that: $[V = pV^* + (1 - p)V^-]$. They'd pay \$0.5, and they'd experience \$0.6. If they were to buy the same option every day for a year, then sometimes they'd win, sometimes they'd lose, but they'd have an average of that. Every time they would pay \$0.5, but that's what they expect to make.

- 6.64 Actually, what happens in practice is, the bank does this calculation, \$0.5, and then they will do the Δ hedging, but they can't sell it for \$0.5 because they would lock in a profit of 0. So they sell it for \$0.55. So they would sell it for \$0.55, do the Δ hedging, make a guaranteed profit of \$0.05. The person who's speculating, who's not hedging, would pay \$0.55 for it, but expect to make \$0.6, an average of \$0.6, which is about, is it 9%? Something like 9%.
- 6.65 So the bank hedges and makes a guaranteed profit of the difference between what they sell it for and what the theoretical value is. The person who's speculating, who's not hedging, has a risky bet. He's exposed to these p's, which, in practice, you don't know. In practice, you know what this is and this is, and from this calculation, well, you need to know what these are. So that's called risk-neutral pricing. And it only works because you can do this Δ hedging.
- 6.66 See, this argument is wrong. This argument is completely wrong. That is not how you value things. It just happens to give the right answer because two wrongs make it right. But if you don't know, if you haven't calculated this: $[\Delta = (V^+ - V^-) / (u - v)S]$, if you do what the people on the risk-neutral planet say, and you haven't discovered Δ hedging, then you're exposed to risk. And the reason why I mentioned credit risk is because you need to be able to hedge whatever is going on to use this argument. It doesn't work if you've got something you can't hedge, and you can't really hedge credit risk.
- 6.67 You've got a company that's issued some bonds or some options or something, and you can hedge against the stock price, but you can't—because the stock price is going up and down— you can't hedge against default because there's no default. Nothing happens. Nothing happens. Nothing happens. Default. You can't hedge against that. So all of this sort of thing does not work in the credit world, for example, although this is still what people do. People still apply. They still take expectations.
- 6.68 And part of that is because it is confusing, and I hope you're confused. I mean that most sincerely. I say this when I'm lecturing to all sorts of mathematicians. At this stage, you should be confused, because this is a confusing thing. Paul said, "Don't take expectations," and now he's saying, "Do take expectations." It's confusing, but you have to take the wrong expectations, not the right expectations, if that makes sense. It's the right wrong expectations. That's what I think I'm saying.
- 6.69 It's one of those things where, when you're studying this, you have to go through this thing. When you're learning anything for the first time, you're learning to play the ukulele or something, and you're strumming along, you're making progress. And there comes a point where suddenly it feels like you've forgotten everything you learned, and it all goes horribly wrong. And you have to go through that, come out the other side, and suddenly it's all very easy. It's like that. You've got to go through a confused stage. So if you leave here today confused, then I've done my job.

- 6.70** One last thing about this is the following, and that is, in practice, this is all about what happens one day before expiration. Usually, we're not one day away from expiration. So what do you do if you're a year away from expiration? How do you value options? It's really very simple. It's very simple. This is the binomial method. So this up-down thing is the binomial model, but the binomial method is for how you then go on to value options using this up-down random walk thing. So this is what is going to happen.
- 6.71** Stock goes from here, goes up to uS , goes down to vS . Then it either goes down to uvS , or up from here, which would be vu , which is the same thing, or it goes down here, v^2S , or up here to u^2S , and then it goes down. That's looking better, isn't it? Something like this. And you build up this tree shape, u^3S . This would be u^2 —sorry, these are all multiplied by S — u^2vS , and so on. This would be uv^2S , and so on. And we do this all the way to expiration.
- 6.72** So what we have at expiration is bits of the tree that look like this, lots of them, because we have lots of time steps. There'll be lots of these. And then we can apply our tree idea, this binomial model idea, to this bit. We know what that is. We know what that is. We know what the risk-neutral probabilities are and so on. We can calculate the option value here. And so we do the same on this bit. We know this payoff. We know this is expiration. This is expiration. We know what, we can calculate the option value there, calculate option value all the way like that, all the way down.
- 6.73** And then, we do the next bit, like that. Since we know the option value here and the stock price here, et cetera, we can calculate this, and we can do it like that. We can calculate that, and we can—and then we can keep—we can go all the way back down the tree. So you build up a tree going this way with all the 100 , 101 , $101 * 1.01^2$, et cetera, et cetera, like that. It gets to the end, and then you work your way back down the tree, these little triangles, one at a time. And so it looks like something that's in here. Let me try this.
- 6.74** It looks like that's one bit of the tree, with the data in yellow, the stock price, the volatility, the interest rate, the strike—C means a call option. Don't worry about it, all this is US. And it's got a u and a v in it and a p' , the risk-neutral probability. It's got a discount factor because the payment is in the future. So that's one-year expiration, but that's just one step. Well, that's rubbish. What we really need to do is something that looks like this, maybe with four steps. I've had to zoom out, but that gives the idea of the tree-like bit.
- 6.75** We solve for that bit to find the option value. It's in blue—it's all color-coordinated—and once we found all the blue down here, we move to this bit of the tree, and then this bit of the tree, and all the way back to now. And we get that an option with the value of the current stock price is \$100, and the strike is \$100, and the volatility is 20%, and the interest rates are 5%, we get a value for the option of \$10.2861. The more steps we have in that tree, the closer we get to a good, accurate number for the option.

- 6.76** Notice there's no mention here, there's no drift of the stock; the stock drift is completely irrelevant. All we need to know is the interest rate and the volatility. And that's why I said earlier about anyone who works in derivatives or options, we don't have to worry about the drift, which is great because it's very hard to measure.
- 6.77** So that was an introduction to options. That was an introduction to the quantitative side of options, the valuation of options. It's not the side of trading options, buy low, sell high. That's it: buy low, sell high. But that's mathematical finance, quantitative finance, financial engineering. There are a lot more types of options besides this thing called the call option, but that's for another day. Thank you. Any questions?
- 6.78** Does this have anything to do with or can this relate to anything with game theory and gambling?
- 6.79** That's harder. You're talking really about knowing what other people are doing. I'm not going to answer the question. The answer is yes. Absolutely, yes. Absolutely, yes. I'm trying to think of an example of anything I've done, but I can tell you one thing that's related to that, and that is, if everybody is doing the same thing, are there any repercussions? That's in the next lecture. Are there any repercussions of people blindly following some kind of formula?
- 6.80** As to game theory, yes, there's lots of research on that thing. I don't know how much it's used in practice. I mean, would you use it in practice in poker terms, knowing what other people are doing?
- 6.81** The question is if everyone is doing the same thing. If everyone's playing a certain way, know that, if I have a hand and this is what's on the table, I'm going to have to bet this amount, then should I be able to maximize on knowing what they're doing?
- 6.82** Absolutely. Yes. In financial terms, it means that if you know what people are doing, can you interact with them personally and take advantage of the herding behavior or stupid behavior, whatever? There's also knowing that everybody is doing—this is maybe easier—knowing what everybody is doing, what effect it has on the market, and just sitting back and taking advantage of the market as a whole. That's the thing I'm going to be talking about tomorrow. It's a very good point. And there's a whole lot of game theory and stuff that's in all of this. And a lot of people who started hedge funds have been like Ed Thorp. He had a hedge fund, and he was effectively a professional gambler to some extent because of his blackjack work.
- 6.83** This leads me on to another thing, which is horse racing, for example. The same ideas can be applied to horse racing. When you place bets on the horses, you get some sort of odds. Well, where do those odds come from? Those odds do not come from the bookmaker thinking that they know the probabilities of a particular horse winning or losing, right? No, that comes from other people having placed bets. And it's the same mathematics here as in the horses. The idea being that people place bets, and the

bookmaker then adjusts the odds, so that whatever happens, generally speaking, they're going to make money.

6.84 And those probabilities, those odds, are like the risk-neutral probabilities we have here. They're not the real probabilities of the horse winning. You might have your own idea of the probability of the horse winning. It'll be completely different from the odds that the bookies got, because he's effectively changing the probabilities to match what the betting is, if that makes sense. So there's a lot of parallel between these things.

Lecture 7: Investor Behavior

At a glance — Investor behavior and behavioral finance.

- 7.1 Last two lectures, a bit more waffly. Certainly, the behavioral finance stuff is a lot more waffly. And then later on, we're doing AI, which is not waffly, but it's not possible to teach much of AI in just one hour. So I'll just give you an overview. But behavioral finance is a great topic, but at the moment, it feels a bit more like it's great for dinner party conversations than actual practical use. But I'm going to do my best to bring together a little bit of behavioral finance and a little bit of mathematics, as you'll see.
- 7.2 So I'm going to kick off with a couple of examples of theories that are popular, which don't make any sense. And I'll explain where they come from. Give a few examples of actual people's behavior, which don't fit in with what the theories are. Then I'll mention a few behavioral studies of late, and by late, I mean, say, the last 50 or so years. This has become a big topic. People come up with all sorts of weird and wonderful experiments to conduct on people to see how they actually respond, whether they are rational or not. And obviously, the answer is, they're not. It would be an awful world if everyone was rational, I think. And then, this is me applying three behavioral ideas to finance. I think they are mostly my examples, in the sense that it's these cases, it's me who's applied them, the ideas, to finance. So that's the original research, if you like.
- 7.3 So first thing is this thing called the efficient market hypothesis. This is just something taken from Wikipedia. "A hypothesis in financial economics," —I don't need these — "that state asset prices reflect all available information." That means that, you see a price in the market. There's nothing secret. Anybody who reads a newspaper would have access to the information that is contained in those prices. There's nothing secret there. And one of the implications of this is that it's impossible to beat the market, because everybody has exactly the same information about the stocks. This leads on to the idea of, potentially, things being random, which we've been talking about all along.
- 7.4 But some people do seem to beat the market consistently. Now, maybe that's a statistical fluke, because there's so many people trying to beat the market, some will. What economists do is, they come up with this, some crazy theory like that. I think it's crazy, but some crazy theory, and then they tweak it and water it down. Actually, if you look into the efficient market hypothesis, you'll find there are different degrees of assumptions, different types of assumptions, getting weaker and weaker, trying to get closer and closer to what's really going on.
- 7.5 Now, that's a common theme. Someone comes up with some wacky theory that sounds great in academia, but then they've never run a grocery store, the economists. Some quote, what's that quote about economic forecasting? It's there just to make astrologists look good? I forget who that was. So that's example one.

- 7.6 And example two—I'll come on to this a bit more later on—is the idea of a utility function, that somehow everybody, whatever you have, any decision you're faced with, you will be able to turn that decision into something, a numerical number. And that numerical, depending on how high that numerical number is, you'll make a decision to do X, Y, or Z. So that's utility theory and utility functions, which I will be doing in a second.
- 7.7 The obvious example, the first obvious example of how people actually behave, would be lotteries. Does anyone invest or gamble, whatever, in lotteries? Yeah, you buy lottery tickets. Now, you know, well, two things. One, there are some cases where it's possible to win in lotteries, because of the size of the pot and the number of tickets that you have to buy; there are occasions. Sometimes you do hear people raising the necessary hundreds of billions of dollars to buy every single lottery ticket. And it seems to work, but that's a heck of a lot of effort. But that doesn't happen very often.
- 7.8 There are things you can do to assist with lotteries. If it's one of these things where you pick some numbers yourself, and the winners get to split the money amongst all the people with the same numbers, the trick there is just to choose the numbers that nobody else is picking. You don't pick numbers up to 31 because people pick birthdays and things like that. So you can find out. I think it's relatively easy to find out which numbers are less popular. They're no more likely to win. But if you do win, you will win more.
- 7.9 Now, lotteries like roulette, you've got a negative expectation, typically, which means that no economist would ever buy a lottery ticket. But then there are times when you might, it would be rational to buy a lottery ticket. If you need to have some life-saving operation, and you're in a country where health care is not free, then the only way you might possibly be able to afford this life-saving operation is if you win a lottery. So it makes perfect sense. But otherwise, it doesn't.
- 7.10 So you can see there's already an example of, "People do buy lottery tickets for rational reasons because they need this life-saving operation." So that's an example of economists, in this example, me, looking for an explanation for why they are being rational. So there's a lot of that going on as well, trying to find the rationality underneath the base of all this.
- 7.11 Here's a great one. The failure experiment. In the late 1990s, there was a competition in The Financial Times. What you had to do to win this competition, you had to write in on a postcard, before the Internet became too big, write on a postcard a number, a number between 1 and 100, something like that. And they would take all these entries, the numbers 1 to 100, the integers 1 to 100, they would find out the average. And then the winner of this competition was the person who predicted two-thirds of that average, not the average, two-thirds. If it was the average, everyone would put in the answer of 50, I guess. But two-thirds of the average?

- 7.12** The way that they were thinking was that, if you submit—well, suppose people submitted 50, then the winner will be the person who submitted the answer 33. Okay, in that case, we'll submit the answer 33. But then the average is going to be 33, and two-thirds of that is 22. Okay, let's submit the answer 22. In which case, the average is going to be about 14. All right, submit the answer 14, and so on. And this goes all the way down to 1 or 0, depending on how you round things.
- 7.13** So there's an example of, if people were totally rational, everybody would submit the answer 1 or 0, depending on how you round up two-thirds, in which case it's not even worth entering the competition. So it was an experiment to see how rational people were, if you like. Now, I was giving a lecture—I was supposed to give a lecture, when I saw this experiment in The Financial Times. So I completely stopped the lecture, because I had the freedom in those days to talk about anything I wanted. And so I thought I'd speak to the class about this experiment and see.
- 7.14** We discussed it. And there were some people in the audience who said, yes, the economically stable solution is 0, so we must submit the answer 0. And I said, "Well, that's not how people behave. Do you think everybody's going to be submitting the answer 0?" So some people insisted it had to be 0. Anyway, I persuaded them to do the following. I persuaded them to all submit the same answer. I paid for the postcards and the postage, and I told them to all submit the number 100.
- 7.15** And the goal of that was to see—well, several things. One, to see if we could move the market to the answer, because no way could 100 win. Even if everybody submitted 100, the answer, the winner would be 67. So, submitting the answer 100, see if we could move the market. In fact, that's one way of looking at it: change the average by submitting this crazy answer. And also, I was kind of secretly hoping to get some kind of honorable mention in The Financial Times when the results came out.
- 7.16** Anyway, both of those came to pass. When the results came out, they mentioned the university, people who all submitted this answer 100. And that was it. No comment about that. I mean, my point was that basically, we were trying to cause trouble to some extent. There goes another rational theory: who causes trouble? What's rational about causing trouble? Anyway, it's rational to me. And they also gave us what the answer was, and the answer was 13. So people had gone through this process in their heads and done some two-thirds of two-thirds. Anyway, the winner was whoever gave the answer 13.
- 7.17** There's a twist in this story, and that is, while all my students were submitting the answer 100, I submitted my answer, which wasn't 100. I was really trying to win. My answer, can you guess what my answer was? My answer wasn't 13. My answer was 12. But all these people submitting the answer 100 on my behalf had moved the answer by one point. In other words, had I not got them to submit the 100, I would have won. Isn't that great? So I think that's a much better story than whatever the prize was.

- 7.18** Another example: diversification. In a previous lecture, we talked about modern portfolio theory and the idea of diversifying, putting your money in lots of different investments to spread your risk. Well, imagine that it's your first job as a trader in a bank. It's your first job out of university. You're being introduced to all the other traders, and you notice that all the other traders seem to be doing the same kind of trade, the same markets, the same—they're all buying the same thing. Everyone seems to be doing the same thing.
- 7.19** You think this is strange. Surely the bank or the hedge fund would be encouraging people to diversify. Very odd. Anyway, so you start work and you think, "It's my first day in the job. I'm going to impress my bosses. So I'm going to—nobody else is doing the diversifying, but I've learned all about it. I'm going to diversify. I'm going to do some different trade. Actually, I think my trade is, my idea for a trade, to buy or sell, is actually better than everybody else. I think all these other traders—I've done the statistics. Their trades are basically a coin toss. My trade, the idea I've got for a trade, is, I think I've got a 75%, let's say, chance of making some money for the bank. So that's what I'm going to do. I'm going to trade, do my investment. I'm more likely to make money than the other traders."
- 7.20** So, what happens? What are the consequences of that? You'd think that'd be good, that I'm going to make lots of money for the bank because my bet, my investment, is going to be more profitable than all the other traders, and I'll be a hero, et cetera, and I'll get a big bonus. Well, there's a problem with that because the only way I'm going to get a bonus is if, at the end of the year—I'm being a bit extreme—if at the end of the year, the hedge fund or bank still exists.
- 7.21** That means that not only do I have to make money to get a bonus, but all of the other traders have to make money, because if they lose, there's no bank. Nobody's getting a bonus. So if I'm uncorrelated with these other traders, the chance of me getting a bonus is 50 %—that's the bank as a whole making money—multiplied by my 75% chance of making money, which comes to 37.5%. So my chance of getting a bonus is actually less than if I tossed a coin, even though I've got a better bet. So, diversification is a great idea in theory, but when you hit the real world, sometimes it's best to just be part of the herd. So, that's an example of what the rational thing to do might be different from what the economic theories say.
- 7.22** All right, back to this thing called expected utility theory, Von Neumann and Morgenstern, 80 years ago. The idea is that we've all got our utility functions. You can see where these people are coming from, because they set everything up as axioms. It's like something to do with geometry, almost, something about "parallel lines never meet" and that kind of thing. Anyway, the axioms are that, between two things, you always have a preference, or maybe they're equal in difference.
- 7.23** Apples, bananas, and A, B, C, carrots. Apples and bananas—which do you prefer? Apples or bananas? So that's the first one. Then the second one is transitivity, which

means if you prefer apples to bananas, or bananas to carrots, then you prefer apples to carrots. It's like with numbers. When you're doing university learning, you go back to learning how to add up. If $x > y$, and $y > z$, then $x > z$.

7.24 Then a thing called continuity, which is essentially saying that if you've got apples, bananas, and carrots, then there is some quantity of apples and carrots that is the same as a banana in terms of preference. And then the fourth one means that if you throw into the pot, if you've got an apple plus D, dog, completely irrelevant to the whole thing, an apple plus a dog is going to be greater than the banana plus the dog. So, throwing a dog into the pot, not literally here, but into the mix, the dog is irrelevant. It shouldn't change things.

7.25 So they said, if these things are satisfied, then you've got a utility function. All of you have got a utility function. Different utility functions, each of you, but you've got one. You didn't know it, but you've got one, and it looks like something like this. I'm not going to label the axes, because the axes could be all sorts of things. But for example, if this were money, how much money you've got, it has this shape, that the more money you've got, getting a little bit extra money becomes less and less important to you.

7.26 The slope sort of falls off, which, incidentally, is a reason why taxes— this is controversial—taxes should decrease as you earn more, so it incentivizes people to work more because eventually you're paying so much in tax. So the UK budget is in a— how long before they take all my money away? Oh, it's exactly a week, my hard-earned money. I'm going to add that adjective, a compound adjective.

7.27 The problem is that this can fail to explain real choice, and I'll give you an example in a second. And of course, it fails the sniff test. It's just not common sense. I mean, what if it changes? One day you like an apple, the next day you just wake up preferring bananas. What if it changes every day? Is it useful at all? People who work in this field, you'd think they would even know their utility functions. So I'd be curious if anybody has ever asked economists what their utility functions are.

7.28 All right. So that goes back to the 1940s. Then the Allais paradox is one of the early examples of showing that the axioms are not valid necessarily. For example, here we go. I'm going to give you two choices. You don't have to answer, but I'll do the maths of this. Choice A is, there's an 11% chance of you getting \$1 million and 89% chance of nothing. So imagine some random number generator, and if we can give the numbers 1 to 100, if it's a number 11 or below, you get a million dollars. Choice B, 10 % chance of getting \$5 million, so slightly smaller chance, but more money, a 90% chance of getting nothing. So you choose A or B.

7.29 And then along comes choice C or D. So I'll ask you that question one day, and then come back the next day; you've forgotten about the first one, and we do the second one, which is, you can have 100 % chance of \$1 million or 10% chance of \$5 million, 89% chance of \$1 million, or 1% chance of 0. So then I'm asking you to choose between

C and D. So you can make a choice between A and B, then you make a choice between C and D.

7.30 Now, a little bit of fiddling with this. I've color-coded this. You can rewrite each of these choices. So you've got choice A and B, and then you've got choice C and D. You can write it in a way that makes the A, B, C, and D look similar. It makes A and C look the same, and B and D look the same. The only difference is the numbers here, 89% chance of doing nothing, as opposed to 89% chance of getting \$1 million. So these are irrelevant because it's the same each time. But the bits in red are exactly the same. The only difference is that between A and B, there's an irrelevant \$0 because it's the same for both, and in the second one, the \$1 million.

7.31 So, in other words, if you're totally rational, if you obey these axioms, you would choose A and C, or B and D. Whereas in the experiments, one-third of people don't. They choose A and D, or B and C. So, one-third of people don't have a utility function. Now, that's a bit strange because I have no idea what the circumstances were when they were asking these questions, because you can imagine, I mean, who could be bothered? Maybe some mathematician would be geeky enough to go through this analysis. Everyone else was going, "Oh, gosh, another one of these crazy psych experiments. I just wanted A and D. I don't know. Get me out of here." They might not be too invested in thinking about it too deeply. Who knows?

7.32 Anyway, then, a little bit later, Kahneman and Tversky came up with this idea of something called prospect theory, which is different from utility theory. It's all about gains and losses, rather than choosing an apple versus a banana. This is interesting because your value function is different depending on whether you're making a profit or losing money.

7.33 Now, this is something that works. This makes a lot more sense. It means that, if you're a trader and you've just made a \$1 million—so let's put it up here, \$1 million—you're happy. But then the next day, you lose \$1 million. You're unhappy, but you're more unhappy than you were happy. You see what I mean? In other words, and there's this ratio of typically 2:1, which is, if you do lose a million dollars as a trader one day, you have to make \$2 million the next day to be happy, to be neutral. I'm going to do the maths of this in a second.

7.34 I don't call this prospect theory or Kahneman-Tversky. I call this Moonraker theory, because it is in James Bond, in Fleming's Moonraker. It's mentioned a couple of times. So, James Bond has been playing bridge against Hugo Drax, who's the baddie in Moonraker, and Hugo Drax has been cheating. So that's why M calls in Bond to play cards, bridge, against Hugo Drax. James Bond comes in with a—he swaps the deck out for one that's stacked in a certain way. Anyway, Hugo Drax loses, I think it's \$15,000 or £15,000, I suspect. He realizes that Bond has cheated and makes various threats.

- 7.35 Anyway, another one, anchoring. Same people. This is the idea that you get attached to a number. These are really great examples. These are really good. If the group was asked to estimate $1 * 2 * 3 * \dots * 8$, and another group, $8 * 7 * \dots * 1$. The same thing—oh, yeah, yes. So the answer is actually $\sim 40,000$. So the two groups were way off, unless I've done my maths wrong, or Excel has done it wrong. The first group, they started $1 * 2 * 3$. So they started off with the number 1. They had the number 1 in their head, and then the number 2 and then number 3. They came up with a much lower estimate for this product than the group who were asked to do the same calculation, but starting with $8 * 7 * \dots$
- 7.36 And here's a funny one. So you spin a wheel. It looks like it could be any number between 1 and 100, but it's set. It can only be 10 or 65. Then, after it's spun, and it's 10 or 65, you get asked: what percentage of African countries are members of the United Nations? And if they saw the number 10, if the number 10 came up, they'd guess 25% of African countries are in the UN. If the number 65 had come up, they'd guess the number 45. Isn't that great? I'll mention this a bit later on as well.
- 7.37 Other things, just in passing. Sunk cost fallacy, the idea of people's tendency to throw good money after bad, which is, in finance, that is an obvious thing. You know this one: 90% of US drivers rate themselves above average.
- 7.38 Herding. Herding. I'll do herding again in a second, but you can imagine what herding is like. But then, as I mentioned with the trader example, joining the bank, maybe herding can make some sense.
- 7.39 Now for some math, which is what you've all come for. As I say, these are things that I've worked on because they're great stories, aren't they? If you're negotiating buying or selling a house or getting a job or something, you're supposed to be the first person to make an offer. You should make an offer. You shouldn't say—or if you want a higher pay, you just say, "I want this much money." You shouldn't get all coy and say, "What do you think I'm worth?" or something like that. You don't do that. You've got to get out there and say a big number; or buying a house, you're going to say a low number. You don't wait until the other side, because everyone gets anchored to that first number.
- 7.40 Anyway, these are great stories, and you can see how they're related to finance in many cases. But I wanted to do proper maths based upon these ideas. One of the first things I did was on anchoring, this idea that you get attached to a certain number. What I've got here are—the top picture is simulations. I don't know, let's say the blue line is a simulation like we did the other day, just numbers pulled out of a hat, normally distributed random numbers, and it looks like that.
- 7.41 Then I plotted—I pulled the numbers out of a hat, which would give me something that looks like this, just one simulation. Then I took the line that joined this dot to this dot and subtracted off a linear thing. So what I ended up with was, had the same

randomness, but it was now flat. So all that took, it was take off a straight line, effectively, just for reasons which will become apparent in a second. So that is a simulation. So that would be the blue line in this picture here.

7.42 Then I just turned it upside down. Now, because we talked about a geometric random walk, if I start off with—yeah, it's going to the details, but one over a lognormal random walk, a geometric random walk, has the same properties as the original. You can see that the red line and the blue line are just mirror images of each other. They both end up at the same place because I made them so. So that's simulated data.

7.43 Then I took some other thing. I don't know what stock it was. I did exactly the same thing. The blue line is the real thing, and the red line is essentially one over. So if it's a genuine geometric random walk, then you get exactly the same look. Now, to me, the red line and the blue line, they've got the same amount of jaggedness. Now, this is just a visual thing. It's the same amount of jaggedness. Yeah, they look very similar, just reflections.

7.44 The bottom two, which are real, are different. I don't know whether you can see it. It's moderately, I think it's moderately obvious. But what I'm seeing is that the blue line, which is the real stock price without turning upside down, goes like this. I'll exaggerate it. It's got this thing going on. Whereas the red line actually looks more like waves about to break. It goes the other way. This is literally turning upside down, like that. So I've exaggerated it there, but you can kind of see that. The red is a bit more rounded, a bit more bouncy at the top.

7.45 Now, when we talked about volatility earlier on— volatility, I had, in my simulations and things, I talked about volatility being 20% or something. No, 19%, I think it was, from the S&P 500 index. I had it as being constant. But when you get into quant finance properly, then people often use fancy models for volatility to capture a certain type of behavior. Now, the popular models often assume volatility is random. That's one popular model. One not-so-popular, but you can do it, is volatility jumps from 10%. Then a week later, it's gone to 20%. It stays at 20%. A month later, it's gone to 35%. Three days later, it drops down to 20% again. So it jumps from one level to another. That's the model.

7.46 There's another model, which is the most popular, and if you remember, the other day, I said that people often choose models because they're user-friendly and tractable, don't necessarily have any relationship to what's going on in practice. Often, people say, instead of having volatility being 20% or something, they say, let's make it a function of stock price and time, so as time changes. Now that makes sense, having different times of the year, you might be coming up to some earnings announcement or something, you might expect more volatility. I don't know. I'm just making that up.

7.47 But the idea of having the volatility depend upon the stock price is to capture something like this behavior, the idea that, when the stock price is low, as in the case of

the blue line, as it's a little bit down, volatility goes up, meaning it's more bouncy and goes back up again. That's not a great description. So you do see that. You do see that, when the stock is lower, volatility can increase. But I don't like that at all. I don't like that at all, because it doesn't fit in with human behavior.

- 7.48 And I think this is a potential for anchoring, in particular. Anchoring to what? I and a couple of co-authors suggested that a more natural thing to anchor to would be the moving average. Now, traders do not have the greatest of memories or concentration, and maybe they only know, "Well, last year, the share price was this. Now it's this." So if you find that the share price ever falls significantly below some average of what the share price has been over the last year or two, that's when they think, "Oh, my gosh, it used to be this level. Now it's much lower. That's a bad sign. The company is doing badly." Volatility increases.
- 7.49 Then, as the stock price increases, or rather goes back to the level of the average, then the stock price hasn't improved; it's just that the average has fallen. People get used to the new level, the new stock price, the new normal. Or if the stock price rises a lot above the average, this company is doing fantastically well, and it's less risky. Anyway, you can have things like moving averages and whatnot.
- 7.50 And what we found was that, if we looked at volatility— the blue line is a measure of actual volatility—we found that a very, very good fit would be some kind of shape like that. This is a relationship between— so this thing here, this ξ , this Greek letter ξ , is the ratio of the stock price to some moving average. And the best fit happened when λ , which is like a half-life for this average, when this λ was about 1, which means whoever was buying and selling shares, had a memory of about a year.
- 7.51 Now, when that happened, when we saw that, I thought that was fantastic, because we could have found something like λ was such that the time scale for memory, for the average, was a thousand years. It could have come out that the best fit was a thousand, time scale of a thousand, which is crazy. That would have been complete nonsense, because no traders have been trading for a thousand years. Or equally, what if it turned out that the memory for the average was a day? Well, equally, that would be a bit silly because, for all their faults, traders do have a memory that's a bit more than a day. So getting the result that λ was about 1 was encouraging.
- 7.52 The problem is the models that follow on from this, there are no nice formulae for valuing complex derivatives. I mentioned Black and Scholes in the previous lecture. They've got nice formulae, one of the reasons why their model is so popular, or indeed, why this model is so popular, because there are nice formulae, and you don't have to do any nasty numerical solutions of things. So it's annoying for someone like me, who's come from all sorts of scientific, mathematical modeling background, to see people are more interested in nice, pretty short formulae than they are in accurate modeling. But that is very much the way things are in quant finance.

- 7.53 Okay, another example: herding. This is another piece of research, and that is—this is interesting. We saw the data for the S&P 500 index on October 19th, 1987. One of the explanations for this is a thing called portfolio insurance. This is the idea, although there were various companies doing this, that you could insure your portfolio against large falls in the market by following this algorithm, if you like. Actually, when I say follow it, there were companies set up specifically to help you insure your portfolio, to do exactly what I'm going to describe now.
- 7.54 So the way it works is, the market is bouncing around as it does. But then if the market falls, you get a little bit worried that maybe it's going to really plummet, and you don't want anything dramatic to happen to your portfolio. So we've got this happening. The market is doing that, but then it starts to fall. Now, what portfolio insurance does is you specify how much you're happy to lose. I'm happy to lose—I don't want to lose any more than, if I started here, I don't want to lose any more than that. That's what you use in your insurance contract or whatever. You say, I want to be sure that that's the most I can lose. So, this level here is very much like the strike price we had in options. In fact, all of this is based on option theory.
- 7.55 What happens is, as the market falls, these portfolio insurers did a calculation using straightforward option theory, like we did in the previous lecture. Remember how we had that Δ , the Δ that tells you how much to sell in order to hedge your portfolio? They calculated the same thing, Δ , how many of a stock you should hold. And what happened is, as the stock or the market as a whole fell, that Δ changes, and it tells you—it's quite logical—it tells you to sell a bit to maintain this hedging thing. What happens if the market falls, and everybody is following the same formula, the same Black-Scholes option pricing Δ formula? I didn't show you. I gave you a hint of the formula yesterday. The formula that tells you to sell more.
- 7.56 Well, the stock falls, you've got to sell something. And what does that do? It causes the stock to fall more. And then you do it. Then you go to your computer, you put in this new share price, which is a bit lower, and it tells you to sell more, and so on. And what happens is with the '87 crash, people have said. I have no idea whether that's true or not.
- 7.57 And so, for example, here is a—this is just a simulation of one of these models that I worked on with a student. So the blue line is the asset, and the red line is the volatility. What happens is, as you—and this is doing a simulation of what's called a—I mentioned very, very briefly that a put option was the option to sell the stock. You don't need to know what I was doing, but I was doing some option calculation here. What happens is, even though the stock on its own would be bouncing around 20%, the effect of people following this formula—you can see what the volatility is doing. It's rising until it gets to a date. And this is the date on which, in our simulation, we said "stop," and then the volatility falls back down to 20%.

- 7.58** Yes, that's the illustration, and you see what happens to the share price, right? The share price collapses. So the volatility, the share price is high. Any sort of fall encourages people selling. Volatility goes up, more falling, and a collapse of the share price, followed by a—once it's all over, the share price is bouncing around 20 % volatility as before.
- 7.59** You get other—there are other effects where people are—you get the opposite effect. Sometimes, depending on what exactly you're hedging, what the formula is that you follow, sometimes you find a strange experience, which is— remember, I mentioned strike price. So you've got a share price. I won't go into the details, but here's some option expiration, and you have strikes. These are the strike prices, the K's. Maybe there's a 100, a 90 strike. So these are strikes for different options: 110, 120. There are options. You find the stock price doing this. And as you get closer to expiration, it does things like that. It just homes in exactly on one of these strike prices because of all the people hedging, doing the opposite. The formula they're using here is, when the stock rises, they sell; when it falls, they buy, and it causes the stocks to just ping, hit that particular stock price. And then, thereafter, the next day, we're back to business as usual. This is all herding behavior, as these things happen.
- 7.60** The final example is prospect theory. So this is applying prospect theory to trading. Now, when I'm lecturing to traders about this thing, I will say, "What's it like losing money?" I think I've only ever had one person give me the correct answer. The correct answer is, "I don't know. I've never lost money." Everyone else has said it's awful, and it is awful. It is gut-wrenchingly awful, losing money. In prospect theory, you have to make up more to compensate for the pain, the agony, of losing money, even if it's not your own money, but it will be, in terms of a bonus.
- 7.61** So, look at this. This is a simulation of a stock price, and it starts off at 100, and over a period of a year, it ends up at 130. So it's a 30% return in one year. That's fantastic. How does that make you feel? The answer is: it depends. What it depends on is how often you look at your portfolio— oh, sorry, how often you look at the stock price. Because the stock price is doing the following. The stock price is doing this.
- 7.62** If you remember when we were doing the simulations, we had the random number, ϕ , out of a hat of normal numbers. We had that multiplied by the σ , the volatility, and we had the growth rate, μ . Well, clearly, the growth is nice and positive because it's going up. But very often, roughly 50% of the time, the stock price has fallen. So: happy, sad, happy, sad, happy, sad, happy, sad if you're looking at every single stock price, every single day. In particular, this: fantastic, awful. So you end up in the same place, but because of this asymmetry, you're miserable.
- 7.63** The best thing to do, of course, is to not look at your portfolio, your stocks, because if you look here and then come back here, fantastic. So when I say it depends, what I mean by it depends, it depends on how frequently you look at your stocks. So, here we

made most of the return, or made all the return in the first few months, and then thereafter, it was mostly flat.

- 7.64 So, for example, if you're looking at your portfolio every month, here I've just done a simple thing. What I've done the following: I had a very—because I want to get formulae, because then people might pay attention if there's a nice formula. So this is your return each day, and I just work with the following. The prospect theory had a curved thing, so an S-shaped thing. I just work with that, that, just to keep it simple, with the ratio of 2:1 in the slope. So if you make money here, you're that happy. If you lose the same amount, you're more unhappy.
- 7.65 And then you've got, on top of this, you've got this normal distribution for what the return actually is. So this is your happiness function—red line, if you like. And then you got a normal distribution on top. To figure out your average happiness, your expected happiness, you multiply these two functions by each other and find the area under the resulting curve. The further to the right this is, the more this hump is in the happy place. This distance is something to do with μ times, let's call it τ , which is a time scale, how frequently, how long you wait before you look at your portfolio again.
- 7.66 Whereas the spread, σ/τ , if τ is small, this is very narrow, but it's in the middle, which means that the negative half dominates the positive half, and you're unhappy. Now, you're not very unhappy because it's all squished up and all the numbers are down here. But you're not very unhappy, but frequently, because τ is very small. So you've got a little bit of unhappiness, but all the time, as opposed to when you—so it's negative overall. If you wait a long time, everything's shifted that way, and you become happy, but not very often.
- 7.67 So what does that look like? Here it is, for example. So this is a simulation of this kind of thing going on. This purple line is looking only once a month at this profit. You're looking, you're observing it here and here and here and so on. Whereas this blue line here, you're looking every single—the ordering of things in the bottom is not very helpful because they're not in size order—that's every day, looking at your PnL every day makes you really miserable. And this is the kind of thing—the math is straightforward. Normal distributions, expectations.
- 7.68 You find that, for example, if you have these numbers, which are not far off what the S&P 500 numbers were, and $\alpha=2$, which just means the slope is twice on the left-hand side than on the right-hand side, you find that there's some period—this is τ —this is the time period, you find that there's some—if you want to be happy, you have to look at this more than every—you've got to wait a few months before looking at your portfolio. And what crops up, again, is the good old Sharpe ratio in this.
- 7.69 I love this picture. So much going on in this picture. This picture is very much the modern portfolio theory picture, with all the yellow dots of the individual stocks. Here we have the expected return. Here's the risk. And what the blue line is—remember we

said that, which portfolio you chose, depending on personal preferences. Well, it turns out that, actually, this pale blue line, this tells you—and this is the scale for the pale blue line, there's a minimum, this blue line drops off here. There's a minimum for, if you invest in this particular portfolio, then that means you can look at your portfolio more frequently than if you're invested in some other portfolio. That's the portfolio that gives you, that's efficient, in that it's on the efficient frontier, but it is better from the perspective of happiness.

7.70 The way you could use this—and this is all recent stuff—the way you could use this is, you're thinking, well, I've got a hedge fund, and I know that every month I have to go to my investors, which is a horrible experience, because it's like having a boss, which is something I've never experienced. And you say, well, I'm going to see them every month, so I'm going to choose my portfolio, so it's most likely to make them happy. I think it's quite cunning, as opposed to the usual reasons people choose portfolios. Because if your investors are happy, they'll leave their money with you. They might give you more money, because they're probably no more rational than anyone else.

7.71 So things to take away: many ideas in economics stem from assumptions of rationality, which I— it's one of those things that—economists, they herd more than anybody else. They should know all about herding. They all have the same crazy, crazy ideas. For them, it's like the trader, the junior trader. There's no incentive for them to do anything too clever. People are irrational, which I think is charming. And to some extent, even irrationality can be modeled. Definitely exploited.

Lecture 8: AI in Finance

At a glance — AI and machine learning in finance.

- 8.1** Machine learning, artificial intelligence. Oh, my god, this is already out of date. And by the time they finish editing these lectures, we'll all be living in caves, fighting off the robots. It's just a matter of how big our cave is. But seriously, this is out of date, and there's no way I can teach what's going on in machine learning and finance. There's going to be current. So I have to just give you some background waffly stuff.
- 8.2** For example, I'm classically trained, don't you know? So I'll tell you about how machine learning is different from classical mathematical modeling. It is completely different. Chalk and cheese. I will give you some of the principal techniques of machine learning. I'll give you some jargon and mathematical concepts, and I will end up with a few examples, which are mostly finance or could be applied to finance, and some of the examples are—what's the word?—wrong in the sense, as you can see. It's me lowering your expectations. But you can learn a lot from when things go wrong. So I'm getting it all wrong, so you don't have to.
- 8.3** All right, the definition from Wikipedia, again, it's something to do with computers and statistics and learning and data. And the key thing, really, is the learning. That's why it's called machine learning. Learning. Learning is key. It's normally what you do if you're trying to model something. We, right at the start, we looked at some, we had some data. We did a little bit of spreadsheet stuff with that data, and we saw bellish-shaped curves, and we had means and standard deviations.
- 8.4** But we didn't really do anything sophisticated with that data. We just looked at expectations, averages, and standard deviations. We didn't look at the nuances of how share price moved one date to the next. But when you're doing machine learning, everything is about data. It's squeezing every last drop of insight or knowledge from that data, as opposed to just eyeballing it a bit. So I made some observations, didn't I, about how, really, all we cared about was the—we didn't care about the level of the stock price. That was the clever bit.
- 8.5** You wouldn't do that if you were doing machine learning. You wouldn't be, nothing clever about what the machine does. So I used that little bit of common sense, and then we ran with it. Sometimes you have a bit of common sense, and you run with it, and you find it all falls apart, and you're back to the drawing board. We write down some differential equation. You might spend months on solving some differential equation that you think makes sense and discover that it doesn't make sense after all, and back to the drawing board, and try and think of something else.
- 8.6** If you're trying to model the relationship between interest rates and inflation, then you might think, well, economists tell me that if they increase interest rates, then inflation goes down. I don't know why, but apparently it does. So I'll do some differential

equations to capture this behavior. And then you find it doesn't make any sense, perhaps.

8.7 Machine learning is different. There are different types of machine learning, which I will talk about. And so what you do is, you have some algorithm, and you feed the data in one end of the algorithm, and it thinks, it learns, and figures things out for itself, and comes up with a model. Details in a second.

8.8 A fascinating bit of history involves the game of noughts and crosses, or tic-tac-toe. And this goes back to a Scotsman, Donald Michie, who used to work during the Second World War. He worked with Alan Turing at Bletchley Park, the spy place on code-cracking, et cetera. Then later on, he, as a professor in Edinburgh, he built something, which in a technical sense might be called a computer, but it wasn't a computer. It was actually a lot of matchboxes, which played the role of a computer in this, and you'll see how it works. It was very clever.

8.9 But this illustrates how something might learn. So this is the first time something learned. And very often, you'll see examples of learning to play games. In the 1980s, video games were always being used as examples of training algorithms. So the way it worked was, he got, I think it was 300 and something, 14—is it 314, 300 and something—304 matchboxes, and on each matchbox, he drew a state of play. State is a technical term.

8.10 So you might find yourself in this position in a game of noughts and crosses. If it's the crosses' turn to play, then as a human, we know, well, we could put a cross there, or we could put a cross there. Yeah. All right. So there are a couple of places we could put a cross, and we will win the game of noughts and crosses.

8.11 Now, that's obvious, and you could imagine writing a piece of code in which you had, if—and then you list there's a nought here, a cross here and da, da, da, da, then put your cross in this square. You can imagine writing a piece of code. It'd be very boring doing that. It would be quite a long piece of code. It'd be very boring, but it would be impractical for anything other than the game of noughts and crosses.

8.12 So what he did was, he got these 304 matchboxes, put these states of play, and there were 304 because there must be 304 possible states, because there's rotational symmetry and reflections and whatnot. So he got it down to 304 possibilities. He then put inside each of these boxes very small beads of different colors. Now, the different colors would represent the possible moves. So here you've got that, that, that, that, that. So it would be that, that.

8.13 Which ones was it? Yeah, so he would have red, green, those three. He'd have, I don't know how many, but let's say a dozen of each of those five colors because those are the five possible places you could put your cross. So you put those beads in the box, and I've got a picture somewhere of his set-up. That's his set-up. And you start playing. You start playing against somebody, and they make their nought somewhere. And he

goes, he says, "Oh, right, which state are we in now?" He goes to the box that has the same picture on it, representing the state that they're in, shakes the box, and randomly picks out a color.

- 8.14** That color—so he puts the box back where it was, and he puts the bead to one side, and that tells him where to put the cross. The player now puts another nought somewhere else. So we're now in a different state. So you go to the matchbox that represents [searching sound], and that tells you where to put the next cross. And then you go through to the end. Now, if you win, then clearly, at the end of this game, you've won noughts and crosses, then these beads led you through a path that was the optimal path. It was a path that worked, shall we say.
- 8.15** So you go back, and you add one or two—I don't know the details of the algorithm—but you go back, and you put beads of the same color back in the box that they came from. So if your first one was a red, then you put the red back in that box, and you put another couple of reds in there as well. On the other hand, if you didn't win, then you take those beads, you take them out, and you throw them away. And then you keep doing that over and over again.
- 8.16** In this way, you will—any winning strategy or play will end up with more beads in that box and fewer of the ones that don't win. So eventually, you get to the situation where, in any particular state, it will only have one color in it, and that means that is the best thing to do for that state. And that's called reinforcement learning, which is different from what you would do if you were doing classical mathematical modeling. You would write that code; you would analyze each position.
- 8.17** Now, people have done things like blackjack, for example. Even I've played around with this reinforcement learning for blackjack, and it works. Blackjack is, you've got two cards, or actually, later on in the game, you've got however many cards you've got. It doesn't matter how many cards you've got. What does it add up to? What's your count? And what's the up card that the dealer has got?
- 8.18** So you can imagine a grid of saying that the dealer's up card is ace, 2, 3, 4, 5, 10. And you've got a count of 12, 13, 14, up to 20, because no one's going to take a card on 21, and everyone's going to take a card on 11, if you know the game of blackjack. So that's about 200 states that you could be in. If you know about blackjack, there's aces counted as 1 or 11, there's hard and soft and all these kind of things, and doubling down, things like that we're not going to talk about. But the idea is, you can imagine playing a game of blackjack, and if you win, then somehow that state, or what you did in that state, is reinforced.
- 8.19** That's completely different from classical mathematical modeling. If you're doing classical mathematical modeling, you might be trying to model fluid mechanics or something. You've got to decide on what are called variables, which are the—you'd have the things called the independent variables. And if it's fluid mechanics, you're

talking about X, Y, and Z to represent where you are in three-dimensional space. And T, time.

- 8.20** And then you'd have dependent variables, which, in the fluid mechanics case, would be the velocity of—that's the speed and the direction—of the fluid particle that is at the point X, Y, and Z. So you've got independent and dependent variables. Okay, we know what we're dealing with. You try not to have too many independent variables. If you don't need it, you won't have the temperature and things like that. But maybe you need it.
- 8.21** Try to keep things as simple as possible. Yeah. "Make things as simple as possible, no simpler," said Einstein, which is so different from machine learning, where you say, "Well, data, I want more data. Give me more data. Give me more data about the—I want a video of fluid moving around something, around a tree. Give me some video," and my algorithm will analyze the video to come up with a model for fluid mechanics. But I need more, I need thousands, millions, billions of videos of fluids flowing around things.
- 8.22** As opposed to, as I mentioned the other day, fluid mechanics. You just sit down, and you say, "What do I know about fluids?" as he swirls his whiskey, his single malt in his glass. "What do I know? Fluids, they're conserved. They never disappear." Sorry. Yes, the conservation of mass. Conservation of momentum. It gives you the equations, right? So you've got to use your brain. So you have main drivers in the system: conservation of something, conservation of mass, conservation of momentum. You don't use that if you're doing machine learning parameters.
- 8.23** In fluid mechanics, you've basically got viscosity, and half the time, you don't even have that, and density. I think that's viscosity, two parameters. You will read in the papers about the training of all these algorithms, how much energy it's using up to train algorithms. Crazy amounts of energy. All a mathematician needs is the single malt and parameters. I used to say, fluid mechanics, you've got viscosity and density.
- 8.24** You will hear about how these machine learning algorithms inside their black box have got billions of parameters. So you wonder, when some black box, some algorithm, is learning about fluid mechanics, has it somewhere inside the black box, which you can't see, does it somehow know about conservation of mass? Does it know about conservation of momentum? Where are the two important parameters? It's so different.
- 8.25** Onto the kind of methodologies that people use in machine learning: there are three principal categories. The first one is supervised learning. One of the things I'm going to tell you about the jargon is about things called feature vectors. Now, assuming you're all on Facebook or something, then there's all sorts of information that is being stored on each of you, what things you're looking at, what ads you respond to, et cetera. So Mark Zuckerberg has a feature vector on each of you.

- 8.26** And what I mean by a feature vector is, a feature vector looks like this. It's a vector, a column—I'm going to talk about column—a column of numbers. The top number will be your date of birth. Then it will have your salary, and then it'll have your zip code, and number of children, and number of pets. There'll be a column of numbers all about you. So that's one person. Everybody has the same thing with date of birth, salary, et cetera, et cetera, shoe size. Going across the top, everyone, it's the same quantity that's being measured. That's the thing called a feature vector.
- 8.27** So when you're doing any analysis, doing any algorithm, training an algorithm, that goes in. But also, you might be trying to predict something—and this is supervised learning—you might be trying to predict, I don't know, where are you going to go on holiday? So, where are you going to go on holiday would play the role of the dependent variable. This is the independent, in classical maths, the dependent, the thing you're trying to predict. So it will have all this data on everybody who is on Facebook about the holidays they've been on.
- 8.28** And then, along comes somebody else who's never been on Facebook before, like me, and very quickly, Mark Zuckerberg will have my feature vector, and then, having been trained on all of you, he will then come up with a prediction for where I should go on holiday. So that's what's supervised learning is. It has the inputs, the inputs, which is the feature vector, variables, and the output is a thing that it's trying to predict effectively.
- 8.29** Unsupervised learning—this is one of my favorite things because it's a bit crazy. You don't know what it's going to do. Unsupervised learning is when, again, there are feature vectors, but you haven't got any—it's looking for relationships between these feature vectors, rather than trying to predict something.
- 8.30** The great example would be, you take your contents of the drawer in your desk, and you ask the algorithm to sort them out. Now, if it was supervised learning, you'd take them out one by one. You'd say, "Ah, that's a paper clip. That's used for holding bits of paper together." So you'd measure the dimensions and the weight and everything of this paper clip and what it's made of, and it would be labeled as "thing for holding bits of paper together." Then there's a stapler and so on. That's putting bits— And then, there's, I don't know, this is a pencil, the dimensions of the pencil, and that goes in the category of "things that you write with," and everything gets categorized like that. I don't know what that is. That would be supervised learning.
- 8.31** Unsupervised learning is where you just say, "Computer, that's the contents of my drawer. Sort it out. You figure out what things are like each other." So that's fun because you don't really know what it's going to do. I'll give some examples of that later on, I think.
- 8.32** Then you have reinforcement learning. This is where the algorithm is learning to do something, which is the tic-tac toe. It's learning to play a game. Another example would

be, does anyone see the AlphaGo movie? It's really depressing. It's probably the most boring title of any movie ever, but it's about Google DeepMind, training an algorithm to play the game of Go. The game of Go is, you have these black and white stones that appear on a 19-by-19 board. You can't address this problem using matchboxes; you'd need more matchboxes than fill the known universe, because of the crazy number of possible states that the game of Go could be in. So they combined different machine learning techniques. But again, you would have the computer playing against itself and learning from how it plays in the game of Go, to converge to some optimal strategy. And AlphaGo is about how a 9th dan professional Go player lost depressingly to the computer.

- 8.33** So within those three principal categories, we have various techniques. So I'll go through these techniques, and then we will do a few examples of these techniques, on real examples. The first one is a thing called k-nearest neighbors. It's a supervised learning technique in which—imagine that this feature vector is—each of these are numbers. So you can think of, if there were just two numbers, height and weight or something, then that's like having X and Y coordinates. So you could visualize X. One number here, another number there, means move this way and this way, there. If it's three, then it's there, there, and it's a certain distance up. It's a three-dimensional thing. Once you're beyond two or three dimensions and you've got a thousand dimensions, then you can't visualize it. It's still a dot in n-dimensional hyperspace.
- 8.34** Now, with supervised learning, let's take the case of just the two dots, a two-dimensional thing. It puts you there, and that is labeled. For example, a silly example would be, you're at a party, and you see these groups of people around, and you walk to one of these groups of people, and where you stand is an X, Y coordinate. So that's your feature vector, where you are standing. You look at the little group that you're chatting to, and each of these people is labeled, because this is a supervised learning technique. It's labeled with: Are they wearing spectacles or not? You see that there are five people around you, and three of them are wearing spectacles.
- 8.35** It's a silly example, but in supervised learning, you stand there, and you look at your nearest neighbors, and you conclude, "I must be, I'm probably wearing spectacles because I've stood in this group." Now, obviously, that's a silly example, but it does physically illustrate why this would work, that obviously there'd have to be some connection between what, the fact that you're standing there and why people are wearing glasses. But in proper examples, that might work. It's very simple, really, really trivial.
- 8.36** You literally look at your nearest neighbors and conclude something from people who have a similar feature vector to you. But of course, it's really, it's a 100-dimensional feature vector. It's in 100-dimensional hyperspace, so you can't visualize it. It might be something to do with dogs. It might be the height, length, breadth, color, weight,

length of the fur, et cetera, of the dog, and then it might be categorized as "it's a poodle." So that would make more sense.

- 8.37** K-means clustering is an unsupervised learning technique, which is similar to the k-nearest neighbors, but it's about—so we have all these dots, these feature vectors, and it tries to look at: can they be grouped together? We'll give an example of this shortly. Now, it's very easy if I just did a few dots on here. I might do something like this. You'd say, yeah, there are two clusters; those are two different types of people. But of course, once you're in four or more dimensions, you can't see it. I will give an example of this later on.
- 8.38** Naive Bayes' classifier, this is a good one. This is often used on speeches and writings, et cetera. It might be used to, Amazon, to conclude from something, a review someone has written, is it a good or a bad review? The way it works is, if somebody says something like, "Property is theft, comrade," what's the chance of that person being left-wing as opposed to right-wing, for example? I'll do an example of that shortly. It uses a thing called Bayes' theorem.
- 8.39** The naive bit is interesting. It's counterintuitive. Naive here means that, when you do the algorithm, you completely ignore the order of words and the context. All that matters is the individual words in the sentence, rather than whether there's a "not" in front of the word; obviously, it would change the meaning completely. But it doesn't matter if you go around just assuming that all these words are independent of each other.
- 8.40** Regression. This is just stats, made-up data, again. Salary of lawyers, obviously, there's a zero missing, versus the age of lawyers. Is there some relationship between the salary of lawyers and the age of lawyers? So you draw some dots, and then you try and fit a line. Sometimes it might be a straight line, sometimes it'll be a curved line, depending on what you're trying to model. That's just basically stats. There's a lot of machine learning, which is stats. It feels like they've rediscovered half of mathematics and given them different names.
- 8.41** Support vector machines, that's another one. It's a supervised technique, but where you've got all these dots, and these dots represent cats, and these dots represent dogs, the physical characteristics of cats and dogs. But you've got so much data. There's so much feature vector for each of these data; the dots are, again, in such high dimensions, you can't visualize it. What you try to do is find a plane, what's called a hyperplane. Can you draw a line between the dots that separates the cats from the dogs?
- 8.42** Decision trees is like the game of—do you know the game of 20 Questions? You might be surprised how many nationalities do not know the game of 20 Questions. For example, I'm thinking of an actor. Who is the actor? So you start off by saying male or female. There was some list of 100 actors I found somewhere, and I'm thinking of one

of them. Male or female? Female. Alive or dead? Dead. You keep going through all these things. The trick to decision trees, it involves information theory and entropy and fun things like that, and lots of algorithms.

8.43 The goal, really, is not to go around randomly guessing people: Judy Garland. No, there goes one of your questions, and you're down now to 99 people. Well, that wasn't really much use. Ideally, what you want to do—and this is an information series— with each question, you want to halve the number of people. If, with every question, you can halve the number of people, one question gets you down from 100 to 50, 25, 12 or 13, six. By about seven questions, you've pinpointed the person. Whereas if you're going to go through all the dead female actresses, then you are unlikely to be right. So you've got to try and—

8.44 When I did this, I was trying to write about this for a book. I was looking for things that divided this 100, a list of 100 that I had, exactly down the middle. And it wasn't alive, dead, male, female. The thing that divided them, 50-50, so the first question you ask—you'll never guess what it is—is, is there an L in their name? That's the first question. That gets you immediately down to 50 people. Little tip there, if you're not playing competitive, after your competitive Snakes and Ladders, Chutes and Ladders.

8.45 Self-organizing maps. I've got an example of this. This is really fun. I like this because it's an unsupervised learning technique. A self-organizing map. We've got all these vectors with 100 different features. We can't visualize this. So, self-organizing maps is a way of visualizing things in two dimensions. Think of a chessboard. And what you do is, each square in the chessboard is seeded with another feature vector, just generated randomly. And what then happens is, each of you, one after another, has to go and stand in the square. It has got a feature vector that is closest to you.

8.46 So this is you, and each square has another, totally random, randomly generated vectors. You just look, and you measure the distance between your vector and each of the vectors, very simple mathematically, and each of these vectors. And then you go and stand in the square that is closest to you. Then something weird happens. The random feature vectors in all of the cells are then, in some weird 1950s sci-fi way, they evolve slowly to be more like you, like Invasion of the Body Snatchers or something.

8.47 And the cell that you are standing in, the square you are standing in, looks more like you. The cells around you, the squares around you, a little bit like you, but not quite as much; and the ones further away, they change a bit to be more like you, but not so much. Then next person goes around, looks at all these cells, and figures out which is the cell that's most like you. And you go and stand in that cell, and the same thing happens.

8.48 Then all of the cells, the feature vectors, these random things, evolve to be a little bit like you. And then we go through everybody, and then we start again. And eventually, it all settles down. And you'll be sitting, you'll be standing in cells that represent you

and people like you; and other people will be in cells or squares further away, meaning they're not at all like you. So suddenly, you've gone from this 100-dimensional feature vector to all standing in little groups that represent what you are like. I'll do an example of that.

- 8.49** Neural networks. This is a really big one, neural networks. This is applying the idea of what goes on in the brain, apparently, to an algorithmic version of this. Again, we start off with various vectors, and we do a lot of—should I draw a picture? Let me draw a picture. Mathematically, this is what happens. And you always do it with little circles. So this is your feature vector. This is the input. So this is your feature vector. And we're trying to predict what salary you deserve. This is your output, which is going to be salary. So this is all your qualifications and your age or whatever goes in here.
- 8.50** And then what we do is, we have what's called a hidden layer of more of these nodes, these cells; and all of these numbers go in here. All these numbers go in here, and they do a little bit of a—they multiply the numbers and add numbers and all sorts of stuff like that, and this goes in here, and more things go in. You see pictures like this, where everything is going into all of these nodes. They do all these—obviously, this is going to take me forever to draw every possible combination, but you get the idea.
- 8.51** And they do something here, which is what happens in the brain, which is, if the signal, if the calculation of all these numbers that go in here is big enough, great. If it's not big enough, it gets ignored. That's kind of like, you get hit on the hand. If it's not very much, you ignore it; but if it's hard enough, you pull your hand away. That's the idea. And then once they've done that, they do more of the same, the numbers going in here, et cetera.
- 8.52** That's what neural networks look like. To me, it's just messy—functions of functions of functions of functions. But it's very, very, very common. And reinforcement learning. This is where the game or whatever, the algorithm is rewarded or punished. That way it learns—I keep doing this—and that way, it learns how to do something.
- 8.53** Jargon, and then I'll do the examples. Jargon. I've talked about data. I've talked about the feature vector. Classification just means that you, well, there are two things. Regression is when you're trying to output a number, like a salary. Classification is when you're trying to output a class of things. For example, you might have a fruit and the dimensions of the fruit, the three dimensions, the color, the weight, et cetera, would be the feature vector, and you're trying to predict, is it an apple, a pear, or a banana, or something? So that would be a classification, rather than a number.
- 8.54** Cost function is very important. This is used in lots and lots of different machine learning techniques. It is a measure of how accurate your model is. For example, in the case of the made-up data of lawyers' salaries versus age, I had these made—I just put these dots in, and I drew a straight line. Well, what straight line captures this

relationship best? What you often do is, you take, how far away is the real data, the dot from my forecast, which is the line?

8.55 And then I take these numbers, that distance, square it, square, square, square. So it doesn't matter which side you're on, because you're squaring. Add up all those—that's like an error. So you'll hear things like ordinary least squares, et cetera. So you measure the distance of the real data from the predicted data, predicted being the line, square up all those errors, add them all together, and then try and minimize that. So find that. Obviously, a line up here is no good, line like that is no good, but some line like that is going to be very good, because it minimizes this error function or cost function.

8.56 Another way of choosing your—so that's all about—because what's that doing? What that does is it chooses the parameters. Remember, I said you have lots of parameters in these models. Here, the parameters are simple. How high up is the line, and what's its slope? So that's only got two parameters. Although, as I say, typically these days, if you're doing some neural networks, go back to the neural networks.

8.57 How many parameters has this got? I've got five things. One, two, three, four, five, six, nine. Each one of these is going to be—so there's five of these, nine of those. So it's going to be 45 lines. So there's going to be 45. You can multiply each of these numbers by some parameter. That's 45 of them. But then you add on another one in each here. That takes us up to 54, so it's 54 here. And then you've got nine going in there, and add a constant. So that's adding a 10, I think. So this is going to have something like 64 parameters. I may be wrong in that, but it's something like 64. Ballpark.

8.58 In practice, though, your feature vector is much bigger. You've got a larger hidden layer. You may have more layers. You may have a bounce, going from layer to layer to layer. You could have thousands of millions of—and when you're talking about something like the game of Go, you're talking billions of parameters, probably. So that's what the cost function is, to try and find those billion parameters that minimize the cost function.

8.59 Here's a fun thing: maximum likelihood estimation. This is a probabilistic thing, but also choosing parameters. For example, suppose you go to a city, and you get into a taxi, and the taxi has a number. Having nothing better to do, you're trying to figure out what's—I know what the number of the taxi is. It's 1234. How many taxis are there in this city? And you make some assumptions, like the positive integers, no numbers missing, no numbers repeated. And so, you think, well, what's the—taxi number 1234, what's the most likely number of taxis in this city?

8.60 It's easy. You just say, suppose there are N taxis. We've got N taxis in the city. What's the chance of me getting into taxi number 1234? Well, there's N taxis. The chance of getting any particular number is $1/N$. Okay, so I want to make that as high as possible. How can I make that as high as possible? I need to make n as small as possible. And what's the smallest possible N ? Well, I know it's going to be at least 1234. So what

makes that probability, this likelihood, the highest is saying that $N=1234$. It seems odd, the chances of me getting the highest-number taxi, but that's the maximum likelihood estimation idea.

- 8.61** Training and testing. Everything has to be trained and tested. What we do, what we often do, is we take our data that we've got, all of you fine people, and all your feature vectors, and put a quarter of you to one side, and 75% are used to train the algorithm, minimize a cost function. And then, we show the algorithm, the people we put to one side, the test people, and we see how well the algorithm does in terms of errors. And you might find something, for example, the algorithm—
- 8.62** So this is the error: as you train more and more, the solid line. Then you hit it with the—you show it the test data, and it does very well. Not as good as training data, but quite good. But often, you might find that it looks like the training data has done very well, minimized the error, but actually, the test is getting worse. And the reason for that is, you might be overfitting. Here's an example where, if you take a simple model like a straight line, it does a really good job of fitting.
- 8.63** But if you think it's not good enough, you want a more complicated model, well, you might have some sort of polynomial of degree whatever. You can get a perfect fit, you can get an error of zero to your training data. But look, does that really—does this dotted line, does that look like a better model? As soon as you start throwing in, you come up with a new number, and it's here. Well, it predicts here. Way off because you've overfitted. Underfitting is obviously the opposite of that.
- 8.64** All right, let's do a few examples of that. First, financial one. Volatility. We've talked about volatility. I've mentioned that often we assume that it's random or jumps, stochastic, et cetera. Here I've got a model which looks at jumping volatility, and I'm going to use k-means clustering, and it's really simple, because k-means clustering is when you're grouping things like this, and you want to figure out how many clusters are there.
- 8.65** What I've done—and this is simple because this is not a 100-dimensional thing; this is actually one-dimensional—I asked the question, let's suppose I've got a mathematical model for options or something, or for portfolios, whatever, that uses volatility that jumps. It's a model for volatility where the volatility is jumping from one state to the next, from low to middle to high, back down again. Let's assume there are three possible states, just because my algorithm, my program, asked me to input three volatilities, three levels.
- 8.66** So what I did was, I took this data, which is data for S&P volatility, and I said: what are the three levels of volatility that best capture this? A low one, a middle one, and high one. Actually, I didn't say low, middle, and high. I just said, "Tell me the three that capture it best." And what it did is, it said the three levels are 9.5%, 18.8%, and 44.3%. It found that, most of the time, 586 out of all the data points I had, the S&P volatility

was around 10%. Then there were quite half as many, around 19%, and then there were a small number at a very high level of volatility. Now that was useful because, if I am going to be talking about what is volatility going to be doing, I can think of three different regimes. Rather than worry too much about the level of volatility, I can think in terms of, three regimes is a good model.

- 8.67** But then, I want to know—and this is where I try to bring in some time dependence— I ask the question: how do I know when I'm going to go from one regime to another? And so I just ask the question: how often, out of all—whenever I'm in cluster one, and the cluster one is, look at the darker line here, the gray line is the actual volatility. The darker line is the nearest cluster. So here, for example, these bits at the bottom are when I'm in the lowest volatility. What's the probability of going to another cluster?
- 8.68** So, I literally looked at this data, and I counted the number of times I went. Clearly, here, it hasn't changed. But then here, this last one moved up to the middle level. I literally counted, out of all the times that you were in the lowest vol regime, how many times do you jump to the next regime? Or even, can you jump from the lowest regime to the highest regime?
- 8.69** What I found was this. I got this table. It's called the transition matrix. It tells you the probability, if you're in cluster one, then 84% of the time, you stay in cluster one. But there's a 16% chance you jump to cluster two, which is the 19-20% one. You never go from cluster one to cluster three. And cluster two, you could go down, stay the same, or there's a small chance of going all the way up there. If you're in the top cluster, the 44%, you're probably going to fall back down. You don't stay in the top cluster for very long, typically one or two days, and then you fall back. Yeah, so you fall back down. So, if you're modeling volatility, that was a very simple model for volatility. Reassuring, when you're in a high volatility regime, you don't stay there very long. So, that was one financial example.
- 8.70** Now, here's a good one. This is the example that goes wrong. Have you seen "The Play That Goes Wrong"? This is the example that goes wrong, but it's useful. I have taken eight speeches—I think it's eight speeches. First of all, disclaimer, you don't take eight speeches. If you're going to analyze speeches, you take thousands, tens of thousands of speeches, hundreds of thousands of speeches, not eight. But this is for illustrative purposes only. This is not meant to be a research thing.
- 8.71** So I took, for example, the famous start of the Communist Manifesto, by that layabout Karl Marx and the poor sucker who funded him, Engels, the start of the Communist Manifesto. "A specter is haunting Europe—the specter of communism", et cetera. So I took these speeches. Well, whose speeches have I got? These are speeches I took. Churchill's famous Beches, JFK's inaugural address. Was that the "ask not what you can do—what the country can do for you"?, et cetera

- 8.72** Tony Benn, you won't know him. He was a left-wing politician. He was one of the ones that had some sort of hereditary title, but he was a very principled man, naive as anything, but principled, and so principled that he gave up his hereditary title to become an MP. Margaret Thatcher, Theresa May, the Prime Minister we had a few years ago, didn't amount to much. You won't know Jeremy Corbyn, thank goodness. And the Donald, his first State of the Union thing. And I've classified these people as left or right, along with Karl Marx.
- 8.73** The way this works is, I looked at all the words that they used, made a list of them, for instance, the left-wing people, left-wing politicians, and the probabilities of the frequency of these words in their speeches. So the left-wing words, they seem to— this is just what happened. They seem to like the word abolish, abolished, abolishing, abolition. They seem to like abolishing things. I'm not saying anything. It's what happened. It's the data. The right-wing words, a bit of addiction in there. But then there's advance, advantage, advantages. There may have been a—yeah. Obviously, I chose which words to—but I wanted to show you the idea.
- 8.74** Anyway, then you take another speech by someone. Oh, yeah, sorry. There's a few tweaks that you have to do. There are some words we just throw out. They're called stop words, like meaningless words, et cetera. And also, there'll be some words that are in speeches, if we're trying to classify a new speech by someone—is it left-wing or right-wing?—then there'll be some words that—Karl Marx hardly ever referred to the internet, for example. So obviously, words like that, we treat them separately.
- 8.75** So I used this probabilistic idea, and I took another speech. I was trying to classify this. So I don't know whether you recognize this, because I've taken out all of the stop words. So this speech goes, "Happy, joined today, down history, greatest demonstration, freedom, history, nation," da da da da da. So that's with all the stop words taken out.
- 8.76** And then, what I did was to pretend that this person was giving this speech, and we were analyzing it word by word. Pretend it's some CEO or something. He's giving a speech, and is he going to say something positive or negative? Should we buy or sell? So we're hanging on literally every single word. And as each word came in, I would modify the probability. So this is what happened. And so, that's a real thing that people do.
- 8.77** As the speech came in, this particular speech, the probability started at 50-50, whether this person is left-wing or right-wing, and he eventually hit a probability. This is the probability of this person being right-wing. Now, I don't know if you know what this person is, if you recognize that speech, but this person is not known for being particularly right-wing. It was Martin Luther King, Jr. So that is where it goes—does it go wrong? What has it done?

- 8.78** Now, obviously, I've only taken eight examples, eight speeches, for my training, which is not enough. But it does illustrate marvelously that it's predicted something. And that's actually one of the great things about machine learning I should have said earlier; it always gives you something. You always get an answer out, always. Whereas if you're doing classical mathematical modeling, you might, on your bits of paper, come up with some wonderful differential equation, and somebody might say, "Yeah, Paul, that's a great differential equation, but you know there's no solution. It violates something or other." Okay, back to the drawing board.
- 8.79** That doesn't happen with machine learning. Something always comes out. But does what come out make sense? I don't know. Maybe he was— shut up, Paul. Or most likely, what it's done, the algorithm's done, maybe it's just picked up that he gave a great speech, whereas he's up—the other politicians, the left-wing politicians I've got in there, Tony Benn, bless his cotton socks, Jeremy Corbyn, they did not give great speeches. So maybe what this has actually done is something unintended.
- 8.80** So that's the lesson of today, is that it always does something, but you don't know whether it's doing what you wanted to do. So you have to go to a lot of trouble. So if I were doing this for real, I'd have a lot more speeches, and I would have done testing. I would have held back some speeches to test it. So I didn't do any of that. I used eight speeches. I trained it, but I didn't do any testing. OK.
- 8.81** Another one with speeches, but this is a regression example. This was fun. So this regression, remember, it's like the lawyer and the salary, the dots and drawing a straight line. What I did here was to take speeches by eight CEOs, and rather than analyze the individual words, there is a famous dictionary called the Loughran-McDonald Dictionary, where it classifies each of these words according to different features.
- 8.82** The features are: is the word, is each individual word, a negative word, a positive word, a word representing uncertainty? Is it something representing litigiousness? That's great for business purposes. Constraining—whatever that is. Superfluous. Interesting. Modal—I forgot what that is. Irregular verbs. Harvard IV-4 is some psychosocial thing. A number of syllables.
- 8.83** So I took their speeches, and I looked at how frequently they were using different words. So the NatWest CEO, 1.1% of his words were negative, 2.28% of his words were positive, and so on. And so that is your feature vector. And then I looked at the return following the speech, the year following the speech. So, are there any clues in the speech about what the return is going to be?
- 8.84** Now, I've only got eight CEOs, a very tiny number, and I've actually got about a dozen or so qualities or numbers in this feature vector. So if I'm looking for a fit, then what it amounts to is having eight equations in a dozen unknowns, which means I'm guaranteed that's big-time overfitting—sorry, big-time overfitting? Yeah, big-time

overfitting. Is that what I mean? Yeah, I've got so many parameters that I'm bound to get perfection.

8.85 So what I did was, I thought, that would be cheating. That's so easy to criticize. So what I did is, I restricted which ones I was going to look at. I only looked at positive, negative, uncertainty, and litigious. So, in other words, I ignored most of these categories, just so I could do a sensible thing where I've got more equations than I've got unknowns, so I'm not guaranteed anything. And then I did this regression. And what I found was this. This is each of these eight companies. This is the actual return, and this is the fitted return. So it's all heading in the right direction.

8.86 And then I put in a new thing. I took a speech by the Credit Suisse CEO. They had collapsed at the time of me doing this, so I thought that was a good—is there anything in the CEO's speech that would lead me to see if—could you tell from the CEO's speech that they were about to collapse? The answer is: Oh, yes. So this one worked, but then equally, if the other one didn't work, maybe this was just luck, but it was really impressive. This was what actually happened: it actually fell 80%. I predicted a 60% fall. I could have made a killing if I'd done this before, but I had no idea it was going to do this. So this is a genuine, proper test of what happened.

8.87 The interesting thing, though, I found in this is, when you look at the parameters—remember the parameter here? These are the things you look at. Those are the parameters that you multiply each of those, whether it's negative, positive, et cetera. They're all negative. You multiply those numbers, which represent the weight, how important it is, in prediction. Every single one is negative, which means that every time a CEO opens their mouth, the stock market falls or the stock falls. Do I believe it? Obviously not. But it's interesting. So just zip it, and your company will be fine. Obviously, if I were doing this for real, I'd have thousands of CEO speeches. I'd do the training, and then it holds them back for testing and so on.

8.88 The final example is self-organizing maps. I've described that. I am going to do the following. I take the constituents of the S&P 500 index and look at the last five years' worth of returns, so return every year for five years. So that is what my thing is. My feature vector has just got five entries in it, and they are all numbers, they're all returns. So it turns out, there are only about 400 and something, because over five years, some companies leave the 500 and other ones come in. So they're only about 400 and something, which stayed for the whole five years.

8.89 And so, rather than doing what we did before, the modern portfolio theory, what I did was, I used self-organizing maps, which means that I tried this thing where you move people around the chessboard. Because I had 476 stocks, I thought, you don't want to have just a two-by-two grid, because that's too coarse, but equally, you don't want to have it too big. So I chose to have 25 cells or nodes.

- 8.90** And what's in here are the numbers of companies that were in each of these cells. So there's some cells—oops, not that pen—some cells which have not very many in, and then some cells which have a lot in. They're very popular. And then I looked at, for example, what was in cell number— this is the one with the most in it. And these are the ticker symbols of those. What is it about those stocks that mean they are very similar to each other? I don't know. This is an unsupervised learning technique, meaning you don't know what's going to come out. You just let the algorithm sort things out the way it sees fit. So there's something about those companies.
- 8.91** Now, obviously, you could use this for diversification purposes. You wouldn't choose all the companies. You wouldn't choose all of those companies in your portfolio. You'd choose companies from different corners of this grid, for example, if you wanted to diversify.
- 8.92** I did look at the sectors that these companies were in, and it was very interesting because you did find, for example, there are 25 cells in all. You did find that things like the energy sector were all in one place. Utilities are only in a couple of places. Real estate were in those particular cells. Financials seem to have two different types of financial and so on. So it has actually found something a bit like sectors. I didn't tell it about sectors, but it found something about sectors. So, actually, of all the things I did here, this is the one I believe the most, I think.
- 8.93** All right. So there we have it. Things to take away: we saw the three main categories of machine learning: unsupervised learning, supervised learning, and reinforcement learning, and lots of different techniques, naive Bayes, classifier, self-organizing maps, regression, et cetera. Very different from classical mathematical modeling, which is sad.
- 8.94** I saw a list a couple of weeks ago for one of these lists of which jobs are going to be made to disappear, thanks to AI. For the first time ever, in any list of anything, mathematicians appeared near the top. They were the third highest, the overlap between what AI can do and what mathematicians can do. I think it's something like, fifth was people who write books. So this is the end. This is my last talk ever, before I'm replaced by a robot. The thing is, be careful. It doesn't always do what you want it to. Thank you very much.